TRIUMPH CHURCH PRODUCTION RIG

Reference

Every page, key, trigger, variable, connection and signal path on the rig

Read off the live rig October 27, 2026 at 10:20 AM · Companion 5.0.7+9763-stable-cec2f88e6f

Contents

1. About this book
2. The deck at a glance
3. Signal flow: video
4. Signal flow: audio
5. Signal flow: control and network
6. Still to trace
7. Page 1: Home
8. Page 2: Power
9. Page 3: PP1
10. Page 4: MA2
11. Page 5: ATEM
12. Page 6: SQ7
13. Page 7: VW
14. Page 8: System
15. Page 9: PTZ
16. Page 10: CAM 1
17. Page 11: CAM 1 Setup
18. Page 12: CAM 3
19. Page 13: CAM 3 Setup
20. Page 14: Tracks
21. Page 15: Worship
22. Page 16: Sermon
23. Page 17: Dante
24. Page 18: Videohub
25. Surfaces
26. Triggers
27. Custom variables
28. Connections
29. Image library
30. Engineering notes: The rig
31. Engineering notes: Cameras
32. Engineering notes: Switcher
33. Engineering notes: Audio
34. Engineering notes: Recording
35. Engineering notes: Checklists
36. Engineering notes: Findings, 2026-08-23

About this book

Everything Companion holds, read off the live rig: 18 pages, 564 controls, 16 triggers, 68 custom variables, 12 connections and 249 library images. It is generated, not written: rebuild it after any change to the rig (`node tools/manual/collect.js`, then `node tools/manual/build.js`, on the AV network).

Positions are written `r1c3`: row 1, column 3, counting from 0 at the top left. Row 0 is the folder row, rows 1–3 are keys, row 4 the touchstrip and row 5 the knobs. The touchstrip and knobs exist only at columns 0, 2, 3, 5, 6 and 8.

No key pictures in this build: the rig could not be reached for them, so each key is drawn as its label on its resting colour. Rebuild on the AV network to add them.

No module labels in this build: actions and feedbacks are named by the modules' internal ids (`program`, not "Set input on Program"). Rebuild on the AV network to add the labels Companion's editor shows.

Credentials never appear: each is named by its 1Password entry in the Recovery Guide.

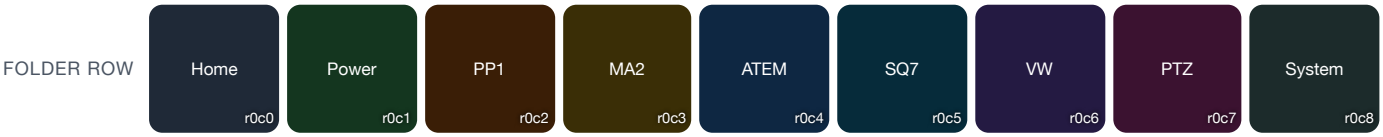
The deck at a glance



Page	Name	Controls	Reached from
1	Home	19	Folder row
2	Power	11	Folder row
3	PP1	40	Folder row
4	MA2	29	Folder row
5	ATEM	34	Folder row
6	SQ7	26	Folder row
7	VW	27	Folder row
8	System	19	Folder row
9	PTZ	17	Folder row
10	CAM 1	46	PTZ r1c2, CAM 1 Setup r1c8
11	CAM 1 Setup	43	PTZ r3c2, CAM 1 r3c3
12	CAM 3	46	PTZ r1c6, CAM 3 Setup r1c8
13	CAM 3 Setup	43	PTZ r3c6, CAM 3 r3c3
14	Tracks	16	PP1 r2c4
15	Worship	43	PTZ r1c4, Sermon r3c4
16	Sermon	44	PTZ r2c4, Worship r3c4
17	Dante	28	Home r1c2, SQ7 r2c8
18	Videohub	33	VW r1c8

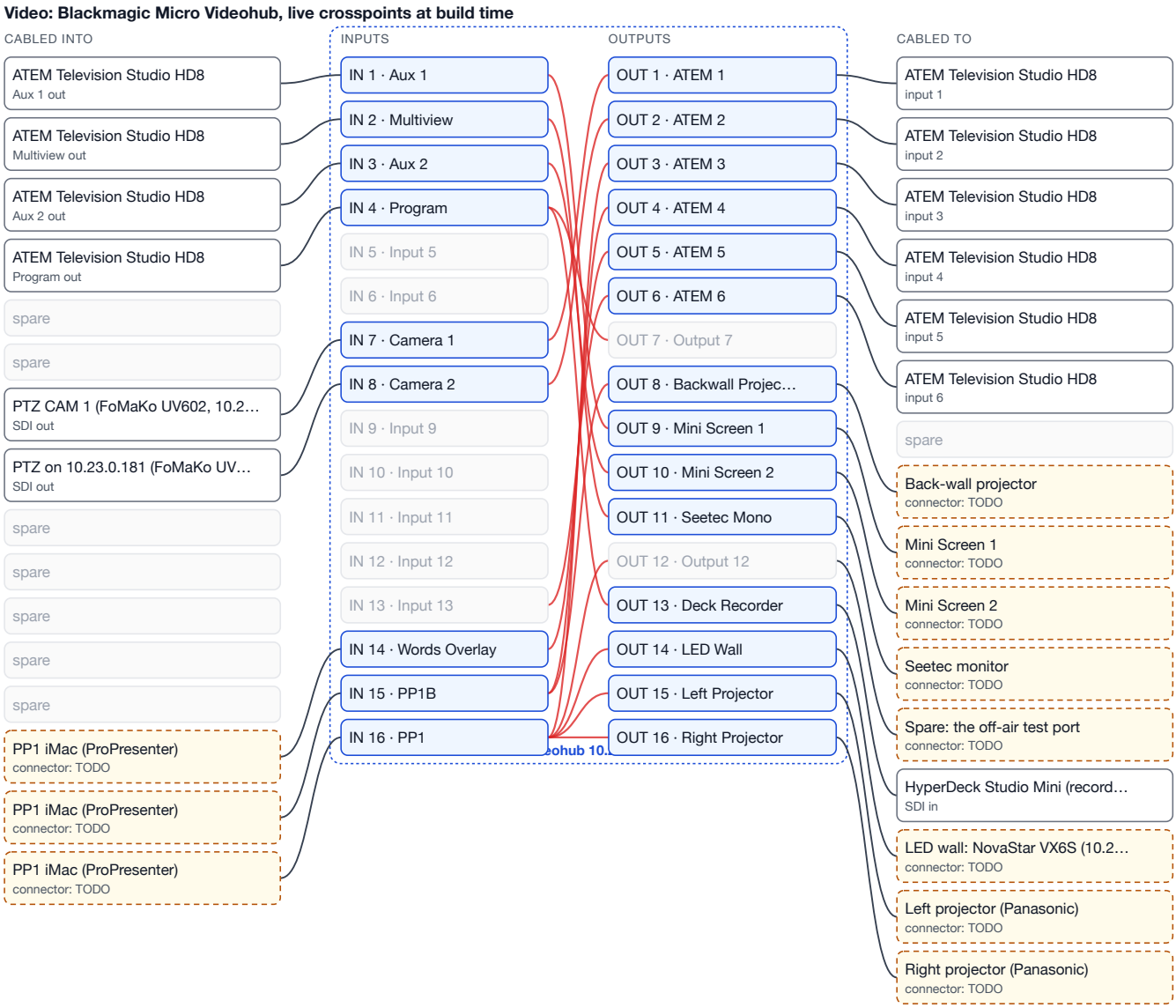
The folder row

The same on 17 of 18 pages, so it is listed once here and left out of each page below.



Signal flow: video

Solid lines are cables, by the names on the router's ports; **coloured lines** are the crosspoints the router reported at build time; dashed boxes are things nobody has yet looked at, listed under Still to trace.



Videohub ports

IN	Label	Cabled from	OUT	Label	Cabled to	Carrying now
1	Aux 1	ATEM Television Studio HD8 · Aux 1 out	1	ATEM 1	ATEM Television Studio HD8 · input 1	IN 7 · Camera 1
2	Multiview	ATEM Television Studio HD8 · Multiview out	2	ATEM 2	ATEM Television Studio HD8 · input 2	IN 8 · Camera 2
3	Aux 2	ATEM Television Studio HD8 · Aux 2 out	3	ATEM 3	ATEM Television Studio HD8 · input 3	IN 16 · PP1
4	Program	ATEM Television Studio HD8 · Program out	4	ATEM 4	ATEM Television Studio HD8 · input 4	IN 15 · PP1B

IN	Label	Cabled from	OUT	Label	Cabled to	Carrying now
5	Input 5	**TODO / spare**	5	ATEM 5	ATEM Television Studio HD8 · input 5	IN 14 · Words Overlay
6	Input 6	**TODO / spare**	6	ATEM 6	ATEM Television Studio HD8 · input 6	IN 13 · Input 13
7	Camera 1	PTZ CAM 1 (FoMaKo UV602, 10.23.0.196) · SDI out	7	Output 7	**TODO / spare**	IN 4 · Program
8	Camera 2	PTZ on 10.23.0.181 (FoMaKo UV602; deck pages "CAM 3", switcher "Cam 2") · SDI out	8	Backwall Projector	Back-wall projector	IN 15 · PP1B
9	Input 9	**TODO / spare**	9	Mini Screen 1	Mini Screen 1	IN 1 · Aux 1
10	Input 10	**TODO / spare**	10	Mini Screen 2	Mini Screen 2	IN 3 · Aux 2
11	Input 11	**TODO / spare**	11	Seetec Mono	Seetec monitor	IN 2 · Multiview
12	Input 12	**TODO / spare**	12	Output 12	Spare: the off-air test port	IN 16 · PP1
13	Input 13	**TODO / spare**	13	Deck Recorder	HyperDeck Studio Mini (recorder, 10.23.0.206) · SDI in	IN 4 · Program
14	Words Overlay	PP1 iMac (ProPresenter)	14	LED Wall	LED wall: NovaStar VX6S (10.23.0.23)	IN 16 · PP1
15	PP1B	PP1 iMac (ProPresenter)	15	Left Projector	Left projector (Panasonic)	IN 16 · PP1
16	PP1	PP1 iMac (ProPresenter)	16	Right Projector	Right projector (Panasonic)	IN 16 · PP1

Switcher inputs

ATEM Television Studio HD8, running **1080p59.94**. The input names are the switcher's own.

Input	Name	Fed by
1	Cam 1	Videohub OUT 1, now carrying IN 7 · Camera 1
2	Cam 2	Videohub OUT 2, now carrying IN 8 · Camera 2
3	PP1	Videohub OUT 3, now carrying IN 16 · PP1
4	PP1B	Videohub OUT 4, now carrying IN 15 · PP1B
5	Words Overlay	Videohub OUT 5, now carrying IN 14 · Words Overlay
6	Camera 6	Videohub OUT 6, now carrying IN 13 · Input 13
7	Camera 7	**not fed by the router: TODO, is anything cabled here?**
8	Camera 8	**not fed by the router: TODO, is anything cabled here?**

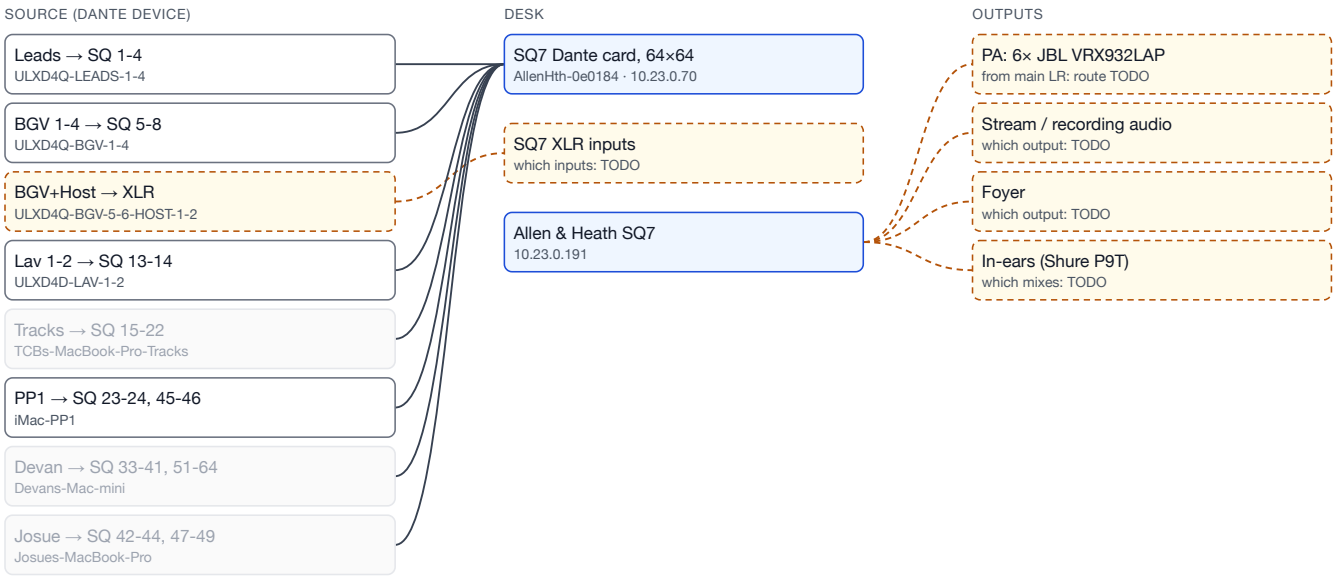
Switcher outputs

Output	Showing now
Auxiliary 1	PP1
Auxiliary 2	PP1B
Webcam Out	Program
Program	Cam 1
Preview	Cam 1

Signal flow: audio

Inputs are the SQ7 Dante card's own subscriptions (`src/dante.js`), the same ones the Dante page checks. The desk's outputs are not readable over the network.

Audio: Dante sources into the SQ7, and the desk's outputs

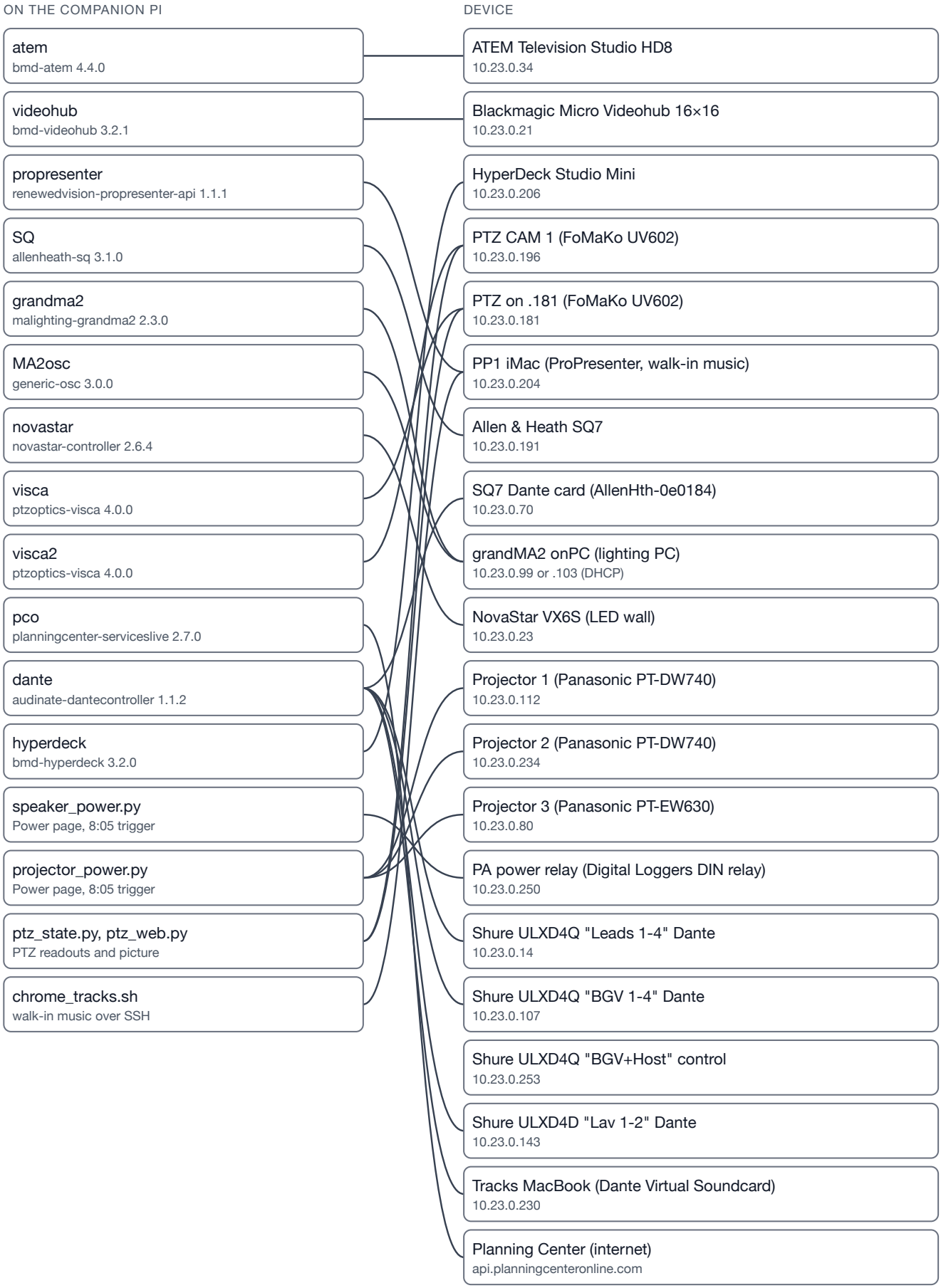


Source	Dante device	Desk inputs	Needed every Sunday
Leads	`ULXD4Q-LEADS-1-4`	SQ 1-4	yes
BGV 1-4	`ULXD4Q-BGV-1-4`	SQ 5-8	yes
BGV+Host	`ULXD4Q-BGV-5-6-HOST-1-2`	XLR (not Dante)	no
Lav 1-2	`ULXD4D-LAV-1-2`	SQ 13-14	yes
Tracks	`TCBs-MacBook-Pro-Tracks`	SQ 15-22	no
PP1	`iMac-PP1`	SQ 23-24, 45-46	yes
Devan	`Devans-Mac-mini`	SQ 33-41, 51-64	no
Josue	`Josues-MacBook-Pro`	SQ 42-44, 47-49	no

Signal flow: control and network

Every device is on the AV LAN, 10.23.0.0/24. Addresses come from DHCP and have drifted before: check them here first when a page goes quiet.

Control: what on the Pi talks to each device



Device	Address	Ports	Talked to by	Source
Companion Pi (Raspberry Pi 5)	`10.23.0.242`	8000 Companion, 80/8080 Service Health board, 22 SSH	everything	read on the Pi
ATEM Television Studio HD8	`10.23.0.34`	UDP 9910 control, TCP 9990 routing	Companion `atem`, board	Companion, board
Blackmagic Micro Videohub 16×16	`10.23.0.21`	TCP 9990	Companion `videohub` (found by Bonjour), board	Companion, board
HyperDeck Studio Mini	`10.23.0.206`	TCP 9993	Companion `hyperdeck`	Companion
PTZ CAM 1 (FoMaKo UV602)	`10.23.0.196`	TCP 5678 VISCA, 80 web API	Companion `visca2`, ptz_state.py, ptz_web.py, board	Companion, board
PTZ on .181 (FoMaKo UV602)	`10.23.0.181`	TCP 5678 VISCA, 80 web API	Companion `visca`, ptz_state.py, ptz_web.py, board	Companion, board
PP1 iMac (ProPresenter, walk-in music)	`10.23.0.204`	1100 ProPresenter API, 22 SSH, 5900 Screen Sharing	Companion `propresenter`, chrome_tracks.sh, board	Companion, board
Allen & Heath SQ7	`10.23.0.191`	TCP 51326	Companion `SQ`, board	Companion, board
SQ7 Dante card (AllenHth-0e0184)	`10.23.0.70`	Dante	Companion `dante`	rig.md Dante
grandMA2 onPC (lighting PC)	`10.23.0.99 or .103 (DHCP)`	Telnet login, OSC 8000 out / 9000 in	Companion `grandma2`, `MA2osc`, board	Companion, board
NovaStar VX6S (LED wall)	`10.23.0.23`	TCP 5200	Companion `novastar`, board	Companion, board
Projector 1 (Panasonic PT-DW740)	`10.23.0.112`	NTCONTROL 1024, PJLink 4352, web 80	projector_power.py	PROTOCOLS.md (2026-07-15)
Projector 2 (Panasonic PT-DW740)	`10.23.0.234`	NTCONTROL 1024, PJLink 4352, web 80	projector_power.py	PROTOCOLS.md (2026-07-15)
Projector 3 (Panasonic PT-EW630)	`10.23.0.80`	NTCONTROL 1024, PJLink 4352, web 80	projector_power.py	PROTOCOLS.md (2026-07-15)
PA power relay (Digital Loggers DIN relay)	`10.23.0.250`	web 80	speaker_power.py	PROTOCOLS.md (2026-07-15)
Shure ULXD4Q "Leads 1-4" Dante	`10.23.0.14`	Dante	Companion `dante`	rig.md Dante
Shure ULXD4Q "BGV 1-4" Dante	`10.23.0.107`	Dante	Companion `dante`	rig.md Dante
Shure ULXD4Q "BGV+Host" control	`10.23.0.253`	2202	nothing (feeds the desk on XLR)	rig.md Dante
Shure ULXD4D "Lav 1-2" Dante	`10.23.0.143`	Dante	Companion `dante`	rig.md Dante
Tracks MacBook (Dante Virtual Soundcard)	`10.23.0.230`	Dante	Companion `dante`	rig.md Dante

Power

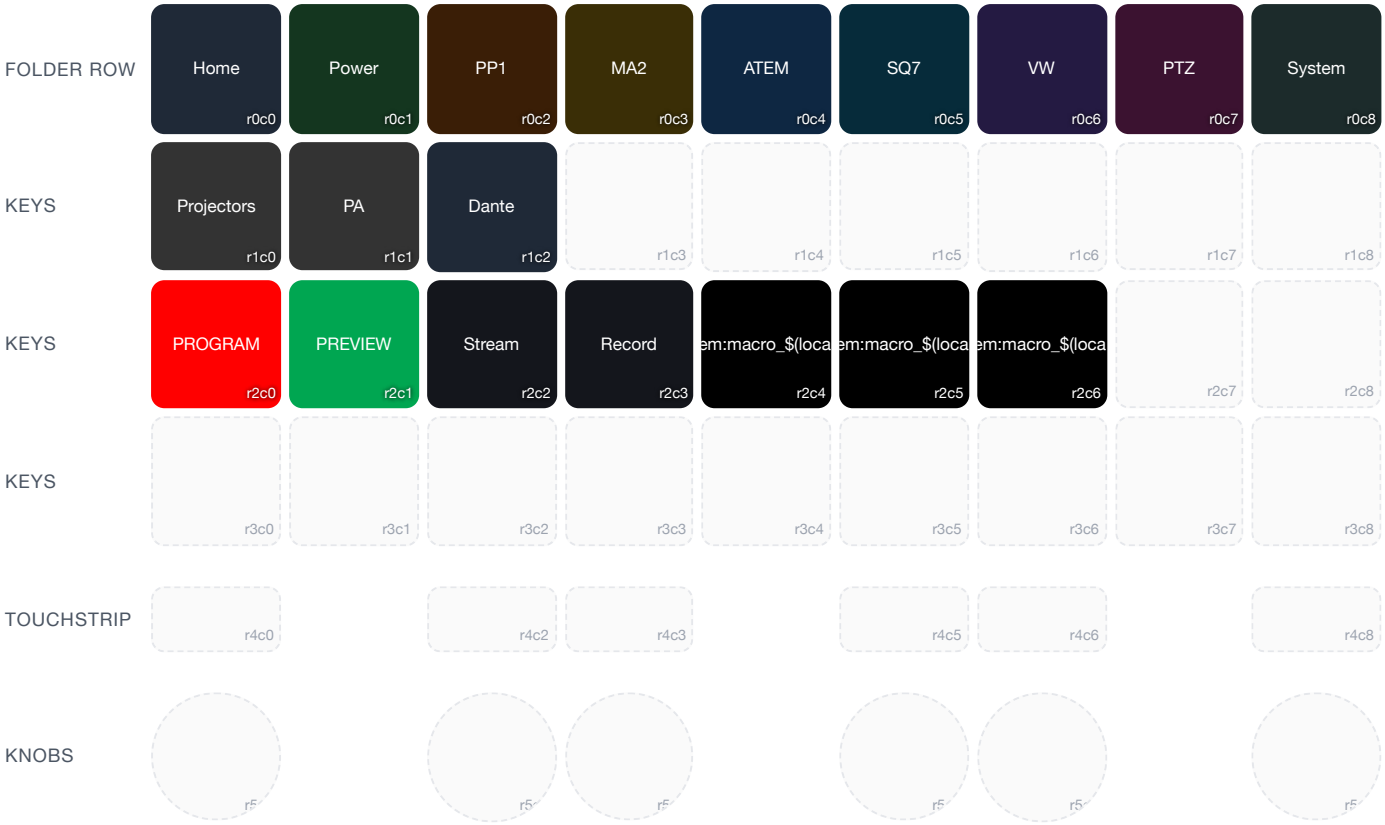
What	How	Switched by
Front-of-house PA	Digital Loggers DIN relay at 10.23.0.250 switches its outlets	Power page, Sunday 8:05 trigger (speaker_power.py)
Projectors (three Panasonic)	Standby on/off over the network (NTCONTROL, PJLink as fallback)	Power page, Sunday 8:05 trigger (projector_power.py)
PTZ cameras	VISCA power on/standby	CAM Setup pages
Companion Pi	Wall power, no UPS (a UPS is item 1 on the 2027 budget)	nothing

Still to trace

Cable runs cannot be read over the network. Each of these needs someone in the room with a torch; record the answer in `src/manual/plant.js` and rebuild.

- ☐ SDI runs: cable, length and any converter between each source and its Videohub input, and each output and its screen
- ☐ Which PP1 iMac outputs (and converters) feed Videohub inputs 14, 15 and 16
- ☐ Where the static JVC GY-HM600 is cabled now that it is off the switcher
- ☐ Audio into the ATEM for the stream, and into the HyperDeck for the recording: which SQ7 output, at what level
- ☐ SQ7 outputs: main LR to the PA amps, the foyer feed, monitor and in-ear sends; the stage box on SLink
- ☐ In-ear transmitters (Shure P9T) and which mixes feed them
- ☐ Network switch ports: which switch and port each device above is on
- ☐ Power circuits and outlets in the rack, and the relay outlet numbers for the PA
- ☐ Rack elevation: what sits in each unit of the rack after the 2026-10-26 move

Page 1: Home



`r1c0` **Projectors**

PRESS

- internal · **custom_variable_set_value**: name `projector_action`; create `yes`; value = `$(internal:custom_projector_state) == "on" ? "off" : "on"`
- internal · **exec**: path `bash -c 'nohup python3 /home/samuelbailey/Desktop/AV_Power_scripts/projector_power.py $(internal:custom_projector_action) >>/tmp/av_power.log 2>&1 &'`; timeout `5000`

CHANGES WHEN

- internal · **variable_value**: variable `custom:projector_state`; op `eq`; value `on`
  → text colour #FFFFFF, colour #00A651, icon projector-ink
- internal · **variable_value**: variable `custom:projector_state`; op `eq`; value `off`
  → text colour #FFFFFF, colour #CC0000, icon projector-paper
- internal · **variable_value**: variable `custom:projector_state`; op `eq`; value `unreachable`
  → text colour #FFFFFF, colour #E67300, icon projector-ink
- internal · **variable_value**: variable `custom:projector_state`; op `eq`; value `mixed`
  → text colour #FFFFFF, colour #E6C000, icon projector-ink

`r1c1` **PA**

PRESS

- internal · **custom_variable_set_value**: name `pa_action`; create `yes`; value = `$(internal:custom_pa_state) == "on" ? "off" : "on"`
- internal · **exec**: path `bash -c 'nohup python3 /home/samuelbailey/Desktop/AV_Power_scripts/speaker_power.py $(internal:custom_pa_action) >>/tmp/av_power.log 2>&1 &'`; timeout `5000`

CHANGES WHEN

- internal · **variable_value**: variable `custom:pa_state`; op `eq`; value `on`
  → text colour #FFFFFF, colour #00A651, icon pa-ink
- internal · **variable_value**: variable `custom:pa_state`; op `eq`; value `off`
  → text colour #FFFFFF, colour #CC0000, icon pa-paper
- internal · **variable_value**: variable `custom:pa_state`; op `eq`; value `unreachable`
  → text colour #FFFFFF, colour #E67300, icon pa-ink
- internal · **variable_value**: variable `custom:pa_state`; op `eq`; value `mixed`
  → text colour #FFFFFF, colour #E6C000, icon pa-ink

r1c2

Dante

dante: opens the Dante page (page 17). Red when a source the desk needs is not reaching it.

PRESS

- internal · **set_page**: surfaceId `self`; page 17

CHANGES WHEN

- internal · **logic_operator**: operation and INVERTED
  → colour #B91C1C
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q–LEADS–1–4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q–LEADS–1–4`; destinationChannelId `01`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q–LEADS–1–4`; destinationChannelId `02`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q–LEADS–1–4`; destinationChannelId `03`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q–LEADS–1–4`; destinationChannelId `04`; destinationDeviceId `AllenHth–0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q–BGV–1–4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q–BGV–1–4`; destinationChannelId `05`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q–BGV–1–4`; destinationChannelId `06`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q–BGV–1–4`; destinationChannelId `07`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q–BGV–1–4`; destinationChannelId `08`; destinationDeviceId `AllenHth–0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4D–LAV–1–2")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4D–LAV–1–2`; destinationChannelId `13`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4D–LAV–1–2`; destinationChannelId `14`; destinationDeviceId `AllenHth–0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "iMac–PP1")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac–PP1`; destinationChannelId `23`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac–PP1`; destinationChannelId `24`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac–PP1`; destinationChannelId `45`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac–PP1`; destinationChannelId `46`; destinationDeviceId `AllenHth–0e0184`

r2c0

PROGRAM

No action: display only.

r2c1

PREVIEW

No action: display only.

r2c2

Stream

STEP 2 · PRESS

- atem · **streamStartStop**: stream `toggle`

CHANGES WHEN

- internal · **check_expression**: expression = `$(this:step) == 2`
  → colour #A16207, label GO LIVE?
- atem · **streamStatus**: state 4
  → colour #B91C1C, label ON AIR

r2c3

Record

record-hyperdeck: starts and stops the HyperDeck. Red with the running time while it records; idle, the time left on the card, amber with NO CARD when there is none. Built by tools/record-hyperdeck.js.

PRESS

- hyperdeck · **recordStopToggle**

CHANGES WHEN

- internal · **check_expression**: expression = `$(hyperdeck:slotId) == "–"`
  → colour #A16207
- hyperdeck · **transport_status**: status `record`
  → colour #B91C1C, label = $(hyperdeck:timecodeHMS)

r2c4

`$(atem:macro_$(local:index))`

PRESS

- atem · **macrorun**: macro = `$(local:index)`; action `runContinue`

CHANGES WHEN

- atem · **macro**: macroIndex = `$(local:index)`; state `isUsed`
  → colour #0000EE, icon macro-run-paper, text colour #FFFFFF
- atem · **macro**: macroIndex = `$(local:index)`; state `isRunning`
  → colour #00EE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isWaiting`
  → colour #EEEE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isRecording`
  → colour #EE0000, icon macro-run-paper, text colour #FFFFFF

r2c5

`$(atem:macro_$(local:index))`

PRESS

- atem · **macrorun**: macro = `$(local:index)`; action `runContinue`

CHANGES WHEN

- atem · **macro**: macroIndex = `$(local:index)`; state `isUsed`
  → colour #0000EE, icon macro-run-paper, text colour #FFFFFF
- atem · **macro**: macroIndex = `$(local:index)`; state `isRunning`
  → colour #00EE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isWaiting`
  → colour #EEEE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isRecording`
  → colour #EE0000, icon macro-run-paper, text colour #FFFFFF

r2c6

`$(atem:macro_$(local:index))`

PRESS

- atem · **macrorun**: macro = `$(local:index)`; action `runContinue`

CHANGES WHEN

- atem · **macro**: macroIndex = `$(local:index)`; state `isUsed`
  → colour #0000EE, icon macro-run-paper, text colour #FFFFFF
- atem · **macro**: macroIndex = `$(local:index)`; state `isRunning`
  → colour #00EE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isWaiting`
  → colour #EEEE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isRecording`
  → colour #EE0000, icon macro-run-paper, text colour #FFFFFF

Page 2: Power



`r1c0` **Projectors**

PRESS

- internal · **custom_variable_set_value**: name `projector_action`; create `yes`; value = `$(internal:custom_projector_state) == "on" ? "off" : "on"`
- internal · **exec**: path `bash -c 'nohup python3 /home/samuelbailey/Desktop/AV_Power_scripts/projector_power.py $(internal:custom_projector_action) >>/tmp/av_power.log 2>&1 &'`; timeout `5000`

CHANGES WHEN

- internal · **variable_value**: variable `custom:projector_state`; op `eq`; value `on`
  → text colour #FFFFFF, colour #00A651, icon projector-ink
- internal · **variable_value**: variable `custom:projector_state`; op `eq`; value `off`
  → text colour #FFFFFF, colour #CC0000, icon projector-paper
- internal · **variable_value**: variable `custom:projector_state`; op `eq`; value `unreachable`
  → text colour #FFFFFF, colour #E67300, icon projector-ink
- internal · **variable_value**: variable `custom:projector_state`; op `eq`; value `mixed`
  → text colour #FFFFFF, colour #E6C000, icon projector-ink

`r1c1` **PA**

PRESS

- internal · **custom_variable_set_value**: name `pa_action`; create `yes`; value = `$(internal:custom_pa_state) == "on" ? "off" : "on"`
- internal · **exec**: path `bash -c 'nohup python3 /home/samuelbailey/Desktop/AV_Power_scripts/speaker_power.py $(internal:custom_pa_action) >>/tmp/av_power.log 2>&1 &'`; timeout `5000`

CHANGES WHEN

- internal · **variable_value**: variable `custom:pa_state`; op `eq`; value `on`
  → text colour #FFFFFF, colour #00A651, icon pa-ink
- internal · **variable_value**: variable `custom:pa_state`; op `eq`; value `off`
  → text colour #FFFFFF, colour #CC0000, icon pa-paper
- internal · **variable_value**: variable `custom:pa_state`; op `eq`; value `unreachable`
  → text colour #FFFFFF, colour #E67300, icon pa-ink
- internal · **variable_value**: variable `custom:pa_state`; op `eq`; value `mixed`
  → text colour #FFFFFF, colour #E6C000, icon pa-ink

Page 3: PP1



r1c0 Clear All

PRESS
- propresenter · **clearLayerOrGroup**: clear_layer_or_group_dropdown `group`; clear_layer_dropdown `slide`; clear_group_id_dropdown `27DDC76F-C385-4E1C-B85F-306238FD8268`

r1c1 Clear Audio

PRESS
- propresenter · **clearLayerOrGroup**: clear_layer_or_group_dropdown `layer`; clear_layer_dropdown `audio`

CHANGES WHEN
- propresenter · **Layer**: layer `audio`
  → text colour #FFC0C0, colour #FF0000, icon clear-audio-ink

r1c2 Clear Messages

PRESS
- propresenter · **clearLayerOrGroup**: clear_layer_or_group_dropdown `layer`; clear_layer_dropdown `messages`

CHANGES WHEN
- propresenter · **Layer**: layer `messages`
  → text colour #FFC0C0, colour #FF0000

r1c3 Clear Props

PRESS
- propresenter · **clearLayerOrGroup**: clear_layer_or_group_dropdown `layer`; clear_layer_dropdown `props`

CHANGES WHEN
- propresenter · **Layer**: layer `props`
  → text colour #FFC0C0, colour #FF0000

r1c4 Clear Announce

PRESS
- propresenter · **clearLayerOrGroup**: clear_layer_or_group_dropdown `layer`; clear_layer_dropdown `announcements`

CHANGES WHEN
- propresenter · **Layer**: layer `announcements`
  → text colour #FFC0C0, colour #FF0000

r1c5

Clear Slide

PRESS
- propresenter · **clearLayerOrGroup**: clear_layer_or_group_dropdown `layer`; clear_layer_dropdown `slide`

CHANGES WHEN
- propresenter · **Layer**: layer `slide`
  → text colour #FFC0C0, colour #FF0000, icon clear-slide-ink

r1c6

Clear Media

PRESS
- propresenter · **clearLayerOrGroup**: clear_layer_or_group_dropdown `layer`; clear_layer_dropdown `media`

CHANGES WHEN
- propresenter · **Layer**: layer `media`
  → text colour #FFC0C0, colour #FF0000

r1c7

Clear Vid Inputs

PRESS
- propresenter · **clearLayerOrGroup**: clear_layer_or_group_dropdown `layer`; clear_layer_dropdown `video_input`

CHANGES WHEN
- propresenter · **Layer**: layer `video_input`
  → text colour #FFC0C0, colour #FF0000

r1c8

Freeze Slide

pp1-freeze: fires the ProPresenter macro "Freeze" (Slide Destination, Stage Only), which holds the audience screens on the current slide. "All Screens" releases it. Built by tools/pp1-freeze.js — rerun it rather than editing here.

PRESS
- propresenter · **marcoIdTrigger**: macro_id_dropdown `3C693319-8EAC-4499-86C3-DE7A13FABCE9`

r2c0

Focus Prev

PRESS
- propresenter · **focusedPresentationOperation**: focused_presentation_operation `focus_previous`

r2c1

Previous

PRESS
- propresenter · **triggerOperation**: trigger_target `presentation`; trigger_next_previous `previous`

r2c2

Next

PRESS
- propresenter · **triggerOperation**: trigger_target `presentation`; trigger_next_previous `next`

r2c3

Pre-Service

Fires the "Pre-Service" section of the "Sunday Service" playlist: item 1, presentation "Pre-Service". ProPresenter triggers by index, so if the playlist changes above it, re-run tools/pp1-preservice.js. Pointed here 2026-10-11 10:07.

PRESS
- propresenter · **specificPlaylistOperation**: specific_playlist_operation `trigger_index`; playlist_id `6551D76D-C095-40B9-94C5-9A97B7CE57FB`; index 1; cue_index `0`

r2c4

Tracks

Opens the Tracks folder (page 14): play/pause, skip and volume for the tracks in Chrome on the PP1 iMac. Green while playing.

PRESS
- internal · **set_page**: surfaceId `self`; page 14
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh state`; timeout `8000`; targetVariable `chrome_tracks`
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh volume`; timeout `8000`; targetVariable `chrome_tracks_volume`
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh title`; timeout `8000`; targetVariable `chrome_tracks_title`

CHANGES WHEN
- internal · **check_expression**: expression `= $(internal:custom_chrome_tracks) == 'Playing'`
  → colour #15803D

r2c5

Stage

Swaps the "Stage Screen" (E668F1AE-30EF-4207-BB51-4EE810A660D9) stage layout between "Full" and "MultiTracks Chords + Lyrics". State comes from the "Poll stage layout" trigger, which polls ProPresenter every 2s into pp1_stage_layout — NOT from the module's own stagescreen_<uuid>_layout variable, which never updates on this rig and made an earlier version of this key only ever set "Full". The press also writes pp1_stage_layout itself so the caption does not lag a press. On any other layout the key names it in amber and the next press returns to "Full".

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_pp1_stage_layout) == "Full"`

  ACTIONS
  - propresenter · **stageDisplayOperation**: stagedisplay_operation `set_layout`; stagescreen_id_dropdown `E668F1AE-30EF-4207-BB51-4EE810A660D9`; stagescreenlayout_id_dropdown `2A86DB5A-8A0A-4E2B-A384-A8806BF95C1E`
  - internal · **custom_variable_set_value**: name `pp1_stage_layout`; create `yes`; value `MultiTracks Chords + Lyrics`

  ELSE_ACTIONS
  - propresenter · **stageDisplayOperation**: stagedisplay_operation `set_layout`; stagescreen_id_dropdown `E668F1AE-30EF-4207-BB51-4EE810A660D9`; stagescreenlayout_id_dropdown `2C3FBE66-55FE-4E4F-BBC0-894FBF8A34DC`
  - internal · **custom_variable_set_value**: name `pp1_stage_layout`; create `yes`; value `Full`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_pp1_stage_layout) != "Full" && $(internal:custom_pp1_stage_layout) != "MultiTracks Chords + Lyrics"`
  → colour #D97706
- internal · **check_expression**: expression = `$(internal:custom_pp1_stage_layout) == "Full"`
  → label Full
- internal · **check_expression**: expression = `$(internal:custom_pp1_stage_layout) == "MultiTracks Chords + Lyrics"`
  → colour #0F766E, label Chords

r2c6

Countdown

Fires the "Countdown" section of the "Sunday Service" playlist: item 3, media "triumph countdown", then starts the ATEM recording (start only, never stops one already running). ProPresenter triggers by index, so if the playlist changes above it, re-run tools/pp1-countdown.js. Pointed here 2026-10-04 10:09.

PRESS

- propresenter · **specificPlaylistOperation**: specific_playlist_operation `trigger_index`; playlist_id `6551D76D-C095-40B9-94C5-9A97B7CE57FB`; index 3; cue_index `0`
- videohub · **route**: source 4; destination 13; ignore_lock `yes`
- hyperdeck · **rec**
- atem · **streamStartStop**: stream `true`

r2c7

All Screens

PRESS

- propresenter · **marcoIdTrigger**: macro_id_dropdown `26024D56-D05A-4E62-A0EC-C650B6E7BDD1`

r2c8

Stage Notes

PRESS

- propresenter · **marcoIdTrigger**: macro_id_dropdown `1805FD23-FAA4-43F2-A75A-CFB571DDB996`

r3c0

Focus Next

PRESS

- propresenter · **focusedPresentationOperation**: focused_presentation_operation `focus_next`

r3c3

Hosting

Hosting — ProPresenter "Main Timer" (8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE). Press sets 00:07:00 and starts it. Also writes "Hosting" to pp1_main_label, which names the strip readout.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE`; timer_operation `set`; timer_increment_value `30`; timer_type `countdown`; timer_duration `00:07:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `start`
- internal · **custom_variable_set_value**: name `pp1_main_label`; create `yes`; value `Hosting`

r3c4

Speaker

Speaker — ProPresenter "Main Timer" (8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE). Press sets 00:45:00 and starts it. Also writes "Speaker" to pp1_main_label, which names the strip readout.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE`; timer_operation `set`; timer_increment_value `30`; timer_type `countdown`; timer_duration `00:45:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `start`
- internal · **custom_variable_set_value**: name `pp1_main_label`; create `yes`; value `Speaker`

r3c5

Worship

Worship — ProPresenter "Segment Countdown" (28FAF666-E93E-4396-B603-1F8B4330DAE5). Press sets 00:25:00 and starts it.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `28FAF666-E93E-4396-B603-1F8B4330DAE5`; timer_operation `set`; timer_increment_value `30`; timer_type `countdown`; timer_duration `00:25:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `start`

r3c6

Hosting

Hosting — ProPresenter "Main Timer" (8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE). Press puts 00:07:00 back on the clock, stopped, whatever was set last. Also writes "Hosting" to pp1_main_label, which names the strip readout.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE`; timer_operation `set`; timer_increment_value `30`; timer_type `countdown`; timer_duration `00:07:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `reset`
- internal · **custom_variable_set_value**: name `pp1_main_label`; create `yes`; value `Hosting`

r3c7

Speaker

Speaker — ProPresenter "Main Timer" (8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE). Press puts 00:45:00 back on the clock, stopped, whatever was set last. Also writes "Speaker" to pp1_main_label, which names the strip readout.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE`; timer_operation `set`; timer_increment_value `30`; timer_type `countdown`; timer_duration `00:45:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `reset`
- internal · **custom_variable_set_value**: name `pp1_main_label`; create `yes`; value `Speaker`

r3c8

Worship

Worship — ProPresenter "Segment Countdown" (28FAF666-E93E-4396-B603-1F8B4330DAE5). Press puts 00:25:00 back on the clock, stopped, whatever was set last.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `28FAF666-E93E-4396-B603-1F8B4330DAE5`; timer_operation `set`; timer_increment_value `30`; timer_type `countdown`; timer_duration `00:25:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `reset`

r4c0 `$(internal:custom_pp1_main_label)`

ProPresenter "Main Timer" (8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE). Display only — no action, so it cannot be restarted by a thumb. The name is $(internal:custom_pp1_main_label), written by whichever preset key last fired this clock.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(propresenter:timer_8F66EEB482F24BFFA8A6DAC0B3EA7AAE_state) == "running"` → colour #14361F
- internal · **check_expression**: expression = `$(propresenter:timer_8F66EEB482F24BFFA8A6DAC0B3EA7AAE_state) == "overran"` → colour #CC0000

r4c2 **Worship**

ProPresenter "Segment Countdown" (28FAF666-E93E-4396-B603-1F8B4330DAE5). Display only — no action, so it cannot be restarted by a thumb. Only the Worship preset fires this clock, so the name is fixed.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(propresenter:timer_28FAF666E93E4396B6031F8B4330DAE5_state) == "running"` → colour #14361F
- internal · **check_expression**: expression = `$(propresenter:timer_28FAF666E93E4396B6031F8B4330DAE5_state) == "overran"` → colour #CC0000

r4c3 `$(internal:custom_pp1_main_label)`

-1 minute on "Main Timer". Applies to the running clock.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE`; timer_operation `increment`; timer_increment_value `-60`; timer_type `countdown`; timer_duration `00:05:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `start`

r4c5 `$(internal:custom_pp1_main_label)`

+1 minute on "Main Timer". Applies to the running clock.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `8F66EEB4-82F2-4BFF-A8A6-DAC0B3EA7AAE`; timer_operation `increment`; timer_increment_value `60`; timer_type `countdown`; timer_duration `00:05:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `start`

r4c6 **Worship**

-1 minute on "Segment Countdown". Applies to the running clock.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `28FAF666-E93E-4396-B603-1F8B4330DAE5`; timer_operation `increment`; timer_increment_value `-60`; timer_type `countdown`; timer_duration `00:05:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `start`

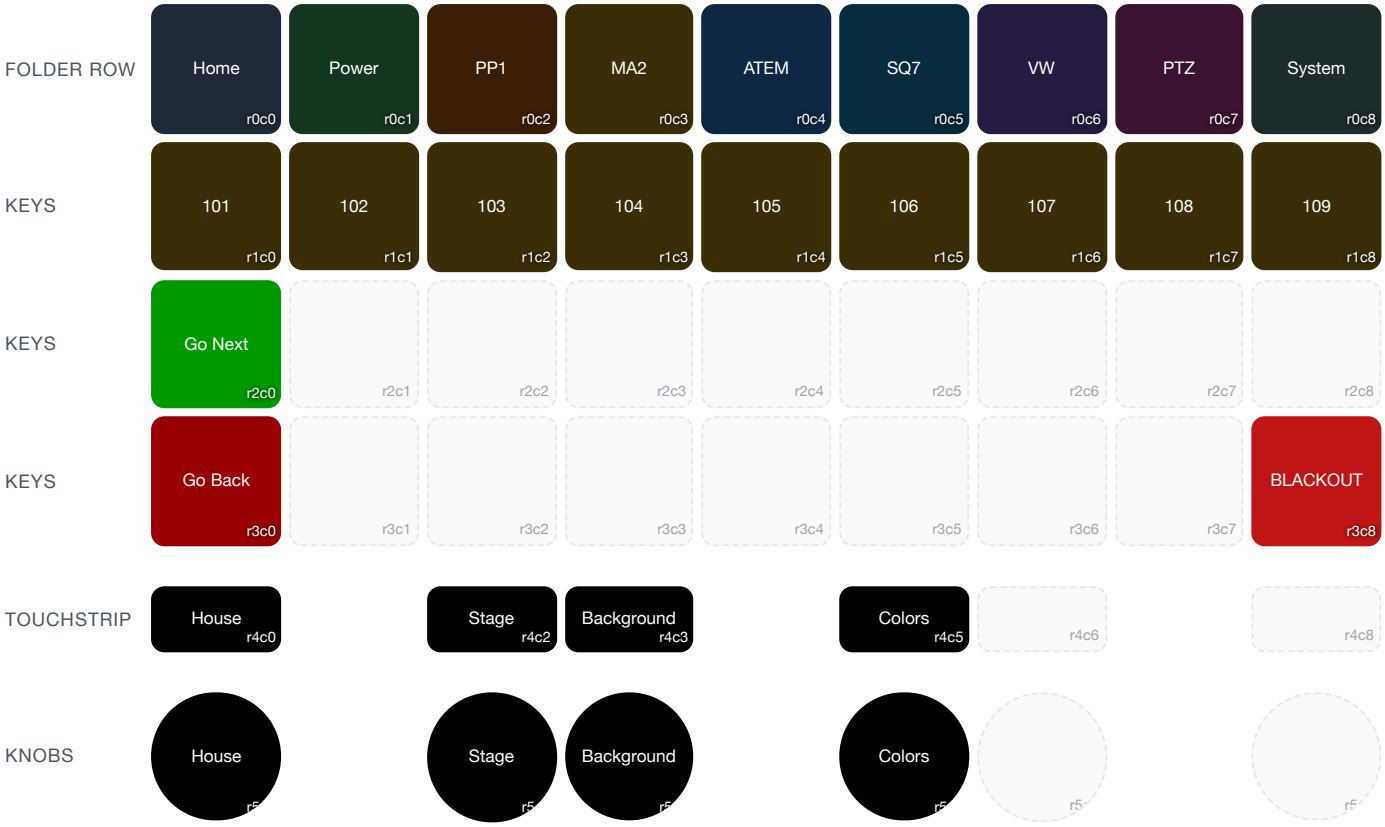
r4c8 **Worship**

+1 minute on "Segment Countdown". Applies to the running clock.

PRESS

- propresenter · **timerOperation**: timer_id_dropdown `28FAF666-E93E-4396-B603-1F8B4330DAE5`; timer_operation `increment`; timer_increment_value `60`; timer_type `countdown`; timer_duration `00:05:00`; timer_time_of_day `09:00:00`; timer_timeperiod `am`; timer_start_time `00:00:00`; timer_allows_overrun `yes`; timer_optional_operation `start`

Page 4: MA2



`r1c0` **101**

Toggle Executor 1.101

PRESS

- grandma2 · **command**: command `Toggle Executor 1.101`

`r1c1` **102**

Toggle Executor 1.102

PRESS

- grandma2 · **command**: command `Toggle Executor 1.102`

`r1c2` **103**

Toggle Executor 1.103

PRESS

- grandma2 · **command**: command `Toggle Executor 1.103`

`r1c3` **104**

Toggle Executor 1.104

PRESS

- grandma2 · **command**: command `Toggle Executor 1.104`

`r1c4` **105**

Toggle Executor 1.105

PRESS

- grandma2 · **command**: command `Toggle Executor 1.105`

`r1c5` **106**

Toggle Executor 1.106

PRESS

- grandma2 · **command**: command `Toggle Executor 1.106`

r1c6

107

Toggle Executor 1.107

PRESS

- grandma2 · **command**: command `Toggle Executor 1.107`

r1c7

108

Toggle Executor 1.108

PRESS

- grandma2 · **command**: command `Toggle Executor 1.108`

r1c8

109

Toggle Executor 1.109

PRESS

- grandma2 · **command**: command `Toggle Executor 1.109`

r2c0

Go Next

PRESS

- grandma2 · **button**: button `11`; dir `true`

r3c0

Go Back

PRESS

- grandma2 · **button**: button `10`; dir `true`

r3c8

BLACKOUT

PRESS

- grandma2 · **command**: command `Blackout`

r4c0

House

No action: display only.

r4c2

Stage

No action: display only.

r4c3

Background

No action: display only.

r4c5

Colors

No action: display only.

r5c0

House

PRESS

- grandma2 · **command**: command `Select Executor 1.1`
- internal · **custom_variable_set_value**: name `selected_exec`; value `1`

TURN LEFT

- internal · **custom_variable_set_value**: name `fader1_level`; value = `max(0, $(this:current) - 5)`
- grandma2 · **command**: command `Executor 1.1 At $(internal:custom_fader1_level)`

TURN RIGHT

- internal · **custom_variable_set_value**: name `fader1_level`; value = `min(100, $(this:current) + 5)`
- grandma2 · **command**: command `Executor 1.1 At $(internal:custom_fader1_level)`

r5c2

Stage

PRESS
- grandma2 · **command**: command `Select Executor 1.2`
- internal · **custom_variable_set_value**: name `selected_exec`; value 2

TURN LEFT
- internal · **custom_variable_set_value**: name `fader2_level`; value = `max(0, $(this:current) - 5)`
- grandma2 · **command**: command `Executor 1.2 At $(internal:custom_fader2_level)`

TURN RIGHT
- internal · **custom_variable_set_value**: name `fader2_level`; value = `min(100, $(this:current) + 5)`
- grandma2 · **command**: command `Executor 1.2 At $(internal:custom_fader2_level)`

r5c3

Background

PRESS
- grandma2 · **command**: command `Select Executor 1.3`
- internal · **custom_variable_set_value**: name `selected_exec`; value 3

TURN LEFT
- internal · **custom_variable_set_value**: name `fader3_level`; value = `max(0, $(this:current) - 5)`
- grandma2 · **command**: command `Executor 1.3 At $(internal:custom_fader3_level)`

TURN RIGHT
- internal · **custom_variable_set_value**: name `fader3_level`; value = `min(100, $(this:current) + 5)`
- grandma2 · **command**: command `Executor 1.3 At $(internal:custom_fader3_level)`

r5c5

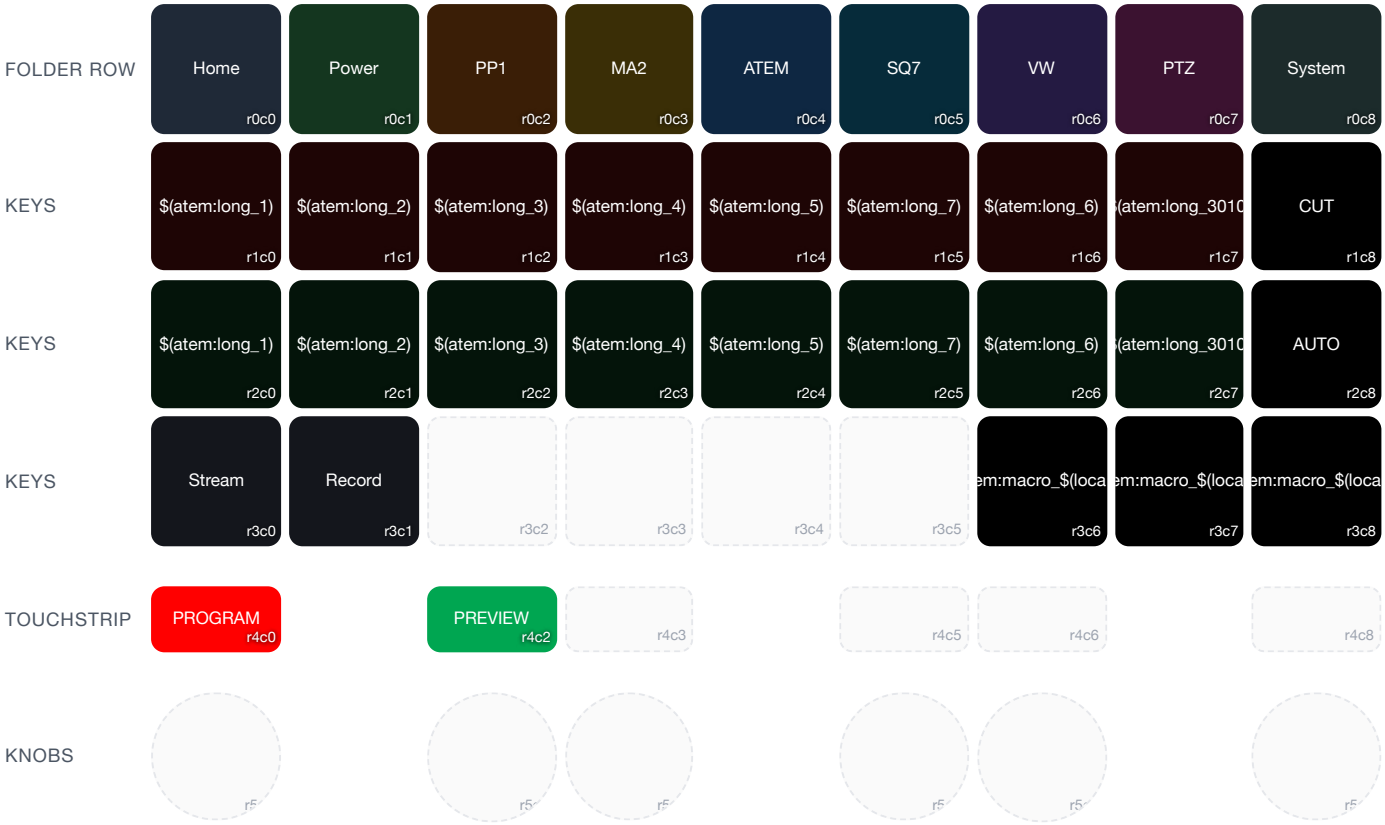
Colors

PRESS
- grandma2 · **command**: command `Select Executor 1.4`
- internal · **custom_variable_set_value**: name `selected_exec`; value 4

TURN LEFT
- internal · **custom_variable_set_value**: name `fader4_level`; value = `max(0, $(this:current) - 5)`
- grandma2 · **command**: command `Executor 1.4 At $(internal:custom_fader4_level)`

TURN RIGHT
- internal · **custom_variable_set_value**: name `fader4_level`; value = `min(100, $(this:current) + 5)`
- grandma2 · **command**: command `Executor 1.4 At $(internal:custom_fader4_level)`

Page 5: ATEM



r1c0 `$(atem:long_1)`

PRESS
- atem · **program**: mixeffect 1; input 1

CHANGES WHEN
- atem · **program**: mixeffect 1; input 1
  → colour #FF0000, text colour #FFFFFF, icon camera-ink

r1c1 `$(atem:long_2)`

PRESS
- atem · **program**: mixeffect 1; input 2

CHANGES WHEN
- atem · **program**: mixeffect 1; input 2
  → colour #FF0000, text colour #FFFFFF, icon camera-ink

r1c2 `$(atem:long_3)`

PRESS
- atem · **program**: mixeffect 1; input 3

CHANGES WHEN
- atem · **program**: mixeffect 1; input 3
  → colour #FF0000, text colour #FFFFFF, icon camera-ink

r1c3 `$(atem:long_4)`

PRESS
- atem · **program**: mixeffect 1; input 4

CHANGES WHEN
- atem · **program**: mixeffect 1; input 4
  → colour #FF0000, text colour #FFFFFF, icon camera-ink

r1c4 `$(atem:long_5)`

PRESS
- atem · **program**: mixeffect 1; input 5

CHANGES WHEN
- atem · **program**: mixeffect 1; input 5
  → colour #FF0000, text colour #FFFFFF, icon message-ink

r1c5 `$(atem:long_7)`

PRESS
- atem · **program**: mixeffect 1; input 7

CHANGES WHEN
- atem · **program**: mixeffect 1; input 7
  → colour #FF0000, text colour #FFFFFF, icon media-ink

r1c6 `$(atem:long_6)`

PRESS
- atem · **program**: mixeffect 1; input 6

CHANGES WHEN
- atem · **program**: mixeffect 1; input 6
  → colour #FF0000, text colour #FFFFFF, icon media-ink

r1c7 `$(atem:long_3010)`

PRESS
- atem · **program**: mixeffect 1; input `3010`

CHANGES WHEN
- atem · **program**: mixeffect 1; input `3010`
  → colour #FF0000, text colour #FFFFFF, icon media-ink

r1c8 **CUT**

PRESS
- atem · **cut**: mixeffect 1

r2c0 `$(atem:long_1)`

PRESS
- atem · **preview**: mixeffect 1; input 1

CHANGES WHEN
- atem · **preview**: mixeffect 1; input 1
  → colour #00A651, text colour #FFFFFF, icon camera-ink

r2c1 `$(atem:long_2)`

PRESS
- atem · **preview**: mixeffect 1; input 2

CHANGES WHEN
- atem · **preview**: mixeffect 1; input 2
  → colour #00A651, text colour #FFFFFF, icon camera-ink

r2c2 `$(atem:long_3)`

PRESS
- atem · **preview**: mixeffect 1; input 3

CHANGES WHEN
- atem · **preview**: mixeffect 1; input 3
  → colour #00A651, text colour #FFFFFF, icon camera-ink

r2c3 `$(atem:long_4)`

PRESS
- atem · **preview**: mixeffect 1; input 4

CHANGES WHEN
- atem · **preview**: mixeffect 1; input 4
  → colour #00A651, text colour #FFFFFF, icon camera-ink

r2c4 `$(atem:long_5)`

PRESS
- atem · **preview**: mixeffect 1; input 5

CHANGES WHEN
- atem · **preview**: mixeffect 1; input 5
  → colour #00A651, text colour #FFFFFF, icon message-ink

r2c5 `$(atem:long_7)`

PRESS
- atem · **preview**: mixeffect 1; input 7

CHANGES WHEN
- atem · **preview**: mixeffect 1; input 7
  → colour #00A651, text colour #FFFFFF, icon media-ink

r2c6 `$(atem:long_6)`

PRESS
- atem · **preview**: mixeffect 1; input 6

CHANGES WHEN
- atem · **preview**: mixeffect 1; input 6
  → colour #00A651, text colour #FFFFFF, icon media-ink

r2c7 `$(atem:long_3010)`

PRESS
- atem · **preview**: mixeffect 1; input `3010`

CHANGES WHEN
- atem · **preview**: mixeffect 1; input `3010`
  → colour #00A651, text colour #FFFFFF, icon media-ink

r2c8 **AUTO**

PRESS
- atem · **auto**: mixeffect 1

CHANGES WHEN
- atem · **inTransition**: mixeffect 1
  → colour #FF0000, text colour #FFFFFF, icon auto-ink

r3c0 **Stream**

STEP 2 · PRESS
- atem · **streamStartStop**: stream `toggle`

CHANGES WHEN
- internal · **check_expression**: expression = `$(this:step) == 2`
  → colour #A16207, label GO LIVE?
- atem · **streamStatus**: state 4
  → colour #B91C1C, label ON AIR

r3c1 **Record**

record-hyperdeck: starts and stops the HyperDeck. Red with the running time while it records; idle, the time left on the card, amber with NO CARD when there is none. Built by tools/record-hyperdeck.js.

PRESS
- hyperdeck · **recordStopToggle**

CHANGES WHEN
- internal · **check_expression**: expression = `$(hyperdeck:slotId) == "–"`
  → colour #A16207
- hyperdeck · **transport_status**: status `record`
  → colour #B91C1C, label = $(hyperdeck:timecodeHMS)

r3c6 `$(atem:macro_$(local:index))`

PRESS
- atem · **macrorun**: macro = `$(local:index)`; action `runContinue`

CHANGES WHEN
- atem · **macro**: macroIndex = `$(local:index)`; state `isUsed`
  → colour #0000EE, icon macro-run-paper, text colour #FFFFFF
- atem · **macro**: macroIndex = `$(local:index)`; state `isRunning`
  → colour #00EE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isWaiting`
  → colour #EEEE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isRecording`
  → colour #EE0000, icon macro-run-paper, text colour #FFFFFF

r3c7 `$(atem:macro_$(local:index))`

PRESS
- atem · **macrorun**: macro = `$(local:index)`; action `runContinue`

CHANGES WHEN
- atem · **macro**: macroIndex = `$(local:index)`; state `isUsed`
  → colour #0000EE, icon macro-run-paper, text colour #FFFFFF
- atem · **macro**: macroIndex = `$(local:index)`; state `isRunning`
  → colour #00EE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isWaiting`
  → colour #EEEE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isRecording`
  → colour #EE0000, icon macro-run-paper, text colour #FFFFFF

r3c8 `$(atem:macro_$(local:index))`

PRESS
- atem · **macrorun**: macro = `$(local:index)`; action `runContinue`

CHANGES WHEN
- atem · **macro**: macroIndex = `$(local:index)`; state `isUsed`
  → colour #0000EE, icon macro-run-paper, text colour #FFFFFF
- atem · **macro**: macroIndex = `$(local:index)`; state `isRunning`
  → colour #00EE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isWaiting`
  → colour #EEEE00, icon macro-run-ink, text colour #000000
- atem · **macro**: macroIndex = `$(local:index)`; state `isRecording`
  → colour #EE0000, icon macro-run-paper, text colour #FFFFFF

r4c0 **PROGRAM**

No action: display only.

r4c2 **PREVIEW**

No action: display only.

Page 6: SQ7



`r1c0`

DCA 1

Toggle mute on DCA 1

PRESS
- SQ · **mute_dca**: strip `0`; mute `0`

CHANGES WHEN
- SQ · **mute_dca**: channel `0`
  → colour #CC0000, icon dca-paper

`r1c1`

DCA 2

Toggle mute on DCA 2

PRESS
- SQ · **mute_dca**: strip 1; mute `0`

CHANGES WHEN
- SQ · **mute_dca**: channel 1
  → colour #CC0000, icon dca-paper

`r1c2`

DCA 3

Toggle mute on DCA 3

PRESS
- SQ · **mute_dca**: strip 2; mute `0`

CHANGES WHEN
- SQ · **mute_dca**: channel 2
  → colour #CC0000, icon dca-paper

`r1c3`

DCA 4

Toggle mute on DCA 4

PRESS
- SQ · **mute_dca**: strip 3; mute `0`

CHANGES WHEN
- SQ · **mute_dca**: channel 3
  → colour #CC0000, icon dca-paper

r1c4

DCA 5

Toggle mute on DCA 5

PRESS
- SQ · **mute_dca**: strip 4; mute `0`

CHANGES WHEN
- SQ · **mute_dca**: channel 4
  → colour #CC0000, icon dca-paper

r1c5

DCA 6

Toggle mute on DCA 6

PRESS
- SQ · **mute_dca**: strip 5; mute `0`

CHANGES WHEN
- SQ · **mute_dca**: channel 5
  → colour #CC0000, icon dca-paper

r1c6

DCA 7

Toggle mute on DCA 7

PRESS
- SQ · **mute_dca**: strip 6; mute `0`

CHANGES WHEN
- SQ · **mute_dca**: channel 6
  → colour #CC0000, icon dca-paper

r1c7

DCA 8

Toggle mute on DCA 8

PRESS
- SQ · **mute_dca**: strip 7; mute `0`

CHANGES WHEN
- SQ · **mute_dca**: channel 7
  → colour #CC0000, icon dca-paper

r1c8

Mute DCAs

PRESS
- SQ · **mute_dca**: strip `0`; mute `0`
- SQ · **mute_dca**: strip 1; mute `0`
- SQ · **mute_dca**: strip 2; mute `0`
- SQ · **mute_dca**: strip 3; mute `0`
- SQ · **mute_dca**: strip 4; mute `0`
- SQ · **mute_dca**: strip 5; mute `0`
- SQ · **mute_dca**: strip 6; mute `0`
- SQ · **mute_dca**: strip 7; mute `0`

r2c4

(no label)

No action: display only.

r2c8

Dante

dante: opens the Dante page (page 17). Red when a source the desk needs is not reaching it.

PRESS

- internal · **set_page**: surfaceId `self`; page 17

CHANGES WHEN

- internal · **logic_operator**: operation and INVERTED
  → colour #B91C1C
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q–LEADS–1–4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q–LEADS–1–4`; destinationChannelId `01`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q–LEADS–1–4`; destinationChannelId `02`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q–LEADS–1–4`; destinationChannelId `03`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q–LEADS–1–4`; destinationChannelId `04`; destinationDeviceId `AllenHth–0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q–BGV–1–4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q–BGV–1–4`; destinationChannelId `05`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q–BGV–1–4`; destinationChannelId `06`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q–BGV–1–4`; destinationChannelId `07`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q–BGV–1–4`; destinationChannelId `08`; destinationDeviceId `AllenHth–0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4D–LAV–1–2")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4D–LAV–1–2`; destinationChannelId `13`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4D–LAV–1–2`; destinationChannelId `14`; destinationDeviceId `AllenHth–0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "iMac–PP1")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac–PP1`; destinationChannelId `23`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac–PP1`; destinationChannelId `24`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac–PP1`; destinationChannelId `45`; destinationDeviceId `AllenHth–0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac–PP1`; destinationChannelId `46`; destinationDeviceId `AllenHth–0e0184`

r4c0

`Stream $(SQ:level_79.17)`

No action: display only.

r4c2

`Foyer $(SQ:level_79.18)`

No action: display only.

r4c5

`MAIN $(SQ:level_79.0)`

No action: display only.

`r5c0`

Stream

PRESS	• SQ · **mute_matrix**: strip `0`; mute `0`
TURN LEFT	• SQ · **matrix_level_output**: input `0`; leveldb `step−3`; fade `0`
TURN RIGHT	• SQ · **matrix_level_output**: input `0`; leveldb `step+3`; fade `0`

`r5c2`

Foyer

PRESS	• SQ · **mute_matrix**: strip 1; mute `0`
TURN LEFT	• SQ · **matrix_level_output**: input 1; leveldb `step−3`; fade `0`
TURN RIGHT	• SQ · **matrix_level_output**: input 1; leveldb `step+3`; fade `0`

`r5c5`

Main

PRESS	• SQ · **mute_lr**: strip `0`; mute `0`
TURN LEFT	• SQ · **lr_level_output**: leveldb `step−3`; fade `0`
TURN RIGHT	• SQ · **lr_level_output**: leveldb `step+3`; fade `0`

Page 7: VW



`r1c0` **Right Proj**

Select Right Projector (output 16) as the routing destination

PRESS
- videohub · **select_destination**: destination 16

CHANGES WHEN
- videohub · **selected_destination**: output 16
  → colour #A78BFA, icon destination-ink

`r1c1` **Left Proj**

Select Left Projector (output 15) as the routing destination

PRESS
- videohub · **select_destination**: destination 15

CHANGES WHEN
- videohub · **selected_destination**: output 15
  → colour #A78BFA, icon destination-ink

`r1c2` **LED Wall**

Select LED Wall (output 14) as the routing destination

PRESS
- videohub · **select_destination**: destination 14

CHANGES WHEN
- videohub · **selected_destination**: output 14
  → colour #A78BFA, icon destination-ink

`r1c3` **Backwall**

Select Backwall Projector (output 8) as the routing destination

PRESS
- videohub · **select_destination**: destination 8

CHANGES WHEN
- videohub · **selected_destination**: output 8
  → colour #A78BFA, icon destination-ink

r1c4

Mini Screen

Select Mini Screen 1 (output 9) as the routing destination

PRESS
- videohub · **select_destination**: destination 9

CHANGES WHEN
- videohub · **selected_destination**: output 9
  → colour #A78BFA, icon destination-ink

r1c8

Video hub

vw: opens the Videohub page (page 18), every input and output on the router

PRESS
- internal · **set_page**: surfaceId `self`; page 18

r2c0

Program

Route Program (input 4) to the selected destination

PRESS
- videohub · **route_source**: source 4; ignore_lock `yes`

CHANGES WHEN
- videohub · **selected_source**: input 4
  → colour #1F6F3A

r2c1

Aux 1

Route Aux 1 (input 1) to the selected destination

PRESS
- videohub · **route_source**: source 1; ignore_lock `yes`

CHANGES WHEN
- videohub · **selected_source**: input 1
  → colour #1F6F3A

r2c2

Aux 2

Route Aux 2 (input 3) to the selected destination

PRESS
- videohub · **route_source**: source 3; ignore_lock `yes`

CHANGES WHEN
- videohub · **selected_source**: input 3
  → colour #1F6F3A

r2c3

Multi view

Route Multiview (input 2) to the selected destination

PRESS
- videohub · **route_source**: source 2; ignore_lock `yes`

CHANGES WHEN
- videohub · **selected_source**: input 2
  → colour #1F6F3A

r3c0

PGM to Proj

PRESS
- videohub · **route**: source 4; destination 16; ignore_lock `yes`
- videohub · **route**: source 4; destination 15; ignore_lock `yes`

CHANGES WHEN
- internal · **variable_value**: variable `videohub:output_16_input_id`; op eq; value 4
  → text colour #FFFFFF, colour #00A651, icon route

r3c1

Aux1 to Proj

PRESS
- videohub · **route**: source 1; destination 16; ignore_lock `yes`
- videohub · **route**: source 1; destination 15; ignore_lock `yes`

CHANGES WHEN
- internal · **variable_value**: variable `videohub:output_16_input_id`; op eq; value 1
  → text colour #FFFFFF, colour #00A651, icon aux

r4c0

`$(videohub:output_16)`

No action: display only.

r4c2

`$(videohub:output_15)`

No action: display only.

r4c3

`$(videohub:output_14)`

No action: display only.

r4c5

`$(videohub:output_8)`

No action: display only.

r4c6

`Wall $(novastar:brite)%`

LED wall brightness, as reported by the VX6S. Display only.

No action: display only.

r5c6

Wall

Turn to adjust LED wall brightness by 5% per detent. Press does nothing.

TURN LEFT
- novastar · **set_brightness**: mode `A`; which 0; value `0`; adj −5

TURN RIGHT
- novastar · **set_brightness**: mode `A`; which 0; value `0`; adj 5

Page 8: System



`r1c0` **CPU Temp**

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_sys_temp) >= 65` → colour #A16207
- internal · **check_expression**: expression = `$(internal:custom_sys_temp) >= 78` → colour #B91C1C

`r1c2` **CPU Load**

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_sys_load) >= 2.5` → colour #A16207
- internal · **check_expression**: expression = `$(internal:custom_sys_load) >= 4` → colour #B91C1C

`r1c4` **Memory**

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_sys_mem) >= 75` → colour #A16207
- internal · **check_expression**: expression = `$(internal:custom_sys_mem) >= 90` → colour #B91C1C

`r1c6` **Disk**

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_sys_disk) >= 80` → colour #A16207
- internal · **check_expression**: expression = `$(internal:custom_sys_disk) >= 92` → colour #B91C1C

r1c8

Power

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_sys_throttle) == "n/a"`
  → colour #A16207
- internal · **check_expression**: expression = `$(internal:custom_sys_throttle) != "OK" && $(internal:custom_sys_throttle) != "n/a"`
  → colour #B91C1C

r2c0

Internet

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_sys_net) != "" && indexOf($(internal:custom_sys_net), "%") >= 0`
  → colour #A16207
- internal · **check_expression**: expression = `$(internal:custom_sys_net) == "DOWN" || $(internal:custom_sys_net) == "n/a"`
  → colour #B91C1C

r2c2

Address

No action: display only.

r2c4

Uptime

No action: display only.

r2c6

Companion

No action: display only.

r2c8

Storage

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_sys_rw) == "READ-ONLY"`
  → colour #CC0000, icon storage-locked
- internal · **check_expression**: expression = `$(internal:custom_sys_rw) == "n/a"`
  → colour #7A5A00

Page 9: PTZ



r1c2 `concat('CAM ', $(internal:custom_ptz2_atem_input))`

Drive CAM 1, the PTZ on ATEM input 1. Goes red when that input is on program and green when it is on preview, so you can see what moving it would cost before you move it.

PRESS
- internal · **set_page**: surfaceId `self`; page `10`

CHANGES WHEN
- internal · **check_expression**: expression = `$(atem:pgm1_input_id) == $(internal:custom_ptz2_atem_input)`
  → colour #FF0000
- internal · **check_expression**: expression = `$(atem:pvw1_input_id) == $(internal:custom_ptz2_atem_input)`
  → colour #00A651
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.online') != "OK"`
  → colour #B91C1C, label = concat('CAM ', $(internal:custom_ptz2_atem_input), ' DOWN')

---

r1c4 **Worship**

Both cameras on one page for worship: their presets, Save, and pan, tilt and zoom by hand.

PRESS
- internal · **set_page**: surfaceId `self`; page 15

r1c6

`concat('CAM ', $(internal:custom_ptz_atem_input))`

Drive CAM 3, the PTZ on ATEM input 3. Goes red when that input is on program and green when it is on preview, so you can see what moving it would cost before you move it.

PRESS
- internal · **set_page**: surfaceId `self`; page 12

CHANGES WHEN
- internal · **check_expression**: expression = `$(atem:pgm1_input_id) == $(internal:custom_ptz_atem_input)`
  → colour #FF0000
- internal · **check_expression**: expression = `$(atem:pvw1_input_id) == $(internal:custom_ptz_atem_input)`
  → colour #00A651
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.online') != "OK"`
  → colour #B91C1C, label = concat('CAM ', $(internal:custom_ptz_atem_input), ' DOWN')

r2c2

`concat(jsonpath($(internal:custom_ptz2_state), '$.zoom'), ' P', $(internal:custom_ptz2_last))`

CAM 1 zoom and the preset it was last sent to, polled every second. Display only.

No action: display only.

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.online') != "OK"`
  → colour #B91C1C, label no answer

r2c4

Sermon

Both cameras on one page for the sermon: tracking, framing, who to follow, and pan, tilt and zoom by hand.

PRESS
- internal · **set_page**: surfaceId `self`; page 16

r2c6

`concat(jsonpath($(internal:custom_ptz_state), '$.zoom'), ' P', $(internal:custom_ptz_last))`

CAM 3 zoom and the preset it was last sent to, polled every second. Display only.

No action: display only.

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.online') != "OK"`
  → colour #B91C1C, label no answer

r3c2

`concat('CAM ', $(internal:custom_ptz2_atem_input), ' Setup')`

Exposure, white balance, backlight, power and tracking for CAM 1.

PRESS
- internal · **set_page**: surfaceId `self`; page 11

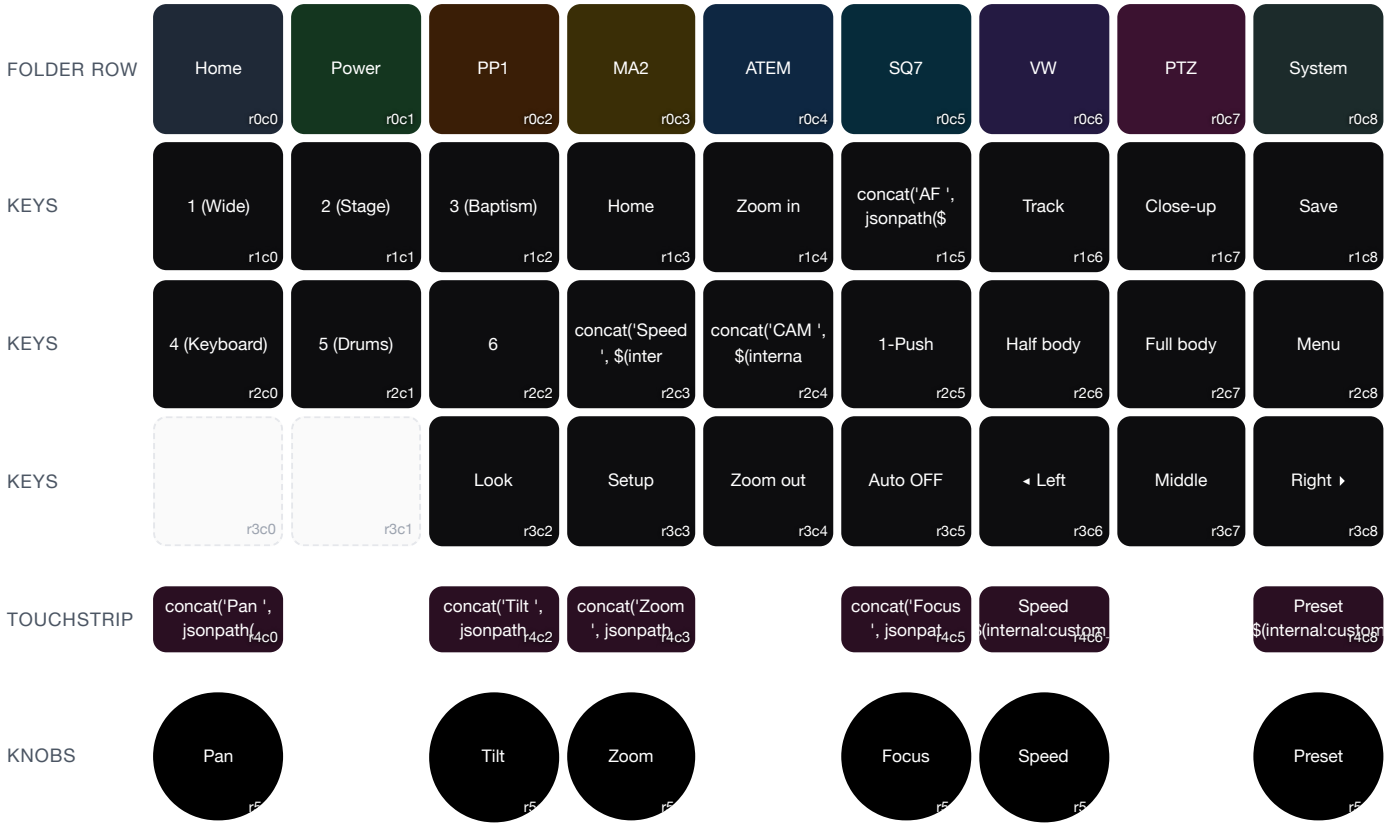
r3c6

`concat('CAM ', $(internal:custom_ptz_atem_input), ' Setup')`

Exposure, white balance, backlight, power and tracking for CAM 3.

PRESS
- internal · **set_page**: surfaceId `self`; page 13

Page 10: CAM 1



`r1c0`

1 (Wide)

Go to preset 1 (Wide), keeping the picture as it is. With Save armed, saves the current shot as preset 1 instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS

  - visca2 · **setPreset**: isText `no`; presetAsNumber 1; presetAsText 1
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 1

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 1
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 1`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 1`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 1, text colour #000000, icon preset-save-ink

r1c1

2 (Stage)

Go to preset 2 (Stage), keeping the picture as it is. With Save armed, saves the current shot as preset 2 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 2; presetAsText 2
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 2

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 2
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 2`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 2`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 2, text colour #000000, icon preset-save-ink

r1c2

3 (Baptism)

Go to preset 3 (Baptism), keeping the picture as it is. With Save armed, saves the current shot as preset 3 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 3; presetAsText 3
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 3

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 3
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 3`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 3`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 3, text colour #000000, icon preset-save-ink

r1c3

Home

Drives pan and tilt to the camera home position.

PRESS

- visca2 · **home**
- internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`

r1c4

Zoom in

Hold to zoom in at the Speed knob's speed; release to stop.

PRESS

- visca2 · **custom**: custom `81 01 04 07 20 FF (zoom in)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_zspeed)`

RELEASE

- visca2 · **zoomS**

r1c5 `concat('AF ', jsonpath($(internal:custom_ptz2_state), '$.focus'))`

Toggles autofocus. Green while the camera is in auto focus.

PRESS
- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto"`

  ACTIONS
  - visca2 · **focusM**: bol 1

  ELSE_ACTIONS
  - visca2 · **focusM**: bol `0`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto"`
  → colour #14361F, icon focus-auto

r1c6 **Track**

Toggles the camera AI auto tracking. Green while on, as the camera reports it.

PRESS
- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.tracking') == "On"`

  ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 track off`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 track on`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.tracking') == "On"`
  → colour #14361F, icon tracking-on, label Track ON

r1c7 **Close-up**

Tracking framing: close-up. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body close`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Close-up"`
  → colour #14361F

r1c8 **Save**

Arms saving: the next preset key or dial press saves the current shot there. Press again to cancel.

PRESS
- internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value = `$(internal:custom_ptz2_armed) == 1 ? 0 : 1`

CHANGES WHEN
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label ARMED, text colour #000000, icon preset-save-ink

`r2c0`

4 (Keyboard)

Go to preset 4 (Keyboard), keeping the picture as it is. With Save armed, saves the current shot as preset 4 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 4; presetAsText 4
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 4

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 4
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 4`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 4`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 4, text colour #000000, icon preset-save-ink

`r2c1`

5 (Drums)

Go to preset 5 (Drums), keeping the picture as it is. With Save armed, saves the current shot as preset 5 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 5; presetAsText 5
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 5

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 5
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 5`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 5`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 5, text colour #000000, icon preset-save-ink

r2c2

6

Go to preset 6, keeping the picture as it is. With Save armed, saves the current shot as preset 6 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 6; presetAsText 6
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 6

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 6
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 6`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 6`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 6, text colour #000000, icon preset-save-ink

r2c3

`concat('Speed ', $(internal:custom_ptz2_speed))`

Steps the drive speed 1 → 10 → 24 → 1 for pan, tilt, zoom and focus. The Speed knob still fine-tunes it.

PRESS

- internal · **custom_variable_set_value**: name `ptz2_speed`; create `yes`; value = `$(internal:custom_ptz2_speed) < 10 ? 10 : ($(internal:custom_ptz2_speed) < 24 ? 24 : 1)`
- internal · **custom_variable_set_value**: name `ptz2_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz2_speed))`
- internal · **custom_variable_set_value**: name `ptz2_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz2_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`

r2c4

`concat('CAM ', $(internal:custom_ptz2_atem_input))`

Stops pan, tilt and zoom, and names the camera this page drives. Closes the camera menu while it is open. Red and LIVE when this camera is on ATEM program, green and PVW on preview (input number in ptz_atem_input).

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.menu') == "On"`

  ACTIONS
  - visca2 · **custom**: custom `81 01 06 06 03 FF (close the camera menu)`

  ELSE_ACTIONS
  - visca2 · **stop**
  - visca2 · **zoomS**

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.menu') == "On"`
  → label EXIT, colour #3A2E06
- internal · **check_expression**: expression = `$(atem:pvw1_input_id) == $(internal:custom_ptz2_atem_input)`
  → colour #00A651, label = concat(concat('CAM ', $(internal:custom_ptz2_atem_input)), ' PVW'), text colour #000000, icon stop-ink
- internal · **check_expression**: expression = `$(atem:pgm1_input_id) == $(internal:custom_ptz2_atem_input)`
  → colour #FF0000, label = concat(concat('CAM ', $(internal:custom_ptz2_atem_input)), ' LIVE'), text colour #FFFFFF, icon stop-paper

r2c5

1-Push

One-push autofocus: the camera focuses once on what it sees now, then holds.

PRESS

- visca2 · **custom**: custom `81 01 04 38 04 FF (one-push autofocus)`

r2c6

Half body

Tracking framing: half body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body half`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Half body"`
  → colour #14361F

r2c7

Full body

Tracking framing: full body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body full`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Full body"`
  → colour #14361F

r2c8

Menu

Opens or closes the camera on-screen menu. While open, the arrows navigate (right enters, left backs out) and STOP closes it.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.menu') == "On"`

  ACTIONS
  - visca2 · **custom**: custom `81 01 06 06 03 FF (close the camera menu)`

  ELSE_ACTIONS
  - visca2 · **custom**: custom `81 01 06 06 02 FF (open the camera menu)`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.menu') == "On"`
  → colour #3A2E06, label Menu OPEN

r3c2

Look

Puts the saved picture settings back (exposure, colour, image, focus, noise reduction) after a preset recall has changed them. Green and "Look set" while the camera is on the saved look. Save look, on the setup page, is what saves it.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 look apply`; timeout `20000`; targetVariable `ptz2_look`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_look), '$.online') == "OK" && jsonpath($(internal:custom_ptz2_state), '$.wb') == jsonpath($(internal:custom_ptz2_look), '$.wb') && jsonpath($(internal:custom_ptz2_state), '$.ae') == jsonpath($(internal:custom_ptz2_look), '$.ae') && jsonpath($(internal:custom_ptz2_state), '$.shutter') == jsonpath($(internal:custom_ptz2_look), '$.shutter') && jsonpath($(internal:custom_ptz2_state), '$.iris') == jsonpath($(internal:custom_ptz2_look), '$.iris') && jsonpath($(internal:custom_ptz2_state), '$.gain') == jsonpath($(internal:custom_ptz2_look), '$.gain') && jsonpath($(internal:custom_ptz2_state), '$.sharp') == jsonpath($(internal:custom_ptz2_look), '$.sharp')`
  → colour #14361F, label Look set
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_look), '$.online') == "OK" && jsonpath($(internal:custom_ptz2_look), '$.left') > 0`
  → colour #3A2E06, label = concat('Look ', jsonpath($(internal:custom_ptz2_look), '$.left'), ' left')
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_look), '$.online') == "DOWN"`
  → colour #B91C1C, label Look FAILED

r3c3

Setup

Opens the PTZ setup page: exposure, white balance, backlight, power and the tracking settings.

PRESS

- internal · **set_page**: surfaceId `self`; page 11

r3c4

Zoom out

Hold to zoom out at the Speed knob's speed; release to stop.

PRESS

- visca2 · **custom**: custom `81 01 04 07 30 FF (zoom out)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_zspeed)`

RELEASE

- visca2 · **zoomS**

r3c5

Auto OFF

Toggles focus and exposure together between automatic and manual (the locked service settings). White balance is left alone. Green while both are auto, amber while only one is. Any press from a not-both-auto state goes to both auto.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto"`

  ACTIONS
  - visca2 · **focusM**: bol 1
  - internal · **wait**: time = `100`
  - visca2 · **expM**: val 1

  ELSE_ACTIONS
  - visca2 · **focusM**: bol `0`
  - internal · **wait**: time = `100`
  - visca2 · **expM**: val `0`

CHANGES WHEN

- internal · **check_expression**: expression = `(jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto" || jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto") && !(jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto")`
  → colour #3A2E06, label Auto PART
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto"`
  → colour #14361F, icon ptz-auto-on, label Auto ON

r3c6

◂ Left

Tracks the person on the left of the frame (the camera is told to take whoever is at a point a sixth of the way across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 select 320 486`; timeout `6000`; targetVariable `ptz2_track`

r3c7

Middle

Tracks the person on the middle of the frame (the camera is told to take whoever is at a point halfway across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 select 960 486`; timeout `6000`; targetVariable `ptz2_track`

r3c8

Right ▸

Tracks the person on the right of the frame (the camera is told to take whoever is at a point five sixths of the way across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 select 1600 486`; timeout `6000`; targetVariable `ptz2_track`

r4c0

`concat('Pan ', jsonpath($(internal:custom_ptz2_state), '$.pan'))`

Pan angle as reported by the camera, polled every second. Turns red if the camera stops answering.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.online') != "OK"`
  → colour #B91C1C, label Camera DOWN

r4c2

`concat('Tilt ', jsonpath($(internal:custom_ptz2_state), '$.tilt'))`

Tilt angle as reported by the camera, polled every second. Display only.

No action: display only.

r4c3

`concat('Zoom ', jsonpath($(internal:custom_ptz2_state), '$.zoom'))`

Zoom position as a percentage of the lens range, polled every second. Display only.

No action: display only.

r4c5

`concat('Focus ', jsonpath($(internal:custom_ptz2_state), '$.focus'))`

Focus mode as reported by the camera. Display only.

No action: display only.

r4c6

`Speed $(internal:custom_ptz2_speed)`

Drive speed for pan, tilt, zoom and focus, 1 (slow) to 24 (fast). Display only.

No action: display only.

r4c8

`Preset $(internal:custom_ptz2_preset)`

The preset the dial is on. Turns red while Save is armed: the next press will overwrite it.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE to $(internal:custom_ptz2_preset), icon preset-save-ink, text colour #000000

r5c0

Pan

Turn to pan at the Speed knob's speed; each detent drives for 150ms. Press stops.

PRESS

- visca2 · **stop**

TURN LEFT

- visca2 · **custom**: custom `81 01 06 01 00 00 01 03 FF (pan left)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

TURN RIGHT

- visca2 · **custom**: custom `81 01 06 01 00 00 02 03 FF (pan right)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

r5c2

Tilt

Turn clockwise to tilt up, anticlockwise down, at the Speed knob's speed. Press stops.

PRESS

- visca2 · **stop**

TURN LEFT

- visca2 · **custom**: custom `81 01 06 01 00 00 03 02 FF (tilt down)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

TURN RIGHT

- visca2 · **custom**: custom `81 01 06 01 00 00 03 01 FF (tilt up)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

r5c3

Zoom

Turn clockwise to zoom in, anticlockwise out. Speed follows the Speed knob. Press stops.

PRESS
- visca2 · **zoomS**

TURN LEFT
- visca2 · **custom**: custom `81 01 04 07 30 FF (zoom out)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_zspeed)`
- internal · **wait**: time = `150`
- visca2 · **zoomS**

TURN RIGHT
- visca2 · **custom**: custom `81 01 04 07 20 FF (zoom in)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_zspeed)`
- internal · **wait**: time = `150`
- visca2 · **zoomS**

r5c5

Focus

Turn to focus by hand (switches the camera to manual focus). Press for one-push autofocus: the camera focuses once and holds it.

PRESS
- visca2 · **custom**: custom `81 01 04 38 04 FF (one-push autofocus)`

TURN LEFT
- visca2 · **focusM**: bol 1
- visca2 · **custom**: custom `81 01 04 08 30 FF (focus near)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_fspeed)`
- internal · **wait**: time = `150`
- visca2 · **custom**: custom `81 01 04 08 00 FF (stop focus)`

TURN RIGHT
- visca2 · **focusM**: bol 1
- visca2 · **custom**: custom `81 01 04 08 20 FF (focus far)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_fspeed)`
- internal · **wait**: time = `150`
- visca2 · **custom**: custom `81 01 04 08 00 FF (stop focus)`

r5c6

Speed

Turn to change drive speed 1-24. Press resets to 12.

PRESS
- internal · **custom_variable_set_value**: name `ptz2_speed`; create `yes`; value 12
- internal · **custom_variable_set_value**: name `ptz2_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz2_speed))`
- internal · **custom_variable_set_value**: name `ptz2_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz2_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`

TURN LEFT
- internal · **custom_variable_set_value**: name `ptz2_speed`; create `yes`; value = `max(1, $(internal:custom_ptz2_speed) - 1)`
- internal · **custom_variable_set_value**: name `ptz2_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz2_speed))`
- internal · **custom_variable_set_value**: name `ptz2_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz2_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`

TURN RIGHT
- internal · **custom_variable_set_value**: name `ptz2_speed`; create `yes`; value = `min(24, $(internal:custom_ptz2_speed) + 1)`
- internal · **custom_variable_set_value**: name `ptz2_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz2_speed))`
- internal · **custom_variable_set_value**: name `ptz2_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz2_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`

r5c8

Preset

Turn to choose a preset 1-254, press to go to it, keeping the picture as it is. With Save armed, press saves the current shot to it instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `yes`; presetAsNumber 1; presetAsText `$(internal:custom_ptz2_preset)`
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value = `$(internal:custom_ptz2_preset)`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value = `$(internal:custom_ptz2_preset)`
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall $(internal:custom_ptz2_preset)`; timeout `20000`; targetVariable `ptz2_recall`

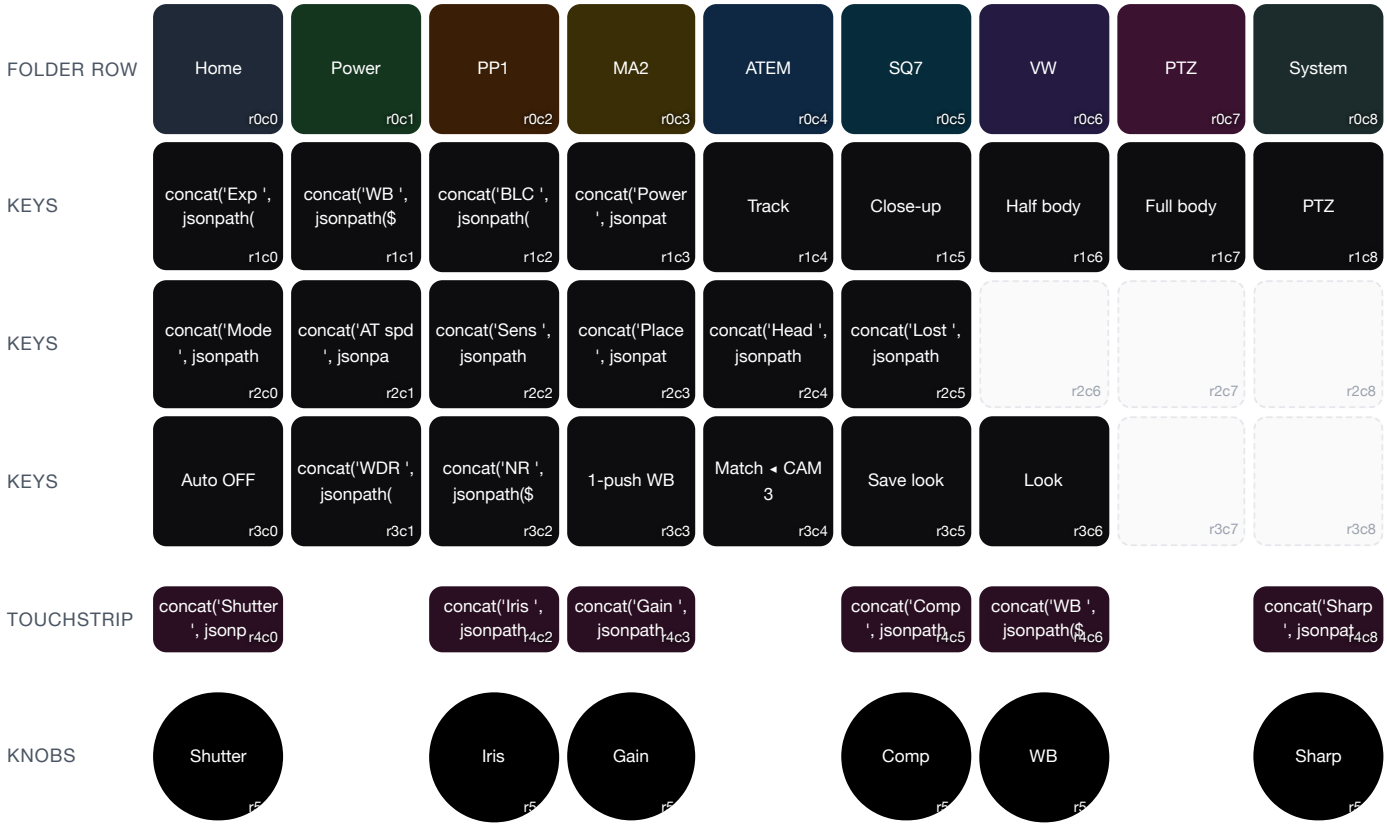
TURN LEFT

- internal · **custom_variable_set_value**: name `ptz2_preset`; create `yes`; value = `$(internal:custom_ptz2_preset) <= 1 ? 254 : $(internal:custom_ptz2_preset) - 1`

TURN RIGHT

- internal · **custom_variable_set_value**: name `ptz2_preset`; create `yes`; value = `$(internal:custom_ptz2_preset) >= 254 ? 1 : $(internal:custom_ptz2_preset) + 1`

Page 11: CAM 1 Setup



`r1c0` `concat('Exp ', jsonpath($(internal:custom_ptz2_state), '$.ae'))`

Steps the exposure mode: Auto, Shutter priority, Iris priority, Manual, then back to Auto.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto"`

  ACTIONS

  - visca2 · **expM**: val 2

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.ae') == "Shutter"`

    actions

    - visca2 · **expM**: val 3

    else_actions

    - internal · **logic_if**

      condition

      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.ae') == "Iris"`

      actions

      - visca2 · **expM**: val 1

      else_actions

      - visca2 · **expM**: val `0`

r1c1 `concat('WB ', jsonpath($(internal:custom_ptz2_state), '$.wb'))`

Toggles white balance between Auto and the colour temperature on the WB dial. The caption is what the camera reports.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wb') == "Auto"`

  ACTIONS

  - internal · **custom_variable_set_value**: name `ptz2_wbcode`; create `yes`; value = `$(internal:custom_ptz2_wbk) == 2400 ? 12 : $(internal:custom_ptz2_wbk) == 2500 ? 13 : $(internal:custom_ptz2_wbk) == 2600 ? 14 : $(internal:custom_ptz2_wbk) == 2700 ? 15 : $(internal:custom_ptz2_wbk) == 2800 ? 16 : $(internal:custom_ptz2_wbk) == 2900 ? 17 : $(internal:custom_ptz2_wbk) == 3000 ? 1 : $(internal:custom_ptz2_wbk) == 3100 ? 18 : $(internal:custom_ptz2_wbk) == 3200 ? 19 : $(internal:custom_ptz2_wbk) == 3300 ? 20 : $(internal:custom_ptz2_wbk) == 3400 ? 21 : $(internal:custom_ptz2_wbk) == 3500 ? 7 : $(internal:custom_ptz2_wbk) == 3600 ? 22 : $(internal:custom_ptz2_wbk) == 3700 ? 23 : $(internal:custom_ptz2_wbk) == 3800 ? 24 : $(internal:custom_ptz2_wbk) == 3900 ? 25 : $(internal:custom_ptz2_wbk) == 4000 ? 2 : $(internal:custom_ptz2_wbk) == 4100 ? 26 : $(internal:custom_ptz2_wbk) == 4200 ? 27 : $(internal:custom_ptz2_wbk) == 4300 ? 28 : $(internal:custom_ptz2_wbk) == 4400 ? 29 : $(internal:custom_ptz2_wbk) == 4500 ? 8 : $(internal:custom_ptz2_wbk) == 4600 ? 30 : $(internal:custom_ptz2_wbk) == 4700 ? 31 : $(internal:custom_ptz2_wbk) == 4800 ? 33 : $(internal:custom_ptz2_wbk) == 4900 ? 34 : $(internal:custom_ptz2_wbk) == 5000 ? 4 : $(internal:custom_ptz2_wbk) == 5100 ? 35 : $(internal:custom_ptz2_wbk) == 5200 ? 36 : $(internal:custom_ptz2_wbk) == 5300 ? 37 : $(internal:custom_ptz2_wbk) == 5400 ? 38 : $(internal:custom_ptz2_wbk) == 5500 ? 9 : $(internal:custom_ptz2_wbk) == 5600 ? 39 : $(internal:custom_ptz2_wbk) == 5700 ? 40 : $(internal:custom_ptz2_wbk) == 5800 ? 41 : $(internal:custom_ptz2_wbk) == 5900 ? 42 : $(internal:custom_ptz2_wbk) == 6000 ? 10 : $(internal:custom_ptz2_wbk) == 6100 ? 43 : $(internal:custom_ptz2_wbk) == 6200 ? 44 : $(internal:custom_ptz2_wbk) == 6300 ? 45 : $(internal:custom_ptz2_wbk) == 6400 ? 46 : $(internal:custom_ptz2_wbk) == 6500 ? 6 : $(internal:custom_ptz2_wbk) == 6600 ? 47 : $(internal:custom_ptz2_wbk) == 6700 ? 48 : $(internal:custom_ptz2_wbk) == 6800 ? 49 : $(internal:custom_ptz2_wbk) == 6900 ? 50 : $(internal:custom_ptz2_wbk) == 7000 ? 11 : $(internal:custom_ptz2_wbk) == 7100 ? 51 : 30`
  - visca2 · **custom**: custom `81 01 04 35 00 FF (white balance mode)`; command_parameters `8,9`; parameter0 `$(internal:custom_ptz2_wbcode)`

  ELSE_ACTIONS

  - visca2 · **wb**: val `automatic`

r1c2 `concat('BLC ', jsonpath($(internal:custom_ptz2_state), '$.backlight'))`

Toggles backlight compensation. Green while on.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.backlight') == "On"`

  ACTIONS

  - visca2 · **custom**: custom `81 01 04 33 03 FF (backlight compensation off)`

  ELSE_ACTIONS

  - visca2 · **custom**: custom `81 01 04 33 02 FF (backlight compensation on)`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.backlight') == "On"`
  → colour #14361F, icon backlight-on

r1c3 `concat('Power ', jsonpath($(internal:custom_ptz2_state), '$.power'))`

Toggles the camera between on and standby. Amber with a red glyph while in standby.

PRESS
- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.power') == "On"`

  ACTIONS
  - visca2 · **power**: bool `off`

  ELSE_ACTIONS
  - visca2 · **power**: bool `on`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.power') == "Standby"`
  → colour #3A2E06, icon ptz-standby, label STANDBY

r1c4 **Track**

Toggles the camera AI auto tracking. Green while on, as the camera reports it.

PRESS
- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.tracking') == "On"`

  ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 track off`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 track on`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.tracking') == "On"`
  → colour #14361F, icon tracking-on, label Track ON

r1c5 **Close-up**

Tracking framing: close-up. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body close`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Close-up"`
  → colour #14361F

r1c6 **Half body**

Tracking framing: half body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body half`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Half body"`
  → colour #14361F

r1c7

Full body

Tracking framing: full body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body full`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Full body"`
  → colour #14361F

r1c8

PTZ

Back to the PTZ run page.

PRESS

- internal · **set_page**: surfaceId `self`; page `10`

r2c0

`concat('Mode ', jsonpath($(internal:custom_ptz2_track), '$.mode'))`

Steps mode through Real-time → Area → Stage → Scene. The caption shows the camera's current value.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.mode') == "Real-time"`

  ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nTrackMode=1`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS
  - internal · **logic_if**
    condition
    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.mode') == "Area"`

    actions
    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nTrackMode=2`; timeout `6000`; targetVariable `ptz2_track`

    else_actions
    - internal · **logic_if**
      condition
      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.mode') == "Stage"`

      actions
      - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nTrackMode=3`; timeout `6000`; targetVariable `ptz2_track`

      else_actions
      - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nTrackMode=0`; timeout `6000`; targetVariable `ptz2_track`

r2c1

`concat('AT spd ', jsonpath($(internal:custom_ptz2_track), '$.speed'))`

Steps at spd through Fast → Middle → Slow. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.speed') == "Fast"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nSpeed=1`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.speed') == "Middle"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nSpeed=2`; timeout `6000`; targetVariable `ptz2_track`

    else_actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nSpeed=0`; timeout `6000`; targetVariable `ptz2_track`

r2c2

`concat('Sens ', jsonpath($(internal:custom_ptz2_track), '$.sensitivity'))`

Steps sens through High → Middle → Low. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.sensitivity') == "High"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nSensitivity=1`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.sensitivity') == "Middle"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nSensitivity=2`; timeout `6000`; targetVariable `ptz2_track`

    else_actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nSensitivity=0`; timeout `6000`; targetVariable `ptz2_track`

r2c3

`concat('Place ', jsonpath($(internal:custom_ptz2_track), '$.placement'))`

Steps place through Left → Centre → Right. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.placement') == "Left"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nCenterPos=1`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.placement') == "Centre"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nCenterPos=2`; timeout `6000`; targetVariable `ptz2_track`

    else_actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nCenterPos=0`; timeout `6000`; targetVariable `ptz2_track`

r2c4

`concat('Head ', jsonpath($(internal:custom_ptz2_track), '$.headroom'))`

Steps head through 5/10 → 6/10 → 7/10 → 8/10 → 9/10. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.headroom') == "5/10"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nHeadRoomRatio=1`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.headroom') == "6/10"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nHeadRoomRatio=2`; timeout `6000`; targetVariable `ptz2_track`

    else_actions

    - internal · **logic_if**

      condition

      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.headroom') == "7/10"`

      actions

      - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nHeadRoomRatio=3`; timeout `6000`; targetVariable `ptz2_track`

      else_actions

      - internal · **logic_if**

        condition

        - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.headroom') == "8/10"`

        actions

        - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nHeadRoomRatio=4`; timeout `6000`; targetVariable `ptz2_track`

        else_actions

        - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nHeadRoomRatio=0`; timeout `6000`; targetVariable `ptz2_track`

r2c5

`concat('Lost ', jsonpath($(internal:custom_ptz2_track), '$.lost'))`

Steps lost through Home → Stay → Preset 0. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.lost') == "Home"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nLostReac=2`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.lost') == "Stay"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nLostReac=4`; timeout `6000`; targetVariable `ptz2_track`

    else_actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 set nLostReac=0`; timeout `6000`; targetVariable `ptz2_track`

r3c0

Auto OFF

Toggles focus and exposure together between automatic and manual (the locked service settings). White balance is left alone. Green while both are auto, amber while only one is. Any press from a not-both-auto state goes to both auto.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto"`

  ACTIONS

  - visca2 · **focusM**: bol 1
  - internal · **wait**: time = `100`
  - visca2 · **expM**: val 1

  ELSE_ACTIONS

  - visca2 · **focusM**: bol `0`
  - internal · **wait**: time = `100`
  - visca2 · **expM**: val `0`

CHANGES WHEN

- internal · **check_expression**: expression = `(jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto" || jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto") && !(jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto")`
  → colour #3A2E06, label Auto PART
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz2_state), '$.ae') == "Auto"`
  → colour #14361F, icon ptz-auto-on, label Auto ON

r3c1 `concat('WDR ', jsonpath($(internal:custom_ptz2_state), '$.wdr'))`

Steps wide dynamic range Off, 1 … 8, Off: how hard the highlights are rolled off before they clip. Green while on. The white-shirt clipping in the docs is what this is for.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') == "Off"`

  ACTIONS
  - visca2 · **custom**: custom `81 01 04 51 00 00 00 01 FF (set WDR level)`

  ELSE_ACTIONS
  - internal · **logic_if**
    condition
    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') == "1"`

    actions
    - visca2 · **custom**: custom `81 01 04 51 00 00 00 02 FF (set WDR level)`

    else_actions
    - internal · **logic_if**
      condition
      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') == "2"`

      actions
      - visca2 · **custom**: custom `81 01 04 51 00 00 00 03 FF (set WDR level)`

      else_actions
      - internal · **logic_if**
        condition
        - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') == "3"`

        actions
        - visca2 · **custom**: custom `81 01 04 51 00 00 00 04 FF (set WDR level)`

        else_actions
        - internal · **logic_if**
          condition
          - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') == "4"`

          actions
          - visca2 · **custom**: custom `81 01 04 51 00 00 00 05 FF (set WDR level)`

          else_actions
          - internal · **logic_if**
            condition
            - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') == "5"`

            actions
            - visca2 · **custom**: custom `81 01 04 51 00 00 00 06 FF (set WDR level)`

            else_actions
            - internal · **logic_if**
              condition
              - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') == "6"`

              actions
              - visca2 · **custom**: custom `81 01 04 51 00 00 00 07 FF (set WDR level)`

else_actions

- internal · **logic_if**

  condition

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') == "7"`

  actions

  - visca2 · **custom**: custom `81 01 04 51 00 00 00 08 FF` (set WDR level)

  else_actions

  - visca2 · **custom**: custom `81 01 04 51 00 00 00 00 FF` (set WDR level)

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wdr') != "Off" && jsonpath($(internal:custom_ptz2_state), '$.wdr') != "--"`
  → colour #14361F

r3c2 `concat('NR ', jsonpath($(internal:custom_ptz2_state), '$.nr'))`

Steps 3D noise reduction Off, 1 … 7, Auto, Off. Green while on. The reference keeps it low: heavy NR smears motion.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') == "Off"`

  ACTIONS

  - visca2 · **custom**: custom `81 01 04 54 01 FF (set noise reduction level)`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') == "1"`

    actions

    - visca2 · **custom**: custom `81 01 04 54 02 FF (set noise reduction level)`

    else_actions

    - internal · **logic_if**

      condition

      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') == "2"`

      actions

      - visca2 · **custom**: custom `81 01 04 54 03 FF (set noise reduction level)`

      else_actions

      - internal · **logic_if**

        condition

        - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') == "3"`

        actions

        - visca2 · **custom**: custom `81 01 04 54 04 FF (set noise reduction level)`

        else_actions

        - internal · **logic_if**

          condition

          - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') == "4"`

          actions

          - visca2 · **custom**: custom `81 01 04 54 05 FF (set noise reduction level)`

          else_actions

          - internal · **logic_if**

            condition

            - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') == "5"`

            actions

            - visca2 · **custom**: custom `81 01 04 54 06 FF (set noise reduction level)`

            else_actions

            - internal · **logic_if**

              condition

              - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') == "6"`

              actions

              - visca2 · **custom**: custom `81 01 04 54 07 FF (set noise reduction level)`

else_actions

- internal · **logic_if**

  condition

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') == "7"`

  actions

  - visca2 · **custom**: custom `81 01 04 54 08 FF` (set noise reduction level)

  else_actions

  - visca2 · **custom**: custom `81 01 04 54 00 FF` (set noise reduction level)

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.nr') != "Off" && jsonpath($(internal:custom_ptz2_state), '$.nr') != "--"`
  → colour #14361F

---

r3c3

1-push WB

Hold a white card at the preaching position, press, and give it a second: the camera measures the white and holds it. The WB readout says 1-Push afterwards. Match the other camera off the same card.

PRESS

- visca2 · **wb**: val `onepush`
- internal · **wait**: time = `300`
- visca2 · **wbOPT**

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.wb') == "1-Push"`
  → colour #14361F

---

r3c4

Match ◂ CAM 3

Copies CAM 3's picture settings (exposure, colour, image, focus, noise reduction; not the flips) onto this camera through the web API. Green and Matched when every setting took, amber with a count if some did not, red if the camera could not be reached. Takes a few seconds.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 match 10.23.0.181`; timeout `20000`; targetVariable `ptz2_match`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_match), '$.online') == "OK" && jsonpath($(internal:custom_ptz2_match), '$.left') > 0`
  → colour #3A2E06, label = concat('Match ', jsonpath($(internal:custom_ptz2_match), '$.left'), ' left')
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_match), '$.online') == "OK" && jsonpath($(internal:custom_ptz2_match), '$.left') == 0`
  → colour #14361F, label Matched
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_match), '$.online') == "DOWN"`
  → colour #B91C1C, label Match FAILED

---

r3c5

Save look

Keeps the picture settings exactly as they are now (exposure, colour, image, focus, noise reduction) as the look that Look puts back. Press when the picture is right. The previous look is kept as .prev.json on the Pi.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 look save`; timeout `20000`; targetVariable `ptz2_look`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_look), '$.online') == "DOWN"`
  → colour #B91C1C, label Save FAILED

---

r3c6

Look

Puts the saved picture settings back (exposure, colour, image, focus, noise reduction) after a preset recall has changed them. Green and "Look set" while the camera is on the saved look. Save look, on the setup page, is what saves it.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 look apply`; timeout `20000`; targetVariable `ptz2_look`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_look), '$.online') == "OK" && jsonpath($(internal:custom_ptz2_state), '$.wb') == jsonpath($(internal:custom_ptz2_look), '$.wb') && jsonpath($(internal:custom_ptz2_state), '$.ae') == jsonpath($(internal:custom_ptz2_look), '$.ae') && jsonpath($(internal:custom_ptz2_state), '$.shutter') == jsonpath($(internal:custom_ptz2_look), '$.shutter') && jsonpath($(internal:custom_ptz2_state), '$.iris') == jsonpath($(internal:custom_ptz2_look), '$.iris') && jsonpath($(internal:custom_ptz2_state), '$.gain') == jsonpath($(internal:custom_ptz2_look), '$.gain') && jsonpath($(internal:custom_ptz2_state), '$.sharp') == jsonpath($(internal:custom_ptz2_look), '$.sharp')`
  → colour #14361F, label Look set
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_look), '$.online') == "OK" && jsonpath($(internal:custom_ptz2_look), '$.left') > 0`
  → colour #3A2E06, label = concat('Look ', jsonpath($(internal:custom_ptz2_look), '$.left'), ' left')
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_look), '$.online') == "DOWN"`
  → colour #B91C1C, label Look FAILED

r4c0

`concat('Shutter ', jsonpath($(internal:custom_ptz2_state), '$.shutter'))`

Shutter speed as the camera reports it. Amber while exposure is not Manual or Shutter priority, when the value is not in effect. Red if the camera stops answering.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.ae') != "Manual" && jsonpath($(internal:custom_ptz2_state), '$.ae') != "Shutter"`
  → colour #3A2E06
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.online') != "OK"`
  → colour #B91C1C, label Camera DOWN

r4c2

`concat('Iris ', jsonpath($(internal:custom_ptz2_state), '$.iris'))`

Iris as the camera reports it. Amber while exposure is not Manual or Iris priority, when the value is not in effect.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.ae') != "Manual" && jsonpath($(internal:custom_ptz2_state), '$.ae') != "Iris"`
  → colour #3A2E06

r4c3

`concat('Gain ', jsonpath($(internal:custom_ptz2_state), '$.gain'))`

Gain 0-15 as the camera reports it. Amber while exposure is not Manual, when the value is not in effect.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!(jsonpath($(internal:custom_ptz2_state), '$.ae') == "Manual")`
  → colour #3A2E06

r4c5

`concat('Comp ', jsonpath($(internal:custom_ptz2_state), '$.expcomp'))`

Exposure compensation as the camera reports it: Off, or an offset in steps either side of 0. Amber while exposure is Manual, when it has no effect.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.ae') == "Manual"`
  → colour #3A2E06

r4c6

`concat('WB ', jsonpath($(internal:custom_ptz2_state), '$.wb'))`

White balance as the camera reports it: a colour temperature, or Auto, Manual or 1-Push. Amber while it is not a temperature, when the dial is not in effect.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `! (includes(jsonpath($(internal:custom_ptz2_state), '$.wb'), "K"))`
  → colour #3A2E06

r4c8

`concat('Sharp ', jsonpath($(internal:custom_ptz2_state), '$.sharp'))`

Sharpness 0-15 as the camera reports it.

No action: display only.

r5c0

Shutter

Turn to step the shutter (switches exposure to Manual). Press returns it to 1/125, the rig's 180° at 59.94.

PRESS

- visca2 · **expM**: val 1
- internal · **wait**: time = `100`
- visca2 · **custom**: custom `81 01 04 4A 00 00 01 05 FF (set shutter)`

TURN LEFT

- visca2 · **expM**: val 1
- visca2 · **custom**: custom `81 01 04 0A 03 FF (shutter slower)`

TURN RIGHT

- visca2 · **expM**: val 1
- visca2 · **custom**: custom `81 01 04 0A 02 FF (shutter faster)`

r5c2

Iris

Turn clockwise to open the iris, anticlockwise to close it (switches exposure to Manual). Press does nothing: there is no standard iris.

TURN LEFT

- visca2 · **expM**: val 1
- visca2 · **custom**: custom `81 01 04 0B 03 FF (iris close)`

TURN RIGHT

- visca2 · **expM**: val 1
- visca2 · **custom**: custom `81 01 04 0B 02 FF (iris open)`

r5c3

Gain

Turn to change gain 0-15 (switches exposure to Manual). Press returns it to 0. The camera has no gain step command, so the deck keeps the dial and sends the value.

PRESS

- internal · **custom_variable_set_value**: name `ptz2_gain`; create `yes`; value `0`
- visca2 · **expM**: val 1
- internal · **wait**: time = `100`
- visca2 · **custom**: custom `81 01 04 4C 00 00 00 00 FF (set gain)`; command_parameters `13,15`; parameter0 `$(internal:custom_ptz2_gain)`

TURN LEFT

- internal · **custom_variable_set_value**: name `ptz2_gain`; create `yes`; value = `max(0, $(internal:custom_ptz2_gain) - 1)`
- visca2 · **expM**: val 1
- internal · **wait**: time = `100`
- visca2 · **custom**: custom `81 01 04 4C 00 00 00 00 FF (set gain)`; command_parameters `13,15`; parameter0 `$(internal:custom_ptz2_gain)`

TURN RIGHT

- internal · **custom_variable_set_value**: name `ptz2_gain`; create `yes`; value = `min(15, $(internal:custom_ptz2_gain) + 1)`
- visca2 · **expM**: val 1
- internal · **wait**: time = `100`
- visca2 · **custom**: custom `81 01 04 4C 00 00 00 00 FF (set gain)`; command_parameters `13,15`; parameter0 `$(internal:custom_ptz2_gain)`

r5c5

Comp

Turn to bias the automatic exposure darker or brighter (switches compensation on). Press switches it off and recentres it. Only applies while exposure is automatic.

PRESS

- visca2 · **custom**: custom `81 01 04 3E 03 FF (exposure compensation off)`
- internal · **wait**: time = `100`
- visca2 · **custom**: custom `81 01 04 4E 00 00 00 07 FF (set exposure compensation)`

TURN LEFT

- visca2 · **custom**: custom `81 01 04 3E 02 FF (exposure compensation on)`
- visca2 · **custom**: custom `81 01 04 0E 03 FF (exposure compensation down)`

TURN RIGHT

- visca2 · **custom**: custom `81 01 04 3E 02 FF (exposure compensation on)`
- visca2 · **custom**: custom `81 01 04 0E 02 FF (exposure compensation up)`

r5c6

WB

Turn to step the colour temperature 100K at a time, 2400-7100K (takes white balance off Auto). Press returns it to 4600K, the stage light.

PRESS

- internal · **custom_variable_set_value**: name ptz2_wbk; create yes; value 4600
- internal · **custom_variable_set_value**: name ptz2_wbcode; create yes; value = $(internal:custom_ptz2_wbk) == 2400 ? 12 : $(internal:custom_ptz2_wbk) == 2500 ? 13 : $(internal:custom_ptz2_wbk) == 2600 ? 14 : $(internal:custom_ptz2_wbk) == 2700 ? 15 : $(internal:custom_ptz2_wbk) == 2800 ? 16 : $(internal:custom_ptz2_wbk) == 2900 ? 17 : $(internal:custom_ptz2_wbk) == 3000 ? 1 : $(internal:custom_ptz2_wbk) == 3100 ? 18 : $(internal:custom_ptz2_wbk) == 3200 ? 19 : $(internal:custom_ptz2_wbk) == 3300 ? 20 : $(internal:custom_ptz2_wbk) == 3400 ? 21 : $(internal:custom_ptz2_wbk) == 3500 ? 7 : $(internal:custom_ptz2_wbk) == 3600 ? 22 : $(internal:custom_ptz2_wbk) == 3700 ? 23 : $(internal:custom_ptz2_wbk) == 3800 ? 24 : $(internal:custom_ptz2_wbk) == 3900 ? 25 : $(internal:custom_ptz2_wbk) == 4000 ? 2 : $(internal:custom_ptz2_wbk) == 4100 ? 26 : $(internal:custom_ptz2_wbk) == 4200 ? 27 : $(internal:custom_ptz2_wbk) == 4300 ? 28 : $(internal:custom_ptz2_wbk) == 4400 ? 29 : $(internal:custom_ptz2_wbk) == 4500 ? 8 : $(internal:custom_ptz2_wbk) == 4600 ? 30 : $(internal:custom_ptz2_wbk) == 4700 ? 31 : $(internal:custom_ptz2_wbk) == 4800 ? 33 : $(internal:custom_ptz2_wbk) == 4900 ? 34 : $(internal:custom_ptz2_wbk) == 5000 ? 4 : $(internal:custom_ptz2_wbk) == 5100 ? 35 : $(internal:custom_ptz2_wbk) == 5200 ? 36 : $(internal:custom_ptz2_wbk) == 5300 ? 37 : $(internal:custom_ptz2_wbk) == 5400 ? 38 : $(internal:custom_ptz2_wbk) == 5500 ? 9 : $(internal:custom_ptz2_wbk) == 5600 ? 39 : $(internal:custom_ptz2_wbk) == 5700 ? 40 : $(internal:custom_ptz2_wbk) == 5800 ? 41 : $(internal:custom_ptz2_wbk) == 5900 ? 42 : $(internal:custom_ptz2_wbk) == 6000 ? 10 : $(internal:custom_ptz2_wbk) == 6100 ? 43 : $(internal:custom_ptz2_wbk) == 6200 ? 44 : $(internal:custom_ptz2_wbk) == 6300 ? 45 : $(internal:custom_ptz2_wbk) == 6400 ? 46 : $(internal:custom_ptz2_wbk) == 6500 ? 6 : $(internal:custom_ptz2_wbk) == 6600 ? 47 : $(internal:custom_ptz2_wbk) == 6700 ? 48 : $(internal:custom_ptz2_wbk) == 6800 ? 49 : $(internal:custom_ptz2_wbk) == 6900 ? 50 : $(internal:custom_ptz2_wbk) == 7000 ? 11 : $(internal:custom_ptz2_wbk) == 7100 ? 51 : 30
- visca2 · **custom**: custom 81 01 04 35 00 FF (white balance mode); command_parameters 8,9; parameter0 $(internal:custom_ptz2_wbcode)

TURN LEFT

- internal · **custom_variable_set_value**: name ptz2_wbk; create yes; value = max(2400, $(internal:custom_ptz2_wbk) - 100)
- internal · **custom_variable_set_value**: name ptz2_wbcode; create yes; value = $(internal:custom_ptz2_wbk) == 2400 ? 12 : $(internal:custom_ptz2_wbk) == 2500 ? 13 : $(internal:custom_ptz2_wbk) == 2600 ? 14 : $(internal:custom_ptz2_wbk) == 2700 ? 15 : $(internal:custom_ptz2_wbk) == 2800 ? 16 : $(internal:custom_ptz2_wbk) == 2900 ? 17 : $(internal:custom_ptz2_wbk) == 3000 ? 1 : $(internal:custom_ptz2_wbk) == 3100 ? 18 : $(internal:custom_ptz2_wbk) == 3200 ? 19 : $(internal:custom_ptz2_wbk) == 3300 ? 20 : $(internal:custom_ptz2_wbk) == 3400 ? 21 : $(internal:custom_ptz2_wbk) == 3500 ? 7 : $(internal:custom_ptz2_wbk) == 3600 ? 22 : $(internal:custom_ptz2_wbk) == 3700 ? 23 : $(internal:custom_ptz2_wbk) == 3800 ? 24 : $(internal:custom_ptz2_wbk) == 3900 ? 25 : $(internal:custom_ptz2_wbk) == 4000 ? 2 : $(internal:custom_ptz2_wbk) == 4100 ? 26 : $(internal:custom_ptz2_wbk) == 4200 ? 27 : $(internal:custom_ptz2_wbk) == 4300 ? 28 : $(internal:custom_ptz2_wbk) == 4400 ? 29 : $(internal:custom_ptz2_wbk) == 4500 ? 8 : $(internal:custom_ptz2_wbk) == 4600 ? 30 : $(internal:custom_ptz2_wbk) == 4700 ? 31 : $(internal:custom_ptz2_wbk) == 4800 ? 33 : $(internal:custom_ptz2_wbk) == 4900 ? 34 : $(internal:custom_ptz2_wbk) == 5000 ? 4 : $(internal:custom_ptz2_wbk) == 5100 ? 35 : $(internal:custom_ptz2_wbk) == 5200 ? 36 : $(internal:custom_ptz2_wbk) == 5300 ? 37 : $(internal:custom_ptz2_wbk) == 5400 ? 38 : $(internal:custom_ptz2_wbk) == 5500 ? 9 : $(internal:custom_ptz2_wbk) == 5600 ? 39 : $(internal:custom_ptz2_wbk) == 5700 ? 40 : $(internal:custom_ptz2_wbk) == 5800 ? 41 : $(internal:custom_ptz2_wbk) == 5900 ? 42 : $(internal:custom_ptz2_wbk) == 6000 ? 10 : $(internal:custom_ptz2_wbk) == 6100 ? 43 : $(internal:custom_ptz2_wbk) == 6200 ? 44 : $(internal:custom_ptz2_wbk) == 6300 ? 45 : $(internal:custom_ptz2_wbk) == 6400 ?

46 : $(internal:custom_ptz2_wbk) == 6500 ? 6 : $(internal:custom_ptz2_wbk) == 6600 ? 47 : $(internal:custom_ptz2_wbk) == 6700 ? 48 : $(internal:custom_ptz2_wbk) == 6800 ? 49 : $(internal:custom_ptz2_wbk) == 6900 ? 50 : $(internal:custom_ptz2_wbk) == 7000 ? 11 : $(internal:custom_ptz2_wbk) == 7100 ? 51 : 30`

- visca2 · **custom**: custom `81 01 04 35 00 FF (white balance mode)`; command_parameters `8,9`; parameter0 `$(internal:custom_ptz2_wbcode)`

TURN RIGHT

- internal · **custom_variable_set_value**: name `ptz2_wbk`; create `yes`; value = `min(7100, $(internal:custom_ptz2_wbk) + 100)`
- internal · **custom_variable_set_value**: name `ptz2_wbcode`; create `yes`; value = `$(internal:custom_ptz2_wbk) == 2400 ? 12 : $(internal:custom_ptz2_wbk) == 2500 ? 13 : $(internal:custom_ptz2_wbk) == 2600 ? 14 : $(internal:custom_ptz2_wbk) == 2700 ? 15 : $(internal:custom_ptz2_wbk) == 2800 ? 16 : $(internal:custom_ptz2_wbk) == 2900 ? 17 : $(internal:custom_ptz2_wbk) == 3000 ? 1 : $(internal:custom_ptz2_wbk) == 3100 ? 18 : $(internal:custom_ptz2_wbk) == 3200 ? 19 : $(internal:custom_ptz2_wbk) == 3300 ? 20 : $(internal:custom_ptz2_wbk) == 3400 ? 21 : $(internal:custom_ptz2_wbk) == 3500 ? 7 : $(internal:custom_ptz2_wbk) == 3600 ? 22 : $(internal:custom_ptz2_wbk) == 3700 ? 23 : $(internal:custom_ptz2_wbk) == 3800 ? 24 : $(internal:custom_ptz2_wbk) == 3900 ? 25 : $(internal:custom_ptz2_wbk) == 4000 ? 2 : $(internal:custom_ptz2_wbk) == 4100 ? 26 : $(internal:custom_ptz2_wbk) == 4200 ? 27 : $(internal:custom_ptz2_wbk) == 4300 ? 28 : $(internal:custom_ptz2_wbk) == 4400 ? 29 : $(internal:custom_ptz2_wbk) == 4500 ? 8 : $(internal:custom_ptz2_wbk) == 4600 ? 30 : $(internal:custom_ptz2_wbk) == 4700 ? 31 : $(internal:custom_ptz2_wbk) == 4800 ? 33 : $(internal:custom_ptz2_wbk) == 4900 ? 34 : $(internal:custom_ptz2_wbk) == 5000 ? 4 : $(internal:custom_ptz2_wbk) == 5100 ? 35 : $(internal:custom_ptz2_wbk) == 5200 ? 36 : $(internal:custom_ptz2_wbk) == 5300 ? 37 : $(internal:custom_ptz2_wbk) == 5400 ? 38 : $(internal:custom_ptz2_wbk) == 5500 ? 9 : $(internal:custom_ptz2_wbk) == 5600 ? 39 : $(internal:custom_ptz2_wbk) == 5700 ? 40 : $(internal:custom_ptz2_wbk) == 5800 ? 41 : $(internal:custom_ptz2_wbk) == 5900 ? 42 : $(internal:custom_ptz2_wbk) == 6000 ? 10 : $(internal:custom_ptz2_wbk) == 6100 ? 43 : $(internal:custom_ptz2_wbk) == 6200 ? 44 : $(internal:custom_ptz2_wbk) == 6300 ? 45 : $(internal:custom_ptz2_wbk) == 6400 ? 46 : $(internal:custom_ptz2_wbk) == 6500 ? 6 : $(internal:custom_ptz2_wbk) == 6600 ? 47 : $(internal:custom_ptz2_wbk) == 6700 ? 48 : $(internal:custom_ptz2_wbk) == 6800 ? 49 : $(internal:custom_ptz2_wbk) == 6900 ? 50 : $(internal:custom_ptz2_wbk) == 7000 ? 11 : $(internal:custom_ptz2_wbk) == 7100 ? 51 : 30`
- visca2 · **custom**: custom `81 01 04 35 00 FF (white balance mode)`; command_parameters `8,9`; parameter0 `$(internal:custom_ptz2_wbcode)`

r5c8

Sharp

Turn to change sharpness 0-15. Press returns it to 0, which is where the reference settings keep it.

PRESS

- visca2 · **custom**: custom `81 01 04 42 00 00 00 00 FF (set sharpness)`

TURN LEFT

- visca2 · **custom**: custom `81 01 04 02 03 FF (sharpness down)`

TURN RIGHT

- visca2 · **custom**: custom `81 01 04 02 02 FF (sharpness up)`

Page 12: CAM 3



`r1c0`

1 (Wide)

Go to preset 1 (Wide), keeping the picture as it is. With Save armed, saves the current shot as preset 1 instead.

PRESS

- internal · **logic_if**

  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 1; presetAsText 1
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 1

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 1
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 1`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 1`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 1, text colour #000000, icon preset-save-ink

r1c1

2 (Drums)

Go to preset 2 (Drums), keeping the picture as it is. With Save armed, saves the current shot as preset 2 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 2; presetAsText 2
  - internal · **custom_variable_set_value**: name `ptz_armed`; create yes; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create yes; value 2

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create yes; value 2
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 2`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 2`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 2, text colour #000000, icon preset-save-ink

r1c2

3 (Keys)

Go to preset 3 (Keys), keeping the picture as it is. With Save armed, saves the current shot as preset 3 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 3; presetAsText 3
  - internal · **custom_variable_set_value**: name `ptz_armed`; create yes; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create yes; value 3

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create yes; value 3
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 3`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 3`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 3, text colour #000000, icon preset-save-ink

r1c3

Home

Drives pan and tilt to the camera home position.

PRESS

- visca · **home**
- internal · **custom_variable_set_value**: name `ptz_last`; create yes

r1c4

Zoom in

Hold to zoom in at the Speed knob's speed; release to stop.

PRESS

- visca · **custom**: custom `81 01 04 07 20 FF (zoom in)`; command_parameters 9; parameter0 `$(internal:custom_ptz_zspeed)`

RELEASE

- visca · **zoomS**

r1c5 `concat('AF ', jsonpath($(internal:custom_ptz_state), '$.focus'))`

Toggles autofocus. Green while the camera is in auto focus.

PRESS
- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto"`

  ACTIONS
  - visca · **focusM**: bol 1

  ELSE_ACTIONS
  - visca · **focusM**: bol `0`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto"`
  → colour #14361F, icon focus-auto

r1c6 **Track**

Toggles the camera AI auto tracking. Green while on, as the camera reports it.

PRESS
- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.tracking') == "On"`

  ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 track off`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 track on`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.tracking') == "On"`
  → colour #14361F, icon tracking-on, label Track ON

r1c7 **Close-up**

Tracking framing: close-up. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body close`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Close-up"`
  → colour #14361F

r1c8 **Save**

Arms saving: the next preset key or dial press saves the current shot there. Press again to cancel.

PRESS
- internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value = `$(internal:custom_ptz_armed) == 1 ? 0 : 1`

CHANGES WHEN
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label ARMED, text colour #000000, icon preset-save-ink

`r2c0`

4 (Stage)

Go to preset 4 (Stage), keeping the picture as it is. With Save armed, saves the current shot as preset 4 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 4; presetAsText 4
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 4

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 4
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 4`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 4`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 4, text colour #000000, icon preset-save-ink

`r2c1`

5 (Bass)

Go to preset 5 (Bass), keeping the picture as it is. With Save armed, saves the current shot as preset 5 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 5; presetAsText 5
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 5

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 5
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 5`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 5`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 5, text colour #000000, icon preset-save-ink

r2c2

6

Go to preset 6, keeping the picture as it is. With Save armed, saves the current shot as preset 6 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 6; presetAsText 6
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 6

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 6
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 6`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 6`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 6, text colour #000000, icon preset-save-ink

r2c3

`concat('Speed ', $(internal:custom_ptz_speed))`

Steps the drive speed 1 → 10 → 24 → 1 for pan, tilt, zoom and focus. The Speed knob still fine-tunes it.

PRESS

- internal · **custom_variable_set_value**: name `ptz_speed`; create `yes`; value = `$(internal:custom_ptz_speed) < 10 ? 10 : ($(internal:custom_ptz_speed) < 24 ? 24 : 1)`
- internal · **custom_variable_set_value**: name `ptz_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz_speed))`
- internal · **custom_variable_set_value**: name `ptz_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`

r2c4

`concat('CAM ', $(internal:custom_ptz_atem_input))`

Stops pan, tilt and zoom, and names the camera this page drives. Closes the camera menu while it is open. Red and LIVE when this camera is on ATEM program, green and PVW on preview (input number in ptz_atem_input).

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.menu') == "On"`

  ACTIONS
  - visca · **custom**: custom `81 01 06 06 03 FF (close the camera menu)`

  ELSE_ACTIONS
  - visca · **stop**
  - visca · **zoomS**

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.menu') == "On"`
  → label EXIT, colour #3A2E06
- internal · **check_expression**: expression = `$(atem:pvw1_input_id) == $(internal:custom_ptz_atem_input)`
  → colour #00A651, label = concat(concat('CAM ', $(internal:custom_ptz_atem_input)), ' PVW'), text colour #000000, icon stop-ink
- internal · **check_expression**: expression = `$(atem:pgm1_input_id) == $(internal:custom_ptz_atem_input)`
  → colour #FF0000, label = concat(concat('CAM ', $(internal:custom_ptz_atem_input)), ' LIVE'), text colour #FFFFFF, icon stop-paper

r2c5

1-Push

One-push autofocus: the camera focuses once on what it sees now, then holds.

PRESS

- visca · **custom**: custom `81 01 04 38 04 FF (one-push autofocus)`

r2c6

Half body

Tracking framing: half body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body half`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Half body"`
  → colour #14361F

r2c7

Full body

Tracking framing: full body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body full`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Full body"`
  → colour #14361F

r2c8

Menu

Opens or closes the camera on-screen menu. While open, the arrows navigate (right enters, left backs out) and STOP closes it.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.menu') == "On"`

  ACTIONS
  - visca · **custom**: custom `81 01 06 06 03 FF (close the camera menu)`

  ELSE_ACTIONS
  - visca · **custom**: custom `81 01 06 06 02 FF (open the camera menu)`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.menu') == "On"`
  → colour #3A2E06, label Menu OPEN

r3c2

Look

Puts the saved picture settings back (exposure, colour, image, focus, noise reduction) after a preset recall has changed them. Green and "Look set" while the camera is on the saved look. Save look, on the setup page, is what saves it.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 look apply`; timeout `20000`; targetVariable `ptz_look`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_look), '$.online') == "OK" && jsonpath($(internal:custom_ptz_state), '$.wb') == jsonpath($(internal:custom_ptz_look), '$.wb') && jsonpath($(internal:custom_ptz_state), '$.ae') == jsonpath($(internal:custom_ptz_look), '$.ae') && jsonpath($(internal:custom_ptz_state), '$.shutter') == jsonpath($(internal:custom_ptz_look), '$.shutter') && jsonpath($(internal:custom_ptz_state), '$.iris') == jsonpath($(internal:custom_ptz_look), '$.iris') && jsonpath($(internal:custom_ptz_state), '$.gain') == jsonpath($(internal:custom_ptz_look), '$.gain') && jsonpath($(internal:custom_ptz_state), '$.sharp') == jsonpath($(internal:custom_ptz_look), '$.sharp')`
  → colour #14361F, label Look set
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_look), '$.online') == "OK" && jsonpath($(internal:custom_ptz_look), '$.left') > 0`
  → colour #3A2E06, label = concat('Look ', jsonpath($(internal:custom_ptz_look), '$.left'), ' left')
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_look), '$.online') == "DOWN"`
  → colour #B91C1C, label Look FAILED

r3c3

Setup

Opens the PTZ setup page: exposure, white balance, backlight, power and the tracking settings.

PRESS

- internal · **set_page**: surfaceId `self`; page 13

r3c4

Zoom out

Hold to zoom out at the Speed knob's speed; release to stop.

PRESS

- visca · **custom**: custom `81 01 04 07 30 FF (zoom out)`; command_parameters 9; parameter0 `$(internal:custom_ptz_zspeed)`

RELEASE

- visca · **zoomS**

r3c5

Auto OFF

Toggles focus and exposure together between automatic and manual (the locked service settings). White balance is left alone. Green while both are auto, amber while only one is. Any press from a not-both-auto state goes to both auto.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto"`

  ACTIONS
  - visca · **focusM**: bol 1
  - internal · **wait**: time = `100`
  - visca · **expM**: val 1

  ELSE_ACTIONS
  - visca · **focusM**: bol `0`
  - internal · **wait**: time = `100`
  - visca · **expM**: val `0`

CHANGES WHEN

- internal · **check_expression**: expression = `(jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto" || jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto") && !(jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto")`
  → colour #3A2E06, label Auto PART
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto"`
  → colour #14361F, icon ptz-auto-on, label Auto ON

r3c6

◂ Left

Tracks the person on the left of the frame (the camera is told to take whoever is at a point a sixth of the way across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 select 320 486`; timeout `6000`; targetVariable `ptz_track`

r3c7

Middle

Tracks the person on the middle of the frame (the camera is told to take whoever is at a point halfway across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 select 960 486`; timeout `6000`; targetVariable `ptz_track`

r3c8

Right ▸

Tracks the person on the right of the frame (the camera is told to take whoever is at a point five sixths of the way across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 select 1600 486`; timeout `6000`; targetVariable `ptz_track`

r4c0

`concat('Pan ', jsonpath($(internal:custom_ptz_state), '$.pan'))`

Pan angle as reported by the camera, polled every second. Turns red if the camera stops answering.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.online') != "OK"`
  → colour #B91C1C, label Camera DOWN

r4c2

`concat('Tilt ', jsonpath($(internal:custom_ptz_state), '$.tilt'))`

Tilt angle as reported by the camera, polled every second. Display only.

No action: display only.

r4c3

`concat('Zoom ', jsonpath($(internal:custom_ptz_state), '$.zoom'))`

Zoom position as a percentage of the lens range, polled every second. Display only.

No action: display only.

r4c5

`concat('Focus ', jsonpath($(internal:custom_ptz_state), '$.focus'))`

Focus mode as reported by the camera. Display only.

No action: display only.

r4c6

`Speed $(internal:custom_ptz_speed)`

Drive speed for pan, tilt, zoom and focus, 1 (slow) to 24 (fast). Display only.

No action: display only.

r4c8

`Preset $(internal:custom_ptz_preset)`

The preset the dial is on. Turns red while Save is armed: the next press will overwrite it.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE to $(internal:custom_ptz_preset), icon preset-save-ink, text colour #000000

r5c0

Pan

Turn to pan at the Speed knob's speed; each detent drives for 150ms. Press stops.

PRESS

- visca · **stop**

TURN LEFT

- visca · **custom**: custom `81 01 06 01 00 00 01 03 FF (pan left)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

TURN RIGHT

- visca · **custom**: custom `81 01 06 01 00 00 02 03 FF (pan right)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

r5c2

Tilt

Turn clockwise to tilt up, anticlockwise down, at the Speed knob's speed. Press stops.

PRESS

- visca · **stop**

TURN LEFT

- visca · **custom**: custom `81 01 06 01 00 00 03 02 FF (tilt down)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

TURN RIGHT

- visca · **custom**: custom `81 01 06 01 00 00 03 01 FF (tilt up)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

r5c3

Zoom

Turn clockwise to zoom in, anticlockwise out. Speed follows the Speed knob. Press stops.

PRESS
- visca · **zoomS**

TURN LEFT
- visca · **custom**: custom `81 01 04 07 30 FF (zoom out)`; command_parameters 9; parameter0 `$(internal:custom_ptz_zspeed)`
- internal · **wait**: time = `150`
- visca · **zoomS**

TURN RIGHT
- visca · **custom**: custom `81 01 04 07 20 FF (zoom in)`; command_parameters 9; parameter0 `$(internal:custom_ptz_zspeed)`
- internal · **wait**: time = `150`
- visca · **zoomS**

r5c5

Focus

Turn to focus by hand (switches the camera to manual focus). Press for one-push autofocus: the camera focuses once and holds it.

PRESS
- visca · **custom**: custom `81 01 04 38 04 FF (one-push autofocus)`

TURN LEFT
- visca · **focusM**: bol 1
- visca · **custom**: custom `81 01 04 08 30 FF (focus near)`; command_parameters 9; parameter0 `$(internal:custom_ptz_fspeed)`
- internal · **wait**: time = `150`
- visca · **custom**: custom `81 01 04 08 00 FF (stop focus)`

TURN RIGHT
- visca · **focusM**: bol 1
- visca · **custom**: custom `81 01 04 08 20 FF (focus far)`; command_parameters 9; parameter0 `$(internal:custom_ptz_fspeed)`
- internal · **wait**: time = `150`
- visca · **custom**: custom `81 01 04 08 00 FF (stop focus)`

r5c6

Speed

Turn to change drive speed 1-24. Press resets to 12.

PRESS
- internal · **custom_variable_set_value**: name `ptz_speed`; create `yes`; value 12
- internal · **custom_variable_set_value**: name `ptz_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz_speed))`
- internal · **custom_variable_set_value**: name `ptz_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`

TURN LEFT
- internal · **custom_variable_set_value**: name `ptz_speed`; create `yes`; value = `max(1, $(internal:custom_ptz_speed) - 1)`
- internal · **custom_variable_set_value**: name `ptz_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz_speed))`
- internal · **custom_variable_set_value**: name `ptz_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`

TURN RIGHT
- internal · **custom_variable_set_value**: name `ptz_speed`; create `yes`; value = `min(24, $(internal:custom_ptz_speed) + 1)`
- internal · **custom_variable_set_value**: name `ptz_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz_speed))`
- internal · **custom_variable_set_value**: name `ptz_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`

r5c8

Preset

Turn to choose a preset 1-254, press to go to it, keeping the picture as it is. With Save armed, press saves the current shot to it instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS

  - visca · **setPreset**: isText `yes`; presetAsNumber 1; presetAsText `$(internal:custom_ptz_preset)`
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value = `$(internal:custom_ptz_preset)`

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value = `$(internal:custom_ptz_preset)`
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall $(internal:custom_ptz_preset)`; timeout `20000`; targetVariable `ptz_recall`

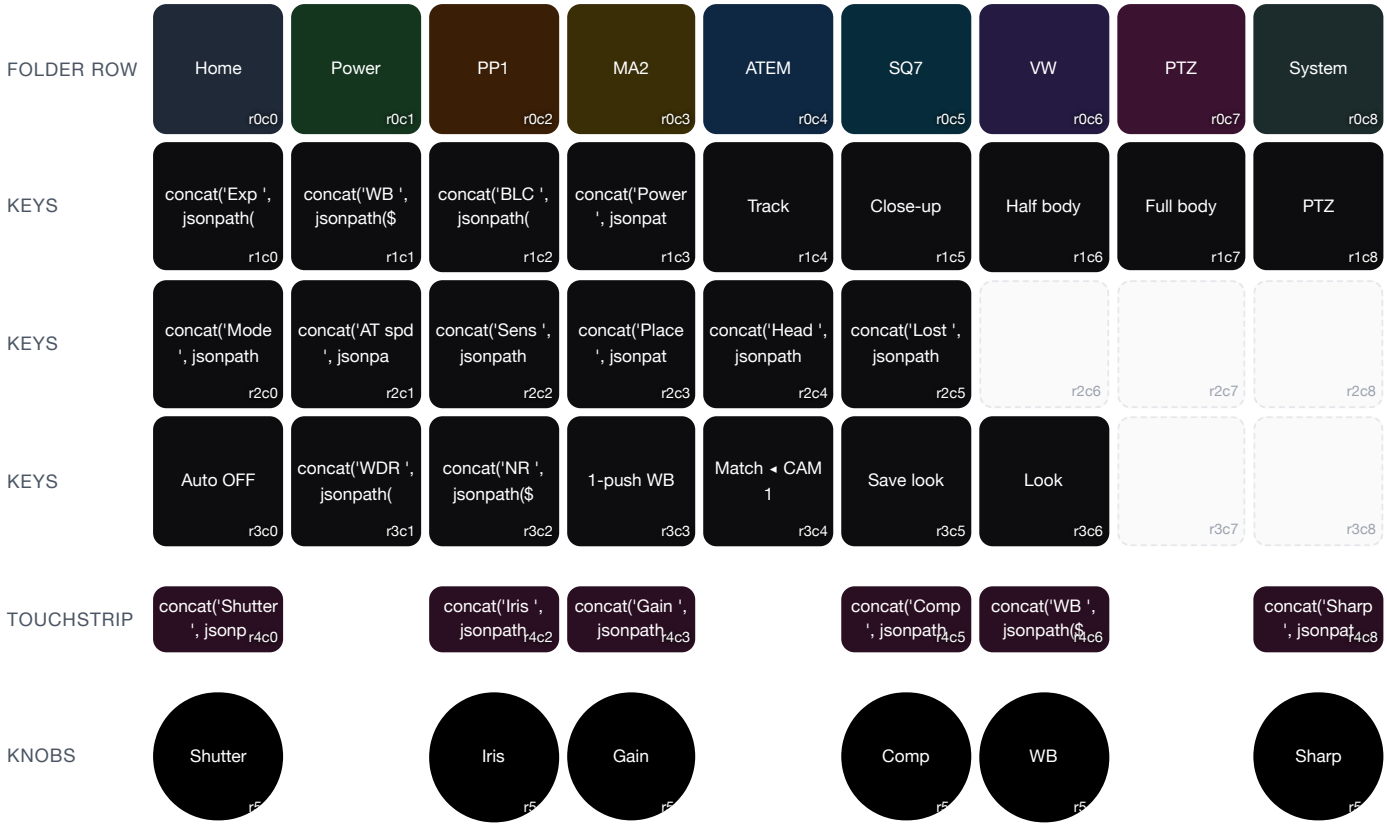
TURN LEFT

- internal · **custom_variable_set_value**: name `ptz_preset`; create `yes`; value = `$(internal:custom_ptz_preset) <= 1 ? 254 : $(internal:custom_ptz_preset) - 1`

TURN RIGHT

- internal · **custom_variable_set_value**: name `ptz_preset`; create `yes`; value = `$(internal:custom_ptz_preset) >= 254 ? 1 : $(internal:custom_ptz_preset) + 1`

Page 13: CAM 3 Setup



`r1c0` `concat('Exp ', jsonpath($(internal:custom_ptz_state), '$.ae'))`

Steps the exposure mode: Auto, Shutter priority, Iris priority, Manual, then back to Auto.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto"`

  ACTIONS

  - visca · **expM**: val 2

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.ae') == "Shutter"`

    actions

    - visca · **expM**: val 3

    else_actions

    - internal · **logic_if**

      condition

      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.ae') == "Iris"`

      actions

      - visca · **expM**: val 1

      else_actions

      - visca · **expM**: val `0`

r1c1

`concat('WB ', jsonpath($(internal:custom_ptz_state), '$.wb'))`

Toggles white balance between Auto and the colour temperature on the WB dial. The caption is what the camera reports.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wb') == "Auto"`

  ACTIONS

  - internal · **custom_variable_set_value**: name `ptz_wbcode`; create `yes`; value = `$(internal:custom_ptz_wbk) == 2400 ? 12 : $(internal:custom_ptz_wbk) == 2500 ? 13 : $(internal:custom_ptz_wbk) == 2600 ? 14 : $(internal:custom_ptz_wbk) == 2700 ? 15 : $(internal:custom_ptz_wbk) == 2800 ? 16 : $(internal:custom_ptz_wbk) == 2900 ? 17 : $(internal:custom_ptz_wbk) == 3000 ? 1 : $(internal:custom_ptz_wbk) == 3100 ? 18 : $(internal:custom_ptz_wbk) == 3200 ? 19 : $(internal:custom_ptz_wbk) == 3300 ? 20 : $(internal:custom_ptz_wbk) == 3400 ? 21 : $(internal:custom_ptz_wbk) == 3500 ? 7 : $(internal:custom_ptz_wbk) == 3600 ? 22 : $(internal:custom_ptz_wbk) == 3700 ? 23 : $(internal:custom_ptz_wbk) == 3800 ? 24 : $(internal:custom_ptz_wbk) == 3900 ? 25 : $(internal:custom_ptz_wbk) == 4000 ? 2 : $(internal:custom_ptz_wbk) == 4100 ? 26 : $(internal:custom_ptz_wbk) == 4200 ? 27 : $(internal:custom_ptz_wbk) == 4300 ? 28 : $(internal:custom_ptz_wbk) == 4400 ? 29 : $(internal:custom_ptz_wbk) == 4500 ? 8 : $(internal:custom_ptz_wbk) == 4600 ? 30 : $(internal:custom_ptz_wbk) == 4700 ? 31 : $(internal:custom_ptz_wbk) == 4800 ? 33 : $(internal:custom_ptz_wbk) == 4900 ? 34 : $(internal:custom_ptz_wbk) == 5000 ? 4 : $(internal:custom_ptz_wbk) == 5100 ? 35 : $(internal:custom_ptz_wbk) == 5200 ? 36 : $(internal:custom_ptz_wbk) == 5300 ? 37 : $(internal:custom_ptz_wbk) == 5400 ? 38 : $(internal:custom_ptz_wbk) == 5500 ? 9 : $(internal:custom_ptz_wbk) == 5600 ? 39 : $(internal:custom_ptz_wbk) == 5700 ? 40 : $(internal:custom_ptz_wbk) == 5800 ? 41 : $(internal:custom_ptz_wbk) == 5900 ? 42 : $(internal:custom_ptz_wbk) == 6000 ? 10 : $(internal:custom_ptz_wbk) == 6100 ? 43 : $(internal:custom_ptz_wbk) == 6200 ? 44 : $(internal:custom_ptz_wbk) == 6300 ? 45 : $(internal:custom_ptz_wbk) == 6400 ? 46 : $(internal:custom_ptz_wbk) == 6500 ? 6 : $(internal:custom_ptz_wbk) == 6600 ? 47 : $(internal:custom_ptz_wbk) == 6700 ? 48 : $(internal:custom_ptz_wbk) == 6800 ? 49 : $(internal:custom_ptz_wbk) == 6900 ? 50 : $(internal:custom_ptz_wbk) == 7000 ? 11 : $(internal:custom_ptz_wbk) == 7100 ? 51 : 30`
  - visca · **custom**: custom `81 01 04 35 00 FF (white balance mode)`; command_parameters `8,9`; parameter0 `$(internal:custom_ptz_wbcode)`

  ELSE_ACTIONS

  - visca · **wb**: val `automatic`

r1c2

`concat('BLC ', jsonpath($(internal:custom_ptz_state), '$.backlight'))`

Toggles backlight compensation. Green while on.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.backlight') == "On"`

  ACTIONS

  - visca · **custom**: custom `81 01 04 33 03 FF (backlight compensation off)`

  ELSE_ACTIONS

  - visca · **custom**: custom `81 01 04 33 02 FF (backlight compensation on)`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.backlight') == "On"`
  → colour #14361F, icon backlight-on

r1c3 `concat('Power ', jsonpath($(internal:custom_ptz_state), '$.power'))`

Toggles the camera between on and standby. Amber with a red glyph while in standby.

PRESS
- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.power') == "On"`

  ACTIONS
  - visca · **power**: bool `off`

  ELSE_ACTIONS
  - visca · **power**: bool `on`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.power') == "Standby"`
  → colour #3A2E06, icon ptz-standby, label STANDBY

r1c4 **Track**

Toggles the camera AI auto tracking. Green while on, as the camera reports it.

PRESS
- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.tracking') == "On"`

  ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 track off`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 track on`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.tracking') == "On"`
  → colour #14361F, icon tracking-on, label Track ON

r1c5 **Close-up**

Tracking framing: close-up. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body close`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Close-up"`
  → colour #14361F

r1c6 **Half body**

Tracking framing: half body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body half`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Half body"`
  → colour #14361F

r1c7

Full body

Tracking framing: full body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body full`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Full body"`
  → colour #14361F

r1c8

PTZ

Back to the PTZ run page.

PRESS

- internal · **set_page**: surfaceId `self`; page 12

r2c0

`concat('Mode ', jsonpath($(internal:custom_ptz_track), '$.mode'))`

Steps mode through Real-time → Area → Stage → Scene. The caption shows the camera's current value.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.mode') == "Real-time"`

  ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nTrackMode=1`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS
  - internal · **logic_if**
    condition
    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.mode') == "Area"`

    actions
    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nTrackMode=2`; timeout `6000`; targetVariable `ptz_track`

    else_actions
    - internal · **logic_if**
      condition
      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.mode') == "Stage"`

      actions
      - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nTrackMode=3`; timeout `6000`; targetVariable `ptz_track`

      else_actions
      - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nTrackMode=0`; timeout `6000`; targetVariable `ptz_track`

r2c1

`concat('AT spd ', jsonpath($(internal:custom_ptz_track), '$.speed'))`

Steps at spd through Fast → Middle → Slow. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.speed') == "Fast"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nSpeed=1`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.speed') == "Middle"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nSpeed=2`; timeout `6000`; targetVariable `ptz_track`

    else_actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nSpeed=0`; timeout `6000`; targetVariable `ptz_track`

r2c2

`concat('Sens ', jsonpath($(internal:custom_ptz_track), '$.sensitivity'))`

Steps sens through High → Middle → Low. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.sensitivity') == "High"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nSensitivity=1`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.sensitivity') == "Middle"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nSensitivity=2`; timeout `6000`; targetVariable `ptz_track`

    else_actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nSensitivity=0`; timeout `6000`; targetVariable `ptz_track`

r2c3

`concat('Place ', jsonpath($(internal:custom_ptz_track), '$.placement'))`

Steps place through Left → Centre → Right. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.placement') == "Left"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nCenterPos=1`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.placement') == "Centre"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nCenterPos=2`; timeout `6000`; targetVariable `ptz_track`

    else_actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nCenterPos=0`; timeout `6000`; targetVariable `ptz_track`

r2c4 `concat('Head ', jsonpath($(internal:custom_ptz_track), '$.headroom'))`

Steps head through 5/10 → 6/10 → 7/10 → 8/10 → 9/10. The caption shows the camera's current value.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.headroom') == "5/10"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nHeadRoomRatio=1`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.headroom') == "6/10"`

    actions

    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nHeadRoomRatio=2`; timeout `6000`; targetVariable `ptz_track`

    else_actions

    - internal · **logic_if**

      condition

      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.headroom') == "7/10"`

      actions

      - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nHeadRoomRatio=3`; timeout `6000`; targetVariable `ptz_track`

      else_actions

      - internal · **logic_if**

        condition

        - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.headroom') == "8/10"`

        actions

        - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nHeadRoomRatio=4`; timeout `6000`; targetVariable `ptz_track`

        else_actions

        - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nHeadRoomRatio=0`; timeout `6000`; targetVariable `ptz_track`

r2c5 `concat('Lost ', jsonpath($(internal:custom_ptz_track), '$.lost'))`

Steps lost through Home → Stay → Preset 0. The caption shows the camera's current value.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.lost') == "Home"`

  ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nLostReac=2`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS
  - internal · **logic_if**
    condition
    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.lost') == "Stay"`

    actions
    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nLostReac=4`; timeout `6000`; targetVariable `ptz_track`

    else_actions
    - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 set nLostReac=0`; timeout `6000`; targetVariable `ptz_track`

r3c0

Auto OFF

Toggles focus and exposure together between automatic and manual (the locked service settings). White balance is left alone. Green while both are auto, amber while only one is. Any press from a not-both-auto state goes to both auto.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto"`

  ACTIONS
  - visca · **focusM**: bol 1
  - internal · **wait**: time = `100`
  - visca · **expM**: val 1

  ELSE_ACTIONS
  - visca · **focusM**: bol `0`
  - internal · **wait**: time = `100`
  - visca · **expM**: val `0`

CHANGES WHEN

- internal · **check_expression**: expression = `(jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto" || jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto") && !(jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto")`
  → colour #3A2E06, label Auto PART
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto" && jsonpath($(internal:custom_ptz_state), '$.ae') == "Auto"`
  → colour #14361F, icon ptz-auto-on, label Auto ON

r3c1 `concat('WDR ', jsonpath($(internal:custom_ptz_state), '$.wdr'))`

Steps wide dynamic range Off, 1 … 8, Off: how hard the highlights are rolled off before they clip. Green while on. The white-shirt clipping in the docs is what this is for.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') == "Off"`

  ACTIONS
  - visca · **custom**: custom `81 01 04 51 00 00 00 01 FF (set WDR level)`

  ELSE_ACTIONS
  - internal · **logic_if**
    condition
    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') == "1"`

    actions
    - visca · **custom**: custom `81 01 04 51 00 00 00 02 FF (set WDR level)`

    else_actions
    - internal · **logic_if**
      condition
      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') == "2"`

      actions
      - visca · **custom**: custom `81 01 04 51 00 00 00 03 FF (set WDR level)`

      else_actions
      - internal · **logic_if**
        condition
        - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') == "3"`

        actions
        - visca · **custom**: custom `81 01 04 51 00 00 00 04 FF (set WDR level)`

        else_actions
        - internal · **logic_if**
          condition
          - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') == "4"`

          actions
          - visca · **custom**: custom `81 01 04 51 00 00 00 05 FF (set WDR level)`

          else_actions
          - internal · **logic_if**
            condition
            - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') == "5"`

            actions
            - visca · **custom**: custom `81 01 04 51 00 00 00 06 FF (set WDR level)`

            else_actions
            - internal · **logic_if**
              condition
              - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') == "6"`

              actions
              - visca · **custom**: custom `81 01 04 51 00 00 00 07 FF (set WDR level)`

else_actions

- internal · **logic_if**

  condition

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') == "7"`

  actions

  - visca · **custom**: custom `81 01 04 51 00 00 00 08 FF` (set WDR level)

  else_actions

  - visca · **custom**: custom `81 01 04 51 00 00 00 00 FF` (set WDR level)

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wdr') != "Off" && jsonpath($(internal:custom_ptz_state), '$.wdr') != "--"`
  → colour #14361F

r3c2 `concat('NR ', jsonpath($(internal:custom_ptz_state), '$.nr'))`

Steps 3D noise reduction Off, 1 … 7, Auto, Off. Green while on. The reference keeps it low: heavy NR smears motion.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') == "Off"`

  ACTIONS

  - visca · **custom**: custom `81 01 04 54 01 FF (set noise reduction level)`

  ELSE_ACTIONS

  - internal · **logic_if**

    condition

    - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') == "1"`

    actions

    - visca · **custom**: custom `81 01 04 54 02 FF (set noise reduction level)`

    else_actions

    - internal · **logic_if**

      condition

      - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') == "2"`

      actions

      - visca · **custom**: custom `81 01 04 54 03 FF (set noise reduction level)`

      else_actions

      - internal · **logic_if**

        condition

        - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') == "3"`

        actions

        - visca · **custom**: custom `81 01 04 54 04 FF (set noise reduction level)`

        else_actions

        - internal · **logic_if**

          condition

          - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') == "4"`

          actions

          - visca · **custom**: custom `81 01 04 54 05 FF (set noise reduction level)`

          else_actions

          - internal · **logic_if**

            condition

            - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') == "5"`

            actions

            - visca · **custom**: custom `81 01 04 54 06 FF (set noise reduction level)`

            else_actions

            - internal · **logic_if**

              condition

              - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') == "6"`

              actions

              - visca · **custom**: custom `81 01 04 54 07 FF (set noise reduction level)`

else_actions

- internal · **logic_if**

  condition

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') == "7"`

  actions

  - visca · **custom**: custom `81 01 04 54 08 FF` (set noise reduction level)

  else_actions

  - visca · **custom**: custom `81 01 04 54 00 FF` (set noise reduction level)

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.nr') != "Off" && jsonpath($(internal:custom_ptz_state), '$.nr') != "--"`
  → colour #14361F

---

r3c3

1-push WB

Hold a white card at the preaching position, press, and give it a second: the camera measures the white and holds it. The WB readout says 1-Push afterwards. Match the other camera off the same card.

PRESS

- visca · **wb**: val `onepush`
- internal · **wait**: time = `300`
- visca · **wbOPT**

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.wb') == "1-Push"`
  → colour #14361F

---

r3c4

Match ◂ CAM 1

Copies CAM 1's picture settings (exposure, colour, image, focus, noise reduction; not the flips) onto this camera through the web API. Green and Matched when every setting took, amber with a count if some did not, red if the camera could not be reached. Takes a few seconds.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 match 10.23.0.196`; timeout `20000`; targetVariable `ptz_match`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_match), '$.online') == "OK" && jsonpath($(internal:custom_ptz_match), '$.left') > 0`
  → colour #3A2E06, label = concat('Match ', jsonpath($(internal:custom_ptz_match), '$.left'), ' left')
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_match), '$.online') == "OK" && jsonpath($(internal:custom_ptz_match), '$.left') == 0`
  → colour #14361F, label Matched
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_match), '$.online') == "DOWN"`
  → colour #B91C1C, label Match FAILED

---

r3c5

Save look

Keeps the picture settings exactly as they are now (exposure, colour, image, focus, noise reduction) as the look that Look puts back. Press when the picture is right. The previous look is kept as .prev.json on the Pi.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 look save`; timeout `20000`; targetVariable `ptz_look`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_look), '$.online') == "DOWN"`
  → colour #B91C1C, label Save FAILED

---

r3c6

Look

Puts the saved picture settings back (exposure, colour, image, focus, noise reduction) after a preset recall has changed them. Green and "Look set" while the camera is on the saved look. Save look, on the setup page, is what saves it.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 look apply`; timeout `20000`; targetVariable `ptz_look`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_look), '$.online') == "OK" && jsonpath($(internal:custom_ptz_state), '$.wb') == jsonpath($(internal:custom_ptz_look), '$.wb') && jsonpath($(internal:custom_ptz_state), '$.ae') == jsonpath($(internal:custom_ptz_look), '$.ae') && jsonpath($(internal:custom_ptz_state), '$.shutter') == jsonpath($(internal:custom_ptz_look), '$.shutter') && jsonpath($(internal:custom_ptz_state), '$.iris') == jsonpath($(internal:custom_ptz_look), '$.iris') && jsonpath($(internal:custom_ptz_state), '$.gain') == jsonpath($(internal:custom_ptz_look), '$.gain') && jsonpath($(internal:custom_ptz_state), '$.sharp') == jsonpath($(internal:custom_ptz_look), '$.sharp')`
  → colour #14361F, label Look set
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_look), '$.online') == "OK" && jsonpath($(internal:custom_ptz_look), '$.left') > 0`
  → colour #3A2E06, label = concat('Look ', jsonpath($(internal:custom_ptz_look), '$.left'), ' left')
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_look), '$.online') == "DOWN"`
  → colour #B91C1C, label Look FAILED

r4c0

`concat('Shutter ', jsonpath($(internal:custom_ptz_state), '$.shutter'))`

Shutter speed as the camera reports it. Amber while exposure is not Manual or Shutter priority, when the value is not in effect. Red if the camera stops answering.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.ae') != "Manual" && jsonpath($(internal:custom_ptz_state), '$.ae') != "Shutter"`
  → colour #3A2E06
- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.online') != "OK"`
  → colour #B91C1C, label Camera DOWN

r4c2

`concat('Iris ', jsonpath($(internal:custom_ptz_state), '$.iris'))`

Iris as the camera reports it. Amber while exposure is not Manual or Iris priority, when the value is not in effect.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.ae') != "Manual" && jsonpath($(internal:custom_ptz_state), '$.ae') != "Iris"`
  → colour #3A2E06

r4c3

`concat('Gain ', jsonpath($(internal:custom_ptz_state), '$.gain'))`

Gain 0-15 as the camera reports it. Amber while exposure is not Manual, when the value is not in effect.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!(jsonpath($(internal:custom_ptz_state), '$.ae') == "Manual")`
  → colour #3A2E06

r4c5

`concat('Comp ', jsonpath($(internal:custom_ptz_state), '$.expcomp'))`

Exposure compensation as the camera reports it: Off, or an offset in steps either side of 0. Amber while exposure is Manual, when it has no effect.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.ae') == "Manual"`
  → colour #3A2E06

r4c6

`concat('WB ', jsonpath($(internal:custom_ptz_state), '$.wb'))`

White balance as the camera reports it: a colour temperature, or Auto, Manual or 1-Push. Amber while it is not a temperature, when the dial is not in effect.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!(includes(jsonpath($(internal:custom_ptz_state), '$.wb'), "K"))`
  → colour #3A2E06

r4c8

`concat('Sharp ', jsonpath($(internal:custom_ptz_state), '$.sharp'))`

Sharpness 0-15 as the camera reports it.

No action: display only.

r5c0

Shutter

Turn to step the shutter (switches exposure to Manual). Press returns it to 1/125, the rig's 180° at 59.94.

PRESS

- visca · **expM**: val 1
- internal · **wait**: time = `100`
- visca · **custom**: custom `81 01 04 4A 00 00 01 05 FF (set shutter)`

TURN LEFT

- visca · **expM**: val 1
- visca · **custom**: custom `81 01 04 0A 03 FF (shutter slower)`

TURN RIGHT

- visca · **expM**: val 1
- visca · **custom**: custom `81 01 04 0A 02 FF (shutter faster)`

r5c2

Iris

Turn clockwise to open the iris, anticlockwise to close it (switches exposure to Manual). Press does nothing: there is no standard iris.

TURN LEFT

- visca · **expM**: val 1
- visca · **custom**: custom `81 01 04 0B 03 FF (iris close)`

TURN RIGHT

- visca · **expM**: val 1
- visca · **custom**: custom `81 01 04 0B 02 FF (iris open)`

r5c3

Gain

Turn to change gain 0-15 (switches exposure to Manual). Press returns it to 0. The camera has no gain step command, so the deck keeps the dial and sends the value.

PRESS

- internal · **custom_variable_set_value**: name `ptz_gain`; create `yes`; value `0`
- visca · **expM**: val 1
- internal · **wait**: time = `100`
- visca · **custom**: custom `81 01 04 4C 00 00 00 00 FF (set gain)`; command_parameters `13,15`; parameter0 `$(internal:custom_ptz_gain)`

TURN LEFT

- internal · **custom_variable_set_value**: name `ptz_gain`; create `yes`; value = `max(0, $(internal:custom_ptz_gain) - 1)`
- visca · **expM**: val 1
- internal · **wait**: time = `100`
- visca · **custom**: custom `81 01 04 4C 00 00 00 00 FF (set gain)`; command_parameters `13,15`; parameter0 `$(internal:custom_ptz_gain)`

TURN RIGHT

- internal · **custom_variable_set_value**: name `ptz_gain`; create `yes`; value = `min(15, $(internal:custom_ptz_gain) + 1)`
- visca · **expM**: val 1
- internal · **wait**: time = `100`
- visca · **custom**: custom `81 01 04 4C 00 00 00 00 FF (set gain)`; command_parameters `13,15`; parameter0 `$(internal:custom_ptz_gain)`

r5c5

Comp

Turn to bias the automatic exposure darker or brighter (switches compensation on). Press switches it off and recentres it. Only applies while exposure is automatic.

PRESS

- visca · **custom**: custom `81 01 04 3E 03 FF (exposure compensation off)`
- internal · **wait**: time = `100`
- visca · **custom**: custom `81 01 04 4E 00 00 00 07 FF (set exposure compensation)`

TURN LEFT

- visca · **custom**: custom `81 01 04 3E 02 FF (exposure compensation on)`
- visca · **custom**: custom `81 01 04 0E 03 FF (exposure compensation down)`

TURN RIGHT

- visca · **custom**: custom `81 01 04 3E 02 FF (exposure compensation on)`
- visca · **custom**: custom `81 01 04 0E 02 FF (exposure compensation up)`

r5c6

WB

Turn to step the colour temperature 100K at a time, 2400-7100K (takes white balance off Auto). Press returns it to 4600K, the stage light.

PRESS

- internal · **custom_variable_set_value**: name ptz_wbk; create yes; value 4600
- internal · **custom_variable_set_value**: name ptz_wbcode; create yes; value = `$(internal:custom_ptz_wbk) == 2400 ? 12 : $(internal:custom_ptz_wbk) == 2500 ? 13 : $(internal:custom_ptz_wbk) == 2600 ? 14 : $(internal:custom_ptz_wbk) == 2700 ? 15 : $(internal:custom_ptz_wbk) == 2800 ? 16 : $(internal:custom_ptz_wbk) == 2900 ? 17 : $(internal:custom_ptz_wbk) == 3000 ? 1 : $(internal:custom_ptz_wbk) == 3100 ? 18 : $(internal:custom_ptz_wbk) == 3200 ? 19 : $(internal:custom_ptz_wbk) == 3300 ? 20 : $(internal:custom_ptz_wbk) == 3400 ? 21 : $(internal:custom_ptz_wbk) == 3500 ? 7 : $(internal:custom_ptz_wbk) == 3600 ? 22 : $(internal:custom_ptz_wbk) == 3700 ? 23 : $(internal:custom_ptz_wbk) == 3800 ? 24 : $(internal:custom_ptz_wbk) == 3900 ? 25 : $(internal:custom_ptz_wbk) == 4000 ? 2 : $(internal:custom_ptz_wbk) == 4100 ? 26 : $(internal:custom_ptz_wbk) == 4200 ? 27 : $(internal:custom_ptz_wbk) == 4300 ? 28 : $(internal:custom_ptz_wbk) == 4400 ? 29 : $(internal:custom_ptz_wbk) == 4500 ? 8 : $(internal:custom_ptz_wbk) == 4600 ? 30 : $(internal:custom_ptz_wbk) == 4700 ? 31 : $(internal:custom_ptz_wbk) == 4800 ? 33 : $(internal:custom_ptz_wbk) == 4900 ? 34 : $(internal:custom_ptz_wbk) == 5000 ? 4 : $(internal:custom_ptz_wbk) == 5100 ? 35 : $(internal:custom_ptz_wbk) == 5200 ? 36 : $(internal:custom_ptz_wbk) == 5300 ? 37 : $(internal:custom_ptz_wbk) == 5400 ? 38 : $(internal:custom_ptz_wbk) == 5500 ? 9 : $(internal:custom_ptz_wbk) == 5600 ? 39 : $(internal:custom_ptz_wbk) == 5700 ? 40 : $(internal:custom_ptz_wbk) == 5800 ? 41 : $(internal:custom_ptz_wbk) == 5900 ? 42 : $(internal:custom_ptz_wbk) == 6000 ? 10 : $(internal:custom_ptz_wbk) == 6100 ? 43 : $(internal:custom_ptz_wbk) == 6200 ? 44 : $(internal:custom_ptz_wbk) == 6300 ? 45 : $(internal:custom_ptz_wbk) == 6400 ? 46 : $(internal:custom_ptz_wbk) == 6500 ? 6 : $(internal:custom_ptz_wbk) == 6600 ? 47 : $(internal:custom_ptz_wbk) == 6700 ? 48 : $(internal:custom_ptz_wbk) == 6800 ? 49 : $(internal:custom_ptz_wbk) == 6900 ? 50 : $(internal:custom_ptz_wbk) == 7000 ? 11 : $(internal:custom_ptz_wbk) == 7100 ? 51 : 30`
- visca · **custom**: custom `81 01 04 35 00 FF (white balance mode)`; command_parameters `8,9`; parameter0 `$(internal:custom_ptz_wbcode)`

TURN LEFT

- internal · **custom_variable_set_value**: name ptz_wbk; create yes; value = `max(2400, $(internal:custom_ptz_wbk) - 100)`
- internal · **custom_variable_set_value**: name ptz_wbcode; create yes; value = `$(internal:custom_ptz_wbk) == 2400 ? 12 : $(internal:custom_ptz_wbk) == 2500 ? 13 : $(internal:custom_ptz_wbk) == 2600 ? 14 : $(internal:custom_ptz_wbk) == 2700 ? 15 : $(internal:custom_ptz_wbk) == 2800 ? 16 : $(internal:custom_ptz_wbk) == 2900 ? 17 : $(internal:custom_ptz_wbk) == 3000 ? 1 : $(internal:custom_ptz_wbk) == 3100 ? 18 : $(internal:custom_ptz_wbk) == 3200 ? 19 : $(internal:custom_ptz_wbk) == 3300 ? 20 : $(internal:custom_ptz_wbk) == 3400 ? 21 : $(internal:custom_ptz_wbk) == 3500 ? 7 : $(internal:custom_ptz_wbk) == 3600 ? 22 : $(internal:custom_ptz_wbk) == 3700 ? 23 : $(internal:custom_ptz_wbk) == 3800 ? 24 : $(internal:custom_ptz_wbk) == 3900 ? 25 : $(internal:custom_ptz_wbk) == 4000 ? 2 : $(internal:custom_ptz_wbk) == 4100 ? 26 : $(internal:custom_ptz_wbk) == 4200 ? 27 : $(internal:custom_ptz_wbk) == 4300 ? 28 : $(internal:custom_ptz_wbk) == 4400 ? 29 : $(internal:custom_ptz_wbk) == 4500 ? 8 : $(internal:custom_ptz_wbk) == 4600 ? 30 : $(internal:custom_ptz_wbk) == 4700 ? 31 : $(internal:custom_ptz_wbk) == 4800 ? 33 : $(internal:custom_ptz_wbk) == 4900 ? 34 : $(internal:custom_ptz_wbk) == 5000 ? 4 : $(internal:custom_ptz_wbk) == 5100 ? 35 : $(internal:custom_ptz_wbk) == 5200 ? 36 : $(internal:custom_ptz_wbk) == 5300 ? 37 : $(internal:custom_ptz_wbk) == 5400 ? 38 : $(internal:custom_ptz_wbk) == 5500 ? 9 : $(internal:custom_ptz_wbk) == 5600 ? 39 : $(internal:custom_ptz_wbk) == 5700 ? 40 : $(internal:custom_ptz_wbk) == 5800 ? 41 : $(internal:custom_ptz_wbk) == 5900 ? 42 : $(internal:custom_ptz_wbk) == 6000 ? 10 : $(internal:custom_ptz_wbk) == 6100 ? 43 : $(internal:custom_ptz_wbk) == 6200 ? 44 : $(internal:custom_ptz_wbk) == 6300 ? 45 : $(internal:custom_ptz_wbk) == 6400 ? 46 : $(internal:custom_ptz_wbk) == 6500 ? 6 : $(internal:custom_ptz_wbk) == 6600 ? 47 : $(internal:custom_ptz_wbk) == 6700 ? 48 : $(internal:custom_ptz_wbk) == 6800 ? 49 : $(internal:custom_ptz_wbk)`

== 6900 ? 50 : $(internal:custom_ptz_wbk) == 7000 ? 11 : $(internal:custom_ptz_wbk) == 7100 ? 51 : 30`

- visca · **custom**: custom `81 01 04 35 00 FF (white balance mode)`; command_parameters `8,9`; parameter0 `$(internal:custom_ptz_wbcode)`

TURN RIGHT

- internal · **custom_variable_set_value**: name `ptz_wbk`; create `yes`; value = `min(7100, $(internal:custom_ptz_wbk) + 100)`
- internal · **custom_variable_set_value**: name `ptz_wbcode`; create `yes`; value = `$(internal:custom_ptz_wbk) == 2400 ? 12 : $(internal:custom_ptz_wbk) == 2500 ? 13 : $(internal:custom_ptz_wbk) == 2600 ? 14 : $(internal:custom_ptz_wbk) == 2700 ? 15 : $(internal:custom_ptz_wbk) == 2800 ? 16 : $(internal:custom_ptz_wbk) == 2900 ? 17 : $(internal:custom_ptz_wbk) == 3000 ? 1 : $(internal:custom_ptz_wbk) == 3100 ? 18 : $(internal:custom_ptz_wbk) == 3200 ? 19 : $(internal:custom_ptz_wbk) == 3300 ? 20 : $(internal:custom_ptz_wbk) == 3400 ? 21 : $(internal:custom_ptz_wbk) == 3500 ? 7 : $(internal:custom_ptz_wbk) == 3600 ? 22 : $(internal:custom_ptz_wbk) == 3700 ? 23 : $(internal:custom_ptz_wbk) == 3800 ? 24 : $(internal:custom_ptz_wbk) == 3900 ? 25 : $(internal:custom_ptz_wbk) == 4000 ? 2 : $(internal:custom_ptz_wbk) == 4100 ? 26 : $(internal:custom_ptz_wbk) == 4200 ? 27 : $(internal:custom_ptz_wbk) == 4300 ? 28 : $(internal:custom_ptz_wbk) == 4400 ? 29 : $(internal:custom_ptz_wbk) == 4500 ? 8 : $(internal:custom_ptz_wbk) == 4600 ? 30 : $(internal:custom_ptz_wbk) == 4700 ? 31 : $(internal:custom_ptz_wbk) == 4800 ? 33 : $(internal:custom_ptz_wbk) == 4900 ? 34 : $(internal:custom_ptz_wbk) == 5000 ? 4 : $(internal:custom_ptz_wbk) == 5100 ? 35 : $(internal:custom_ptz_wbk) == 5200 ? 36 : $(internal:custom_ptz_wbk) == 5300 ? 37 : $(internal:custom_ptz_wbk) == 5400 ? 38 : $(internal:custom_ptz_wbk) == 5500 ? 9 : $(internal:custom_ptz_wbk) == 5600 ? 39 : $(internal:custom_ptz_wbk) == 5700 ? 40 : $(internal:custom_ptz_wbk) == 5800 ? 41 : $(internal:custom_ptz_wbk) == 5900 ? 42 : $(internal:custom_ptz_wbk) == 6000 ? 10 : $(internal:custom_ptz_wbk) == 6100 ? 43 : $(internal:custom_ptz_wbk) == 6200 ? 44 : $(internal:custom_ptz_wbk) == 6300 ? 45 : $(internal:custom_ptz_wbk) == 6400 ? 46 : $(internal:custom_ptz_wbk) == 6500 ? 6 : $(internal:custom_ptz_wbk) == 6600 ? 47 : $(internal:custom_ptz_wbk) == 6700 ? 48 : $(internal:custom_ptz_wbk) == 6800 ? 49 : $(internal:custom_ptz_wbk) == 6900 ? 50 : $(internal:custom_ptz_wbk) == 7000 ? 11 : $(internal:custom_ptz_wbk) == 7100 ? 51 : 30`
- visca · **custom**: custom `81 01 04 35 00 FF (white balance mode)`; command_parameters `8,9`; parameter0 `$(internal:custom_ptz_wbcode)`

`r5c8`

Sharp

Turn to change sharpness 0-15. Press returns it to 0, which is where the reference settings keep it.

PRESS

- visca · **custom**: custom `81 01 04 42 00 00 00 00 FF (set sharpness)`

TURN LEFT

- visca · **custom**: custom `81 01 04 02 03 FF (sharpness down)`

TURN RIGHT

- visca · **custom**: custom `81 01 04 02 02 FF (sharpness up)`

Page 14: Tracks



r2c3

Previous

YouTube Music's back button on the PP1 iMac: restarts the song, or the one before if it has only just begun. Via /home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh.

PRESS
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh prev`; timeout `8000`; targetVariable `chrome_tracks_title`

r2c4

Play / Pause

Play/pause the tracks in Chrome on the PP1 iMac, via /home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh. Green while playing.

PRESS
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh toggle`; timeout `8000`; targetVariable `chrome_tracks`

CHANGES WHEN
- internal · **check_expression**: expression = `$(internal:custom_chrome_tracks) == 'Playing'` → colour #15803D

r2c5

Skip

Skip to the next track in YouTube Music on the PP1 iMac, via /home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh.

PRESS
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh next`; timeout `8000`; targetVariable `chrome_tracks_title`

r3c4

Fade Out

Fade the tracks to silence over four seconds, pause, and put the volume back for next time. Does nothing if they are already paused. Via /home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh.

PRESS
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh fade`; timeout `15000`; targetVariable `chrome_tracks`

r4c3

`$(internal:custom_chrome_tracks_title)`

The song YouTube Music is on, as of opening the folder or the last Skip/Previous.

No action: display only.

r4c5

`Vol $(internal:custom_chrome_tracks_volume)`

The tracks' volume on YouTube Music's own 0-100 slider, as the knob last left it.

No action: display only.

r5c5

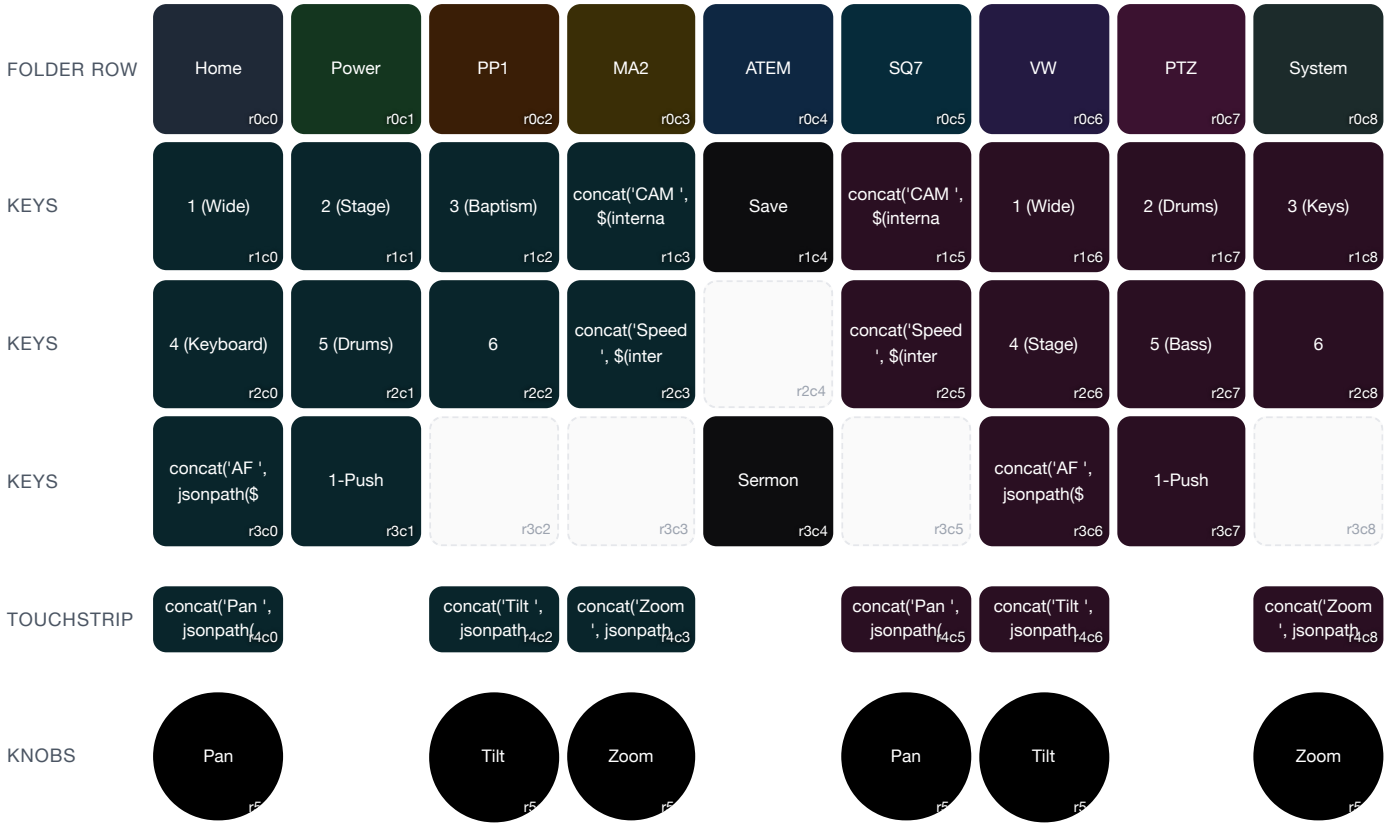
Volume

Turn to move YouTube Music's volume slider on the PP1 iMac to the next multiple of 5 per detent. Press does nothing.

TURN LEFT
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh down`; timeout `8000`; targetVariable `chrome_tracks_volume`

TURN RIGHT
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh up`; timeout `8000`; targetVariable `chrome_tracks_volume`

Page 15: Worship



`r1c0`

1 (Wide)

Go to preset 1 (Wide), keeping the picture as it is. With Save armed, saves the current shot as preset 1 instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS

  - visca2 · **setPreset**: isText `no`; presetAsNumber 1; presetAsText 1
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 1
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 1
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 1`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 1`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 1, text colour #000000, icon preset-save-ink

r1c1

2 (Stage)

Go to preset 2 (Stage), keeping the picture as it is. With Save armed, saves the current shot as preset 2 instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS

  - visca2 · **setPreset**: isText `no`; presetAsNumber 2; presetAsText 2
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 2
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 2
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 2`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 2`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 2, text colour #000000, icon preset-save-ink

r1c2

3 (Baptism)

Go to preset 3 (Baptism), keeping the picture as it is. With Save armed, saves the current shot as preset 3 instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS

  - visca2 · **setPreset**: isText `no`; presetAsNumber 3; presetAsText 3
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 3
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 3
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 3`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 3`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 3, text colour #000000, icon preset-save-ink

r1c3 `concat('CAM ', $(internal:custom_ptz2_atem_input))`

Stops pan, tilt and zoom, and names the camera this page drives. Closes the camera menu while it is open. Red and LIVE when this camera is on ATEM program, green and PVW on preview (input number in ptz_atem_input).

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.menu') == "On"`

  ACTIONS
  - visca2 · **custom**: custom `81 01 06 06 03 FF (close the camera menu)`

  ELSE_ACTIONS
  - visca2 · **stop**
  - visca2 · **zoomS**

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.menu') == "On"`
  → label EXIT, colour #3A2E06
- internal · **check_expression**: expression = `$(atem:pvw1_input_id) == $(internal:custom_ptz2_atem_input)`
  → colour #00A651, label = concat(concat('CAM ', $(internal:custom_ptz2_atem_input)), ' PVW'), text colour #000000, icon stop-ink
- internal · **check_expression**: expression = `$(atem:pgm1_input_id) == $(internal:custom_ptz2_atem_input)`
  → colour #FF0000, label = concat(concat('CAM ', $(internal:custom_ptz2_atem_input)), ' LIVE'), text colour #FFFFFF, icon stop-paper

r1c4 **Save**

Arms saving on both cameras: the next preset key pressed saves the current shot there, on its own camera, and disarms both. Press again to cancel.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1 || $(internal:custom_ptz_armed) == 1`

  ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value 1
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value 1

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1 || $(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label ARMED, text colour #000000, icon preset-save-ink

r1c5 `concat('CAM ', $(internal:custom_ptz_atem_input))`

Stops pan, tilt and zoom, and names the camera this page drives. Closes the camera menu while it is open. Red and LIVE when this camera is on ATEM program, green and PVW on preview (input number in ptz_atem_input).

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.menu') == "On"`

  ACTIONS
  - visca · **custom**: custom `81 01 06 06 03 FF (close the camera menu)`

  ELSE_ACTIONS
  - visca · **stop**
  - visca · **zoomS**

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.menu') == "On"`
  → label EXIT, colour #3A2E06
- internal · **check_expression**: expression = `$(atem:pvw1_input_id) == $(internal:custom_ptz_atem_input)`
  → colour #00A651, label = concat(concat('CAM ', $(internal:custom_ptz_atem_input)), ' PVW'), text colour #000000, icon stop-ink
- internal · **check_expression**: expression = `$(atem:pgm1_input_id) == $(internal:custom_ptz_atem_input)`
  → colour #FF0000, label = concat(concat('CAM ', $(internal:custom_ptz_atem_input)), ' LIVE'), text colour #FFFFFF, icon stop-paper

r1c6 **1 (Wide)**

Go to preset 1 (Wide), keeping the picture as it is. With Save armed, saves the current shot as preset 1 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText `no`; presetAsNumber 1; presetAsText 1
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 1
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 1
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 1`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 1`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 1, text colour #000000, icon preset-save-ink

r1c7

2 (Drums)

Go to preset 2 (Drums), keeping the picture as it is. With Save armed, saves the current shot as preset 2 instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS

  - visca · **setPreset**: isText no; presetAsNumber 2; presetAsText 2
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 2
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 2
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 2`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 2`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 2, text colour #000000, icon preset-save-ink

r1c8

3 (Keys)

Go to preset 3 (Keys), keeping the picture as it is. With Save armed, saves the current shot as preset 3 instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS

  - visca · **setPreset**: isText no; presetAsNumber 3; presetAsText 3
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 3
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 3
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 3`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 3`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 3, text colour #000000, icon preset-save-ink

r2c0

4 (Keyboard)

Go to preset 4 (Keyboard), keeping the picture as it is. With Save armed, saves the current shot as preset 4 instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS

  - visca2 · **setPreset**: isText `no`; presetAsNumber 4; presetAsText 4
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 4
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 4
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 4`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 4`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 4, text colour #000000, icon preset-save-ink

r2c1

5 (Drums)

Go to preset 5 (Drums), keeping the picture as it is. With Save armed, saves the current shot as preset 5 instead.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS

  - visca2 · **setPreset**: isText `no`; presetAsNumber 5; presetAsText 5
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 5
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS

  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 5
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 5`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 5`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 5, text colour #000000, icon preset-save-ink

r2c2

6

Go to preset 6, keeping the picture as it is. With Save armed, saves the current shot as preset 6 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 6; presetAsText 6
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 6
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 6
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 6`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 6`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 6, text colour #000000, icon preset-save-ink

r2c3

`concat('Speed ', $(internal:custom_ptz2_speed))`

Steps the drive speed 1 → 10 → 24 → 1 for pan, tilt, zoom and focus. The Speed knob still fine-tunes it.

PRESS

- internal · **custom_variable_set_value**: name `ptz2_speed`; create `yes`; value = `$(internal:custom_ptz2_speed) < 10 ? 10 : ($(internal:custom_ptz2_speed) < 24 ? 24 : 1)`
- internal · **custom_variable_set_value**: name `ptz2_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz2_speed))`
- internal · **custom_variable_set_value**: name `ptz2_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz2_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz2_speed) * 7 / 24)))`

r2c5

`concat('Speed ', $(internal:custom_ptz_speed))`

Steps the drive speed 1 → 10 → 24 → 1 for pan, tilt, zoom and focus. The Speed knob still fine-tunes it.

PRESS

- internal · **custom_variable_set_value**: name `ptz_speed`; create `yes`; value = `$(internal:custom_ptz_speed) < 10 ? 10 : ($(internal:custom_ptz_speed) < 24 ? 24 : 1)`
- internal · **custom_variable_set_value**: name `ptz_tspeed`; create `yes`; value = `min(20, $(internal:custom_ptz_speed))`
- internal · **custom_variable_set_value**: name `ptz_zspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`
- internal · **custom_variable_set_value**: name `ptz_fspeed`; create `yes`; value = `max(1, min(7, round($(internal:custom_ptz_speed) * 7 / 24)))`

r2c6

4 (Stage)

Go to preset 4 (Stage), keeping the picture as it is. With Save armed, saves the current shot as preset 4 instead.

PRESS

- internal · **logic_if**

  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 4; presetAsText 4
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 4
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 4
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 4`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 4`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 4, text colour #000000, icon preset-save-ink

r2c7

5 (Bass)

Go to preset 5 (Bass), keeping the picture as it is. With Save armed, saves the current shot as preset 5 instead.

PRESS

- internal · **logic_if**

  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 5; presetAsText 5
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 5
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 5
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 5`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 5`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 5, text colour #000000, icon preset-save-ink

r2c8

6

Go to preset 6, keeping the picture as it is. With Save armed, saves the current shot as preset 6 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 6; presetAsText 6
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 6
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 6
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 6`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 6`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 6, text colour #000000, icon preset-save-ink

r3c0

`concat('AF ', jsonpath($(internal:custom_ptz2_state), '$.focus'))`

Toggles autofocus. Green while the camera is in auto focus.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto"`

  ACTIONS
  - visca2 · **focusM**: bol 1

  ELSE_ACTIONS
  - visca2 · **focusM**: bol `0`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.focus') == "Auto"`
  → colour #14361F, icon focus-auto

r3c1

1-Push

One-push autofocus: the camera focuses once on what it sees now, then holds.

PRESS

- visca2 · **custom**: custom `81 01 04 38 04 FF (one-push autofocus)`

r3c4

Sermon

Both cameras on one page for the sermon: tracking, framing, who to follow, and pan, tilt and zoom by hand.

PRESS

- internal · **set_page**: surfaceId `self`; page 16

r3c6

`concat('AF ', jsonpath($(internal:custom_ptz_state), '$.focus'))`

Toggles autofocus. Green while the camera is in auto focus.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto"`

  ACTIONS

  - visca · **focusM**: bol 1

  ELSE_ACTIONS

  - visca · **focusM**: bol `0`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.focus') == "Auto"`
  → colour #14361F, icon focus-auto

r3c7

1-Push

One-push autofocus: the camera focuses once on what it sees now, then holds.

PRESS

- visca · **custom**: custom `81 01 04 38 04 FF (one-push autofocus)`

r4c0

`concat('Pan ', jsonpath($(internal:custom_ptz2_state), '$.pan'))`

Pan angle as reported by the camera, polled every second. Turns red if the camera stops answering.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.online') != "OK"`
  → colour #B91C1C, label Camera DOWN

r4c2

`concat('Tilt ', jsonpath($(internal:custom_ptz2_state), '$.tilt'))`

Tilt angle as reported by the camera, polled every second. Display only.

No action: display only.

r4c3

`concat('Zoom ', jsonpath($(internal:custom_ptz2_state), '$.zoom'))`

Zoom position as a percentage of the lens range, polled every second. Display only.

No action: display only.

r4c5

`concat('Pan ', jsonpath($(internal:custom_ptz_state), '$.pan'))`

Pan angle as reported by the camera, polled every second. Turns red if the camera stops answering.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.online') != "OK"`
  → colour #B91C1C, label Camera DOWN

r4c6

`concat('Tilt ', jsonpath($(internal:custom_ptz_state), '$.tilt'))`

Tilt angle as reported by the camera, polled every second. Display only.

No action: display only.

r4c8

`concat('Zoom ', jsonpath($(internal:custom_ptz_state), '$.zoom'))`

Zoom position as a percentage of the lens range, polled every second. Display only.

No action: display only.

`r5c0`

Pan

Turn to pan at the Speed knob's speed; each detent drives for 150ms. Press stops.

PRESS
- visca2 · **stop**

TURN LEFT
- visca2 · **custom**: custom `81 01 06 01 00 00 01 03 FF (pan left)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

TURN RIGHT
- visca2 · **custom**: custom `81 01 06 01 00 00 02 03 FF (pan right)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

`r5c2`

Tilt

Turn clockwise to tilt up, anticlockwise down, at the Speed knob's speed. Press stops.

PRESS
- visca2 · **stop**

TURN LEFT
- visca2 · **custom**: custom `81 01 06 01 00 00 03 02 FF (tilt down)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

TURN RIGHT
- visca2 · **custom**: custom `81 01 06 01 00 00 03 01 FF (tilt up)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

`r5c3`

Zoom

Turn clockwise to zoom in, anticlockwise out. Speed follows the Speed knob. Press stops.

PRESS
- visca2 · **zoomS**

TURN LEFT
- visca2 · **custom**: custom `81 01 04 07 30 FF (zoom out)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_zspeed)`
- internal · **wait**: time = `150`
- visca2 · **zoomS**

TURN RIGHT
- visca2 · **custom**: custom `81 01 04 07 20 FF (zoom in)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_zspeed)`
- internal · **wait**: time = `150`
- visca2 · **zoomS**

`r5c5`

Pan

Turn to pan at the Speed knob's speed; each detent drives for 150ms. Press stops.

PRESS
- visca · **stop**

TURN LEFT
- visca · **custom**: custom `81 01 06 01 00 00 01 03 FF (pan left)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

TURN RIGHT
- visca · **custom**: custom `81 01 06 01 00 00 02 03 FF (pan right)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

r5c6

Tilt

Turn clockwise to tilt up, anticlockwise down, at the Speed knob's speed. Press stops.

PRESS
- visca · **stop**

TURN LEFT
- visca · **custom**: custom `81 01 06 01 00 00 03 02 FF (tilt down)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

TURN RIGHT
- visca · **custom**: custom `81 01 06 01 00 00 03 01 FF (tilt up)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

r5c8

Zoom

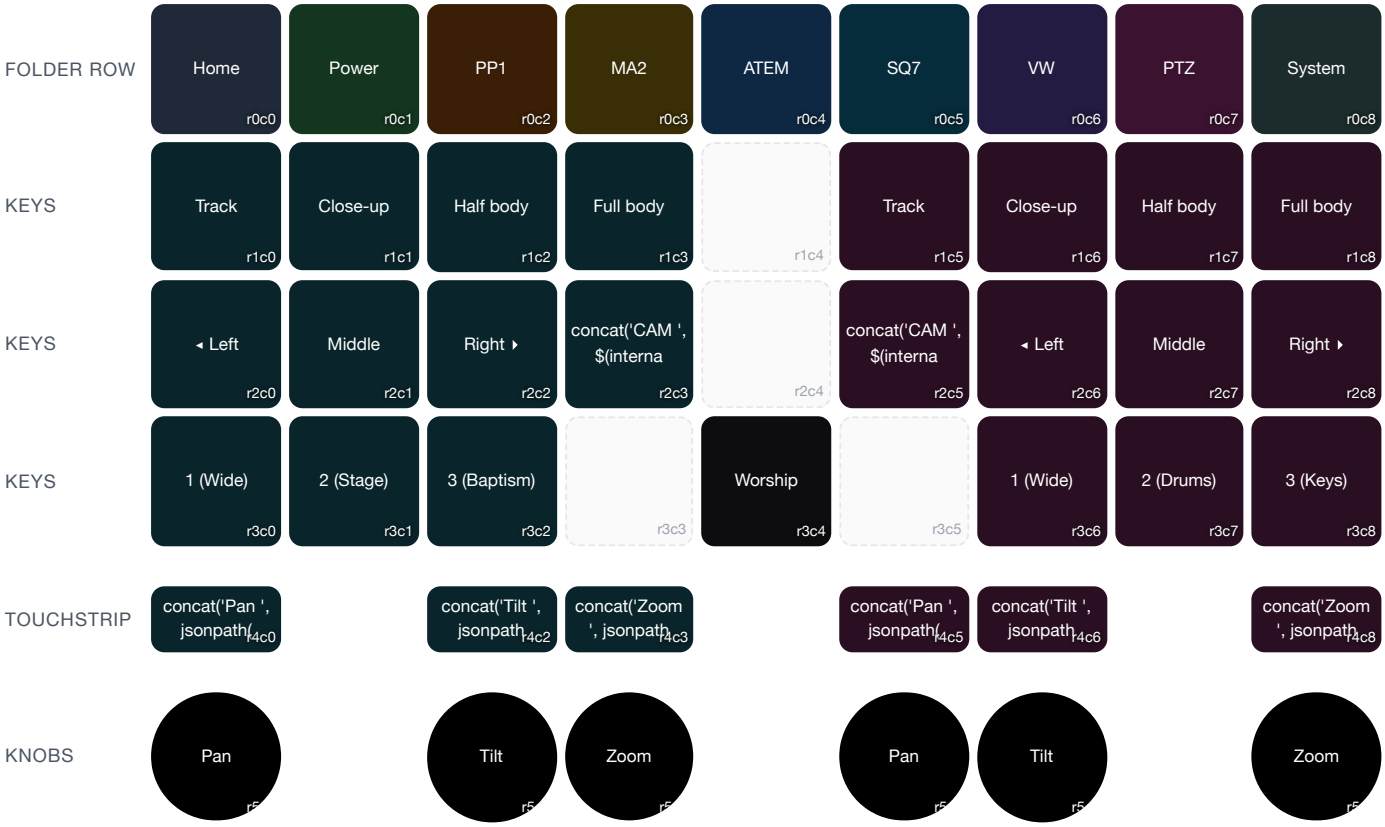
Turn clockwise to zoom in, anticlockwise out. Speed follows the Speed knob. Press stops.

PRESS
- visca · **zoomS**

TURN LEFT
- visca · **custom**: custom `81 01 04 07 30 FF (zoom out)`; command_parameters 9; parameter0 `$(internal:custom_ptz_zspeed)`
- internal · **wait**: time = `150`
- visca · **zoomS**

TURN RIGHT
- visca · **custom**: custom `81 01 04 07 20 FF (zoom in)`; command_parameters 9; parameter0 `$(internal:custom_ptz_zspeed)`
- internal · **wait**: time = `150`
- visca · **zoomS**

Page 16: Sermon



`r1c0`

Track

Toggles the camera AI auto tracking. Green while on, as the camera reports it.

PRESS

- internal · **logic_if**

  CONDITION

  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.tracking') == "On"`

  ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 track off`; timeout `6000`; targetVariable `ptz2_track`

  ELSE_ACTIONS

  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 track on`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.tracking') == "On"`
  → colour #14361F, icon tracking-on, label Track ON

`r1c1`

Close-up

Tracking framing: close-up. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body close`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Close-up"`
  → colour #14361F

r1c2

Half body

Tracking framing: half body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body half`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Half body"`
  → colour #14361F

r1c3

Full body

Tracking framing: full body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 body full`; timeout `6000`; targetVariable `ptz2_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_track), '$.body') == "Full body"`
  → colour #14361F

r1c5

Track

Toggles the camera AI auto tracking. Green while on, as the camera reports it.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.tracking') == "On"`

  ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 track off`; timeout `6000`; targetVariable `ptz_track`

  ELSE_ACTIONS
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 track on`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.tracking') == "On"`
  → colour #14361F, icon tracking-on, label Track ON

r1c6

Close-up

Tracking framing: close-up. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body close`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Close-up"`
  → colour #14361F

r1c7

Half body

Tracking framing: half body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body half`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Half body"`
  → colour #14361F

r1c8

Full body

Tracking framing: full body. Lit while selected. (Ignored by the camera in Stage mode.)

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 body full`; timeout `6000`; targetVariable `ptz_track`

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_track), '$.body') == "Full body"`
  → colour #14361F

r2c0

◂ Left

Tracks the person on the left of the frame (the camera is told to take whoever is at a point a sixth of the way across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 select 320 486`; timeout `6000`; targetVariable `ptz2_track`

r2c1

Middle

Tracks the person on the middle of the frame (the camera is told to take whoever is at a point halfway across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 select 960 486`; timeout `6000`; targetVariable `ptz2_track`

r2c2

Right ▸

Tracks the person on the right of the frame (the camera is told to take whoever is at a point five sixths of the way across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 select 1600 486`; timeout `6000`; targetVariable `ptz2_track`

r2c3

`concat('CAM ', $(internal:custom_ptz2_atem_input))`

Stops pan, tilt and zoom, and names the camera this page drives. Closes the camera menu while it is open. Red and LIVE when this camera is on ATEM program, green and PVW on preview (input number in ptz_atem_input).

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.menu') == "On"`

  ACTIONS
  - visca2 · **custom**: custom `81 01 06 06 03 FF (close the camera menu)`

  ELSE_ACTIONS
  - visca2 · **stop**
  - visca2 · **zoomS**

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.menu') == "On"`
  → label EXIT, colour #3A2E06
- internal · **check_expression**: expression = `$(atem:pvw1_input_id) == $(internal:custom_ptz2_atem_input)`
  → colour #00A651, label = concat(concat('CAM ', $(internal:custom_ptz2_atem_input)), ' PVW'), text colour #000000, icon stop-ink
- internal · **check_expression**: expression = `$(atem:pgm1_input_id) == $(internal:custom_ptz2_atem_input)`
  → colour #FF0000, label = concat(concat('CAM ', $(internal:custom_ptz2_atem_input)), ' LIVE'), text colour #FFFFFF, icon stop-paper

r2c5 `concat('CAM ', $(internal:custom_ptz_atem_input))`

Stops pan, tilt and zoom, and names the camera this page drives. Closes the camera menu while it is open. Red and LIVE when this camera is on ATEM program, green and PVW on preview (input number in ptz_atem_input).

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.menu') == "On"`

  ACTIONS
  - visca · **custom**: custom `81 01 06 06 03 FF (close the camera menu)`

  ELSE_ACTIONS
  - visca · **stop**
  - visca · **zoomS**

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.menu') == "On"`
  → label EXIT, colour #3A2E06
- internal · **check_expression**: expression = `$(atem:pvw1_input_id) == $(internal:custom_ptz_atem_input)`
  → colour #00A651, label = concat(concat('CAM ', $(internal:custom_ptz_atem_input)), ' PVW'), text colour #000000, icon stop-ink
- internal · **check_expression**: expression = `$(atem:pgm1_input_id) == $(internal:custom_ptz_atem_input)`
  → colour #FF0000, label = concat(concat('CAM ', $(internal:custom_ptz_atem_input)), ' LIVE'), text colour #FFFFFF, icon stop-paper

r2c6 **◂ Left**

Tracks the person on the left of the frame (the camera is told to take whoever is at a point a sixth of the way across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 select 320 486`; timeout `6000`; targetVariable `ptz_track`

r2c7 **Middle**

Tracks the person on the middle of the frame (the camera is told to take whoever is at a point halfway across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 select 960 486`; timeout `6000`; targetVariable `ptz_track`

r2c8 **Right ▸**

Tracks the person on the right of the frame (the camera is told to take whoever is at a point five sixths of the way across, at torso height). Works while tracking is on.

PRESS

- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 select 1600 486`; timeout `6000`; targetVariable `ptz_track`

r3c0

1 (Wide)

Go to preset 1 (Wide), keeping the picture as it is. With Save armed, saves the current shot as preset 1 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 1; presetAsText 1
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 1
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 1
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 1`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 1`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 1, text colour #000000, icon preset-save-ink

r3c1

2 (Stage)

Go to preset 2 (Stage), keeping the picture as it is. With Save armed, saves the current shot as preset 2 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 2; presetAsText 2
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 2
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 2
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 2`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 2`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 2, text colour #000000, icon preset-save-ink

r3c2

3 (Baptism)

Go to preset 3 (Baptism), keeping the picture as it is. With Save armed, saves the current shot as preset 3 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`

  ACTIONS
  - visca2 · **setPreset**: isText `no`; presetAsNumber 3; presetAsText 3
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 3
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz2_last`; create `yes`; value 3
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 recall 3`; timeout `20000`; targetVariable `ptz2_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz2_last) == 3`
  → box0.borderWidth 6, border #22D3EE
- internal · **check_expression**: expression = `$(internal:custom_ptz2_armed) == 1`
  → colour #B91C1C, label SAVE 3, text colour #000000, icon preset-save-ink

r3c4

Worship

Both cameras on one page for worship: their presets, Save, and pan, tilt and zoom by hand.

PRESS

- internal · **set_page**: surfaceId `self`; page 15

r3c6

1 (Wide)

Go to preset 1 (Wide), keeping the picture as it is. With Save armed, saves the current shot as preset 1 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText `no`; presetAsNumber 1; presetAsText 1
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 1
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 1
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 1`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 1`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 1, text colour #000000, icon preset-save-ink

r3c7

2 (Drums)

Go to preset 2 (Drums), keeping the picture as it is. With Save armed, saves the current shot as preset 2 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 2; presetAsText 2
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 2
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 2
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 2`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 2`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 2, text colour #000000, icon preset-save-ink

r3c8

3 (Keys)

Go to preset 3 (Keys), keeping the picture as it is. With Save armed, saves the current shot as preset 3 instead.

PRESS

- internal · **logic_if**
  CONDITION
  - internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`

  ACTIONS
  - visca · **setPreset**: isText no; presetAsNumber 3; presetAsText 3
  - internal · **custom_variable_set_value**: name `ptz_armed`; create `yes`; value `0`
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 3
  - internal · **custom_variable_set_value**: name `ptz2_armed`; create `yes`; value `0`

  ELSE_ACTIONS
  - internal · **custom_variable_set_value**: name `ptz_last`; create `yes`; value 3
  - internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 recall 3`; timeout `20000`; targetVariable `ptz_recall`

CHANGES WHEN

- internal · **check_expression**: expression = `$(internal:custom_ptz_last) == 3`
  → box0.borderWidth 6, border #F472B6
- internal · **check_expression**: expression = `$(internal:custom_ptz_armed) == 1`
  → colour #B91C1C, label SAVE 3, text colour #000000, icon preset-save-ink

r4c0

`concat('Pan ', jsonpath($(internal:custom_ptz2_state), '$.pan'))`

Pan angle as reported by the camera, polled every second. Turns red if the camera stops answering.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz2_state), '$.online') != "OK"`
  → colour #B91C1C, label Camera DOWN

r4c2

`concat('Tilt ', jsonpath($(internal:custom_ptz2_state), '$.tilt'))`

Tilt angle as reported by the camera, polled every second. Display only.

No action: display only.

r4c3

`concat('Zoom ', jsonpath($(internal:custom_ptz2_state), '$.zoom'))`

Zoom position as a percentage of the lens range, polled every second. Display only.

No action: display only.

r4c5

`concat('Pan ', jsonpath($(internal:custom_ptz_state), '$.pan'))`

Pan angle as reported by the camera, polled every second. Turns red if the camera stops answering.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `jsonpath($(internal:custom_ptz_state), '$.online') != "OK"`
  → colour #B91C1C, label Camera DOWN

r4c6

`concat('Tilt ', jsonpath($(internal:custom_ptz_state), '$.tilt'))`

Tilt angle as reported by the camera, polled every second. Display only.

No action: display only.

r4c8

`concat('Zoom ', jsonpath($(internal:custom_ptz_state), '$.zoom'))`

Zoom position as a percentage of the lens range, polled every second. Display only.

No action: display only.

r5c0

Pan

Turn to pan at the Speed knob's speed; each detent drives for 150ms. Press stops.

PRESS

- visca2 · **stop**

TURN LEFT

- visca2 · **custom**: custom `81 01 06 01 00 00 01 03 FF (pan left)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

TURN RIGHT

- visca2 · **custom**: custom `81 01 06 01 00 00 02 03 FF (pan right)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

r5c2

Tilt

Turn clockwise to tilt up, anticlockwise down, at the Speed knob's speed. Press stops.

PRESS

- visca2 · **stop**

TURN LEFT

- visca2 · **custom**: custom `81 01 06 01 00 00 03 02 FF (tilt down)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

TURN RIGHT

- visca2 · **custom**: custom `81 01 06 01 00 00 03 01 FF (tilt up)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz2_speed)`; parameter1 `$(internal:custom_ptz2_tspeed)`
- internal · **wait**: time = `150`
- visca2 · **stop**

r5c3

Zoom

Turn clockwise to zoom in, anticlockwise out. Speed follows the Speed knob. Press stops.

PRESS
- visca2 · **zoomS**

TURN LEFT
- visca2 · **custom**: custom `81 01 04 07 30 FF (zoom out)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_zspeed)`
- internal · **wait**: time = `150`
- visca2 · **zoomS**

TURN RIGHT
- visca2 · **custom**: custom `81 01 04 07 20 FF (zoom in)`; command_parameters 9; parameter0 `$(internal:custom_ptz2_zspeed)`
- internal · **wait**: time = `150`
- visca2 · **zoomS**

r5c5

Pan

Turn to pan at the Speed knob's speed; each detent drives for 150ms. Press stops.

PRESS
- visca · **stop**

TURN LEFT
- visca · **custom**: custom `81 01 06 01 00 00 01 03 FF (pan left)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

TURN RIGHT
- visca · **custom**: custom `81 01 06 01 00 00 02 03 FF (pan right)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

r5c6

Tilt

Turn clockwise to tilt up, anticlockwise down, at the Speed knob's speed. Press stops.

PRESS
- visca · **stop**

TURN LEFT
- visca · **custom**: custom `81 01 06 01 00 00 03 02 FF (tilt down)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

TURN RIGHT
- visca · **custom**: custom `81 01 06 01 00 00 03 01 FF (tilt up)`; command_parameters `8,9;10,11`; parameter0 `$(internal:custom_ptz_speed)`; parameter1 `$(internal:custom_ptz_tspeed)`
- internal · **wait**: time = `150`
- visca · **stop**

r5c8

Zoom

Turn clockwise to zoom in, anticlockwise out. Speed follows the Speed knob. Press stops.

PRESS
- visca · **zoomS**

TURN LEFT
- visca · **custom**: custom `81 01 04 07 30 FF (zoom out)`; command_parameters 9; parameter0 `$(internal:custom_ptz_zspeed)`
- internal · **wait**: time = `150`
- visca · **zoomS**

TURN RIGHT
- visca · **custom**: custom `81 01 04 07 20 FF (zoom in)`; command_parameters 9; parameter0 `$(internal:custom_ptz_zspeed)`
- internal · **wait**: time = `150`
- visca · **zoomS**

Page 17: Dante



r1c0

Leads SQ 1-4

dante: ULXD4Q-LEADS-1-4 into AllenHth-0e0184 (SQ 1-4). Read-only. Green: on the network and every route subscribed. Amber: on the network, a route down. Red: not on the network.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4")`
  → colour #B91C1C, label Leads Offline
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4")`
  → colour #A16207, label Leads Not routed
- internal · **logic_operator**: operation and
  → colour #15803D, label Leads SQ 1-4
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `01`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `02`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `03`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `04`; destinationDeviceId `AllenHth-0e0184`

r1c1

BGV 1-4 SQ 5-8

dante: ULXD4Q-BGV-1-4 into AllenHth-0e0184 (SQ 5-8). Read-only. Green: on the network and every route subscribed. Amber: on the network, a route down. Red: not on the network.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4")`
  → colour #B91C1C, label BGV 1-4 Offline
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4")`
  → colour #A16207, label BGV 1-4 Not routed
- internal · **logic_operator**: operation and
  → colour #15803D, label BGV 1-4 SQ 5-8
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `05`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `06`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `07`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `08`; destinationDeviceId `AllenHth-0e0184`

r1c2

BGV+Host SQ 9-12

dante: ULXD4Q-BGV-5-6-HOST-1-2 into AllenHth-0e0184 (SQ 9-12). Read-only. Green: on the network and every route subscribed. Amber: on the network, a route down. Slate "On XLR": not on the network, which is normal for this one.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "ULXD4Q-BGV-5-6-HOST-1-2")`
  → colour #1F2937, label BGV+Host On XLR
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-BGV-5-6-HOST-1-2")`
  → colour #A16207, label BGV+Host Not routed
- internal · **logic_operator**: operation and
  → colour #15803D, label BGV+Host SQ 9-12
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-BGV-5-6-HOST-1-2")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q-BGV-5-6-HOST-1-2`; destinationChannelId `09`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q-BGV-5-6-HOST-1-2`; destinationChannelId `10`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q-BGV-5-6-HOST-1-2`; destinationChannelId 11; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q-BGV-5-6-HOST-1-2`; destinationChannelId 12; destinationDeviceId `AllenHth-0e0184`

r1c3

Lav 1-2 SQ 13-14

dante: ULXD4D-LAV-1-2 into AllenHth-0e0184 (SQ 13-14). Read-only. Green: on the network and every route subscribed. Amber: on the network, a route down. Red: not on the network.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2")`
  → colour #B91C1C, label Lav 1-2 Offline
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2")`
  → colour #A16207, label Lav 1-2 Not routed
- internal · **logic_operator**: operation and
  → colour #15803D, label Lav 1-2 SQ 13-14
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4D-LAV-1-2`; destinationChannelId 13; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4D-LAV-1-2`; destinationChannelId 14; destinationDeviceId `AllenHth-0e0184`

r1c4

Tracks SQ 15-22

dante: TCBs-MacBook-Pro-Tracks into AllenHth-0e0184 (SQ 15-22). Read-only. Green: on the network and every route subscribed. Amber: on the network, a route down. Slate "Not here": not on the network, which is normal for this one.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "TCBs-MacBook-Pro-Tracks")`
  → colour #1F2937, label Tracks Not here
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "TCBs-MacBook-Pro-Tracks")`
  → colour #A16207, label Tracks Not routed
- internal · **logic_operator**: operation and
  → colour #15803D, label Tracks SQ 15-22
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "TCBs-MacBook-Pro-Tracks")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `TCBs-MacBook-Pro-Tracks`; destinationChannelId 15; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `TCBs-MacBook-Pro-Tracks`; destinationChannelId 16; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `TCBs-MacBook-Pro-Tracks`; destinationChannelId 17; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `TCBs-MacBook-Pro-Tracks`; destinationChannelId 18; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `05`; sourceDeviceName `TCBs-MacBook-Pro-Tracks`; destinationChannelId 19; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `06`; sourceDeviceName `TCBs-MacBook-Pro-Tracks`; destinationChannelId 20; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `07`; sourceDeviceName `TCBs-MacBook-Pro-Tracks`; destinationChannelId 21; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `08`; sourceDeviceName `TCBs-MacBook-Pro-Tracks`; destinationChannelId 22; destinationDeviceId `AllenHth-0e0184`

r1c5

PP1 SQ 23-24 45-46

dante: iMac-PP1 into AllenHth-0e0184 (SQ 23-24, 45-46). Read-only. Green: on the network and every route subscribed. Amber: on the network, a route down. Red: not on the network.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "iMac-PP1")`
  → colour #B91C1C, label PP1 Offline
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "iMac-PP1")`
  → colour #A16207, label PP1 Not routed
- internal · **logic_operator**: operation and
  → colour #15803D, label PP1 SQ 23-24 45-46
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "iMac-PP1")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac-PP1`; destinationChannelId `23`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac-PP1`; destinationChannelId `24`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac-PP1`; destinationChannelId `45`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac-PP1`; destinationChannelId `46`; destinationDeviceId `AllenHth-0e0184`

r1c6

Devan SQ 33-41 51-64

dante: Devans-Mac-mini into AllenHth-0e0184 (SQ 33-41, 51-64). Read-only. Green: on the network and every route subscribed. Amber: on the network, a route down. Slate "Not here": not on the network, which is normal for this one.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "Devans-Mac-mini")`
  → colour #1F2937, label Devan Not here
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "Devans-Mac-mini")`
  → colour #A16207, label Devan Not routed
- internal · **logic_operator**: operation and
  → colour #15803D, label Devan SQ 33-41 51-64

  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "Devans-Mac-mini")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 33; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 34; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 35; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 36; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `05`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 37; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `06`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 38; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `07`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 39; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `08`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 40; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `09`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 41; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 19; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 51; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `20`; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 52; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 21; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 53; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 22; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 54; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 23; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 55; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 24; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 56; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 25; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 57; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 26; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 58; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 27; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 59; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 28; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 60; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 29; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 61; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName 30; sourceDeviceName `Devans-Mac-mini`; destinationChannelId 62; destinationDeviceId `AllenHth-0e0184`

- dante · **routing_bg_manual**: sourceChannelName 31; sourceDeviceName Devans-Mac-mini; destinationChannelId 63; destinationDeviceId AllenHth-0e0184
- dante · **routing_bg_manual**: sourceChannelName 32; sourceDeviceName Devans-Mac-mini; destinationChannelId 64; destinationDeviceId AllenHth-0e0184

r1c7

Josue SQ 42-44 47-49

dante: Josues-MacBook-Pro into AllenHth-0e0184 (SQ 42-44, 47-49). Read-only. Green: on the network and every route subscribed. Amber: on the network, a route down. Slate "Not here": not on the network, which is normal for this one.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "Josues-MacBook-Pro")`
  → colour #1F2937, label Josue Not here
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "Josues-MacBook-Pro")`
  → colour #A16207, label Josue Not routed
- internal · **logic_operator**: operation and
  → colour #15803D, label Josue SQ 42-44 47-49
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "Josues-MacBook-Pro")`
  - dante · **routing_bg_manual**: sourceChannelName 01; sourceDeviceName Josues-MacBook-Pro; destinationChannelId 42; destinationDeviceId AllenHth-0e0184
  - dante · **routing_bg_manual**: sourceChannelName 02; sourceDeviceName Josues-MacBook-Pro; destinationChannelId 43; destinationDeviceId AllenHth-0e0184
  - dante · **routing_bg_manual**: sourceChannelName 03; sourceDeviceName Josues-MacBook-Pro; destinationChannelId 44; destinationDeviceId AllenHth-0e0184
  - dante · **routing_bg_manual**: sourceChannelName 06; sourceDeviceName Josues-MacBook-Pro; destinationChannelId 47; destinationDeviceId AllenHth-0e0184
  - dante · **routing_bg_manual**: sourceChannelName 07; sourceDeviceName Josues-MacBook-Pro; destinationChannelId 48; destinationDeviceId AllenHth-0e0184
  - dante · **routing_bg_manual**: sourceChannelName 08; sourceDeviceName Josues-MacBook-Pro; destinationChannelId 49; destinationDeviceId AllenHth-0e0184

r1c8

Routing

dante: green when every required source is on the network with all its routes into AllenHth-0e0184 subscribed; red otherwise. Sources the desk does not need over Dante (BGV+Host, Tracks, Devan, Josue) are not counted. Read-only.

No action: display only.

CHANGES WHEN

- internal · **logic_operator**: operation and INVERTED
  → colour #B91C1C, label Routing Not normal
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `01`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `02`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `03`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `04`; destinationDeviceId `AllenHth-0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `05`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `06`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `07`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `08`; destinationDeviceId `AllenHth-0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4D-LAV-1-2`; destinationChannelId `13`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4D-LAV-1-2`; destinationChannelId `14`; destinationDeviceId `AllenHth-0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "iMac-PP1")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac-PP1`; destinationChannelId `23`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac-PP1`; destinationChannelId `24`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac-PP1`; destinationChannelId `45`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac-PP1`; destinationChannelId `46`; destinationDeviceId `AllenHth-0e0184`
- internal · **logic_operator**: operation and
  → colour #15803D, label Routing Normal
  DEFAULT
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4")`
  - dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `01`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `02`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `03`; destinationDeviceId `AllenHth-0e0184`
  - dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q-LEADS-1-4`; destinationChannelId `04`; destinationDeviceId `AllenHth-0e0184`
  - internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4")`

- dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `05`; destinationDeviceId `AllenHth-0e0184`
- dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `06`; destinationDeviceId `AllenHth-0e0184`
- dante · **routing_bg_manual**: sourceChannelName `03`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `07`; destinationDeviceId `AllenHth-0e0184`
- dante · **routing_bg_manual**: sourceChannelName `04`; sourceDeviceName `ULXD4Q-BGV-1-4`; destinationChannelId `08`; destinationDeviceId `AllenHth-0e0184`
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2")`
- dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `ULXD4D-LAV-1-2`; destinationChannelId 13; destinationDeviceId `AllenHth-0e0184`
- dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `ULXD4D-LAV-1-2`; destinationChannelId 14; destinationDeviceId `AllenHth-0e0184`
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "iMac-PP1")`
- dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac-PP1`; destinationChannelId 23; destinationDeviceId `AllenHth-0e0184`
- dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac-PP1`; destinationChannelId 24; destinationDeviceId `AllenHth-0e0184`
- dante · **routing_bg_manual**: sourceChannelName `01`; sourceDeviceName `iMac-PP1`; destinationChannelId 45; destinationDeviceId `AllenHth-0e0184`
- dante · **routing_bg_manual**: sourceChannelName `02`; sourceDeviceName `iMac-PP1`; destinationChannelId 46; destinationDeviceId `AllenHth-0e0184`

`r2c0`

`arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4") ? concat(round($(dante:ULXD4Q-LEADS-1-4_sr) / 100) / 10, " kHz / ", $(dante:ULXD4Q-LEADS-1-4_latency), " ms") : "-"`

dante: Leads (ULXD4Q-LEADS-1-4) sample rate and Dante latency, or "-" when it is not on the network. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4") && $(dante:ULXD4Q-LEADS-1-4_sr) != 48000`
  → colour #B91C1C

`r2c1`

`arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4") ? concat(round($(dante:ULXD4Q-BGV-1-4_sr) / 100) / 10, " kHz / ", $(dante:ULXD4Q-BGV-1-4_latency), " ms") : "-"`

dante: BGV 1-4 (ULXD4Q-BGV-1-4) sample rate and Dante latency, or "-" when it is not on the network. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4") && $(dante:ULXD4Q-BGV-1-4_sr) != 48000`
  → colour #B91C1C

`r2c2`

`arrayIncludes($(dante:devices), "ULXD4Q-BGV-5-6-HOST-1-2") ? concat(round($(dante:ULXD4Q-BGV-5-6-HOST-1-2_sr) / 100) / 10, " kHz / ", $(dante:ULXD4Q-BGV-5-6-HOST-1-2_latency), " ms") : "-"`

dante: BGV+Host (ULXD4Q-BGV-5-6-HOST-1-2) sample rate and Dante latency, or "-" when it is not on the network. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4Q-BGV-5-6-HOST-1-2") && $(dante:ULXD4Q-BGV-5-6-HOST-1-2_sr) != 48000`
  → colour #B91C1C

---

r2c3

`arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2") ? concat(round($(dante:ULXD4D-LAV-1-2_sr) / 100) / 10, " kHz / ", $(dante:ULXD4D-LAV-1-2_latency), " ms") : "–"`

dante: Lav 1-2 (ULXD4D-LAV-1-2) sample rate and Dante latency, or "-" when it is not on the network. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2") && $(dante:ULXD4D-LAV-1-2_sr) != 48000`
  → colour #B91C1C

r2c4

`arrayIncludes($(dante:devices), "TCBs-MacBook-Pro-Tracks") ? concat(round($(dante:TCBs-MacBook-Pro-Tracks_sr) / 100) / 10, " kHz / ", $(dante:TCBs-MacBook-Pro-Tracks_latency), " ms") : "–"`

dante: Tracks (TCBs-MacBook-Pro-Tracks) sample rate and Dante latency, or "-" when it is not on the network. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "TCBs-MacBook-Pro-Tracks") && $(dante:TCBs-MacBook-Pro-Tracks_sr) != 48000`
  → colour #B91C1C

r2c5

`arrayIncludes($(dante:devices), "iMac-PP1") ? concat(round($(dante:iMac-PP1_sr) / 100) / 10, " kHz / ", $(dante:iMac-PP1_latency), " ms") : "–"`

dante: PP1 (iMac-PP1) sample rate and Dante latency, or "-" when it is not on the network. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "iMac-PP1") && $(dante:iMac-PP1_sr) != 48000`
  → colour #B91C1C

r2c6

`arrayIncludes($(dante:devices), "Devans-Mac-mini") ? concat(round($(dante:Devans-Mac-mini_sr) / 100) / 10, " kHz / ", $(dante:Devans-Mac-mini_latency), " ms") : "–"`

dante: Devan (Devans-Mac-mini) sample rate and Dante latency, or "-" when it is not on the network. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "Devans-Mac-mini") && $(dante:Devans-Mac-mini_sr) != 48000`
  → colour #B91C1C

r2c7

`arrayIncludes($(dante:devices), "Josues-MacBook-Pro") ? concat(round($(dante:Josues-MacBook-Pro_sr) / 100) / 10, " kHz / ", $(dante:Josues-MacBook-Pro_latency), " ms") : "–"`

dante: Josue (Josues-MacBook-Pro) sample rate and Dante latency, or "-" when it is not on the network. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "Josues-MacBook-Pro") && $(dante:Josues-MacBook-Pro_sr) != 48000`
  → colour #B91C1C

r2c8

SQ7 card

dante: the SQ7's Dante card (AllenHth-0e0184), which every route on this page lands on, and its sample rate and latency. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!arrayIncludes($(dante:devices), "AllenHth-0e0184")`
  → colour #B91C1C, label SQ7 card Offline
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "AllenHth-0e0184")`
  → colour #15803D, label = concat("SQ7 card\n", arrayIncludes($(dante:devices), "AllenHth-0e0184") ? concat(round($(dante:AllenHth-0e0184_sr) / 100) / 10, " kHz / ", $(dante:AllenHth-0e0184_latency), " ms") : "-")

r3c0

Sample rate

dante: green when every Dante device present runs 48 kHz, red when one differs or the SQ7 card is missing. Read-only.

No action: display only.

CHANGES WHEN

- internal · **check_expression**: expression = `!(arrayIncludes($(dante:devices), "AllenHth-0e0184") && (!arrayIncludes($(dante:devices), "AllenHth-0e0184") || $(dante:AllenHth-0e0184_sr) == 48000) && (!arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4") || $(dante:ULXD4Q-LEADS-1-4_sr) == 48000) && (!arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4") || $(dante:ULXD4Q-BGV-1-4_sr) == 48000) && (!arrayIncludes($(dante:devices), "ULXD4Q-BGV-5-6-HOST-1-2") || $(dante:ULXD4Q-BGV-5-6-HOST-1-2_sr) == 48000) && (!arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2") || $(dante:ULXD4D-LAV-1-2_sr) == 48000) && (!arrayIncludes($(dante:devices), "TCBs-MacBook-Pro-Tracks") || $(dante:TCBs-MacBook-Pro-Tracks_sr) == 48000) && (!arrayIncludes($(dante:devices), "iMac-PP1") || $(dante:iMac-PP1_sr) == 48000) && (!arrayIncludes($(dante:devices), "Devans-Mac-mini") || $(dante:Devans-Mac-mini_sr) == 48000) && (!arrayIncludes($(dante:devices), "Josues-MacBook-Pro") || $(dante:Josues-MacBook-Pro_sr) == 48000))`
  → colour #B91C1C, label Sample rate Mismatch
- internal · **check_expression**: expression = `arrayIncludes($(dante:devices), "AllenHth-0e0184") && (!arrayIncludes($(dante:devices), "AllenHth-0e0184") || $(dante:AllenHth-0e0184_sr) == 48000) && (!arrayIncludes($(dante:devices), "ULXD4Q-LEADS-1-4") || $(dante:ULXD4Q-LEADS-1-4_sr) == 48000) && (!arrayIncludes($(dante:devices), "ULXD4Q-BGV-1-4") || $(dante:ULXD4Q-BGV-1-4_sr) == 48000) && (!arrayIncludes($(dante:devices), "ULXD4Q-BGV-5-6-HOST-1-2") || $(dante:ULXD4Q-BGV-5-6-HOST-1-2_sr) == 48000) && (!arrayIncludes($(dante:devices), "ULXD4D-LAV-1-2") || $(dante:ULXD4D-LAV-1-2_sr) == 48000) && (!arrayIncludes($(dante:devices), "TCBs-MacBook-Pro-Tracks") || $(dante:TCBs-MacBook-Pro-Tracks_sr) == 48000) && (!arrayIncludes($(dante:devices), "iMac-PP1") || $(dante:iMac-PP1_sr) == 48000) && (!arrayIncludes($(dante:devices), "Devans-Mac-mini") || $(dante:Devans-Mac-mini_sr) == 48000) && (!arrayIncludes($(dante:devices), "Josues-MacBook-Pro") || $(dante:Josues-MacBook-Pro_sr) == 48000)`
  → colour #15803D, label Sample rate 48 kHz

Page 18: Videohub



`r0c0` **OUT 1**

videohub: select output 1 as the destination

PRESS
- videohub · **select_destination**: destination 1

CHANGES WHEN
- videohub · **selected_destination**: output 1
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r0c1` **OUT 2**

videohub: select output 2 as the destination

PRESS
- videohub · **select_destination**: destination 2

CHANGES WHEN
- videohub · **selected_destination**: output 2
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r0c2` **OUT 3**

videohub: select output 3 as the destination

PRESS
- videohub · **select_destination**: destination 3

CHANGES WHEN
- videohub · **selected_destination**: output 3
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r0c3` **OUT 4**

videohub: select output 4 as the destination

PRESS
- videohub · **select_destination**: destination 4

CHANGES WHEN
- videohub · **selected_destination**: output 4
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r0c4`

OUT 5

videohub: select output 5 as the destination

PRESS
- videohub · **select_destination**: destination 5

CHANGES WHEN
- videohub · **selected_destination**: output 5
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r0c5`

OUT 6

videohub: select output 6 as the destination

PRESS
- videohub · **select_destination**: destination 6

CHANGES WHEN
- videohub · **selected_destination**: output 6
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r0c6`

OUT 7

videohub: select output 7 as the destination

PRESS
- videohub · **select_destination**: destination 7

CHANGES WHEN
- videohub · **selected_destination**: output 7
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r0c7`

OUT 8

videohub: select output 8 as the destination

PRESS
- videohub · **select_destination**: destination 8

CHANGES WHEN
- videohub · **selected_destination**: output 8
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r0c8`

VW

videohub: back to VW (page 7)

PRESS
- internal · **set_page**: surfaceId `self`; page 7

`r1c0`

OUT 9

videohub: select output 9 as the destination

PRESS
- videohub · **select_destination**: destination 9

CHANGES WHEN
- videohub · **selected_destination**: output 9
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r1c1`

OUT 10

videohub: select output 10 as the destination

PRESS
- videohub · **select_destination**: destination `10`

CHANGES WHEN
- videohub · **selected_destination**: output `10`
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

`r1c2`

OUT 11

videohub: select output 11 as the destination

PRESS
- videohub · **select_destination**: destination 11

CHANGES WHEN
- videohub · **selected_destination**: output 11
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

r1c3

OUT 12

videohub: select output 12 as the destination

PRESS
- videohub · **select_destination**: destination 12

CHANGES WHEN
- videohub · **selected_destination**: output 12
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

r1c4

OUT 13

videohub: select output 13 as the destination

PRESS
- videohub · **select_destination**: destination 13

CHANGES WHEN
- videohub · **selected_destination**: output 13
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

r1c5

OUT 14

videohub: select output 14 as the destination

PRESS
- videohub · **select_destination**: destination 14

CHANGES WHEN
- videohub · **selected_destination**: output 14
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

r1c6

OUT 15

videohub: select output 15 as the destination

PRESS
- videohub · **select_destination**: destination 15

CHANGES WHEN
- videohub · **selected_destination**: output 15
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

r1c7

OUT 16

videohub: select output 16 as the destination

PRESS
- videohub · **select_destination**: destination 16

CHANGES WHEN
- videohub · **selected_destination**: output 16
  → colour #A78BFA, colour #111827, colour #111827, colour #111827

r2c0

IN 1

videohub: route input 1 to the selected output

PRESS
- videohub · **route_source**: source 1; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 1
  → colour #1F6F3A, colour #FFFFFF

r2c1

IN 2

videohub: route input 2 to the selected output

PRESS
- videohub · **route_source**: source 2; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 2
  → colour #1F6F3A, colour #FFFFFF

r2c2

IN 3

videohub: route input 3 to the selected output

PRESS
- videohub · **route_source**: source 3; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 3
  → colour #1F6F3A, colour #FFFFFF

r2c3

IN 4

videohub: route input 4 to the selected output

PRESS
- videohub · **route_source**: source 4; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 4
  → colour #1F6F3A, colour #FFFFFF

r2c4

IN 5

videohub: route input 5 to the selected output

PRESS
- videohub · **route_source**: source 5; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 5
  → colour #1F6F3A, colour #FFFFFF

r2c5

IN 6

videohub: route input 6 to the selected output

PRESS
- videohub · **route_source**: source 6; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 6
  → colour #1F6F3A, colour #FFFFFF

r2c6

IN 7

videohub: route input 7 to the selected output

PRESS
- videohub · **route_source**: source 7; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 7
  → colour #1F6F3A, colour #FFFFFF

r2c7

IN 8

videohub: route input 8 to the selected output

PRESS
- videohub · **route_source**: source 8; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 8
  → colour #1F6F3A, colour #FFFFFF

r3c0

IN 9

videohub: route input 9 to the selected output

PRESS
- videohub · **route_source**: source 9; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 9
  → colour #1F6F3A, colour #FFFFFF

r3c1

IN 10

videohub: route input 10 to the selected output

PRESS
- videohub · **route_source**: source 10; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 10
  → colour #1F6F3A, colour #FFFFFF

r3c2

IN 11

videohub: route input 11 to the selected output

PRESS
- videohub · **route_source**: source 11; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 11
  → colour #1F6F3A, colour #FFFFFF

r3c3

IN 12

videohub: route input 12 to the selected output

PRESS
- videohub · **route_source**: source 12; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 12
  → colour #1F6F3A, colour #FFFFFF

r3c4

IN 13

videohub: route input 13 to the selected output

PRESS
- videohub · **route_source**: source 13; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 13
  → colour #1F6F3A, colour #FFFFFF

r3c5

IN 14

videohub: route input 14 to the selected output

PRESS
- videohub · **route_source**: source 14; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 14
  → colour #1F6F3A, colour #FFFFFF

r3c6

IN 15

videohub: route input 15 to the selected output

PRESS
- videohub · **route_source**: source 15; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 15
  → colour #1F6F3A, colour #FFFFFF

r3c7

IN 16

videohub: route input 16 to the selected output

PRESS
- videohub · **route_source**: source 16; ignore_lock yes

CHANGES WHEN
- videohub · **selected_source**: input 16
  → colour #1F6F3A, colour #FFFFFF

Surfaces

The Stream Decks Companion drives, by serial number. A replacement deck of the same model is picked up as a new surface: give it the same settings.

Deck	Serial	Brightness	Starts on
Elgato Stream Deck +	`A00WA31410205Q`	100%	the page it was last on
Elgato Stream Deck + XL	`AD4MA619138NSK`	100%	the page it was last on

Companion also lists 21 × Service Health deck, 4 × timing probe: software surfaces (the Service Health board's Deck tab opens one per browser).

Triggers

Poll PA state

WHEN
- Every 10 s

DOES
- internal · **exec**: path `python3 "/home/samuelbailey/Desktop/AV_Power_scripts/speaker_power.py" state`; timeout `15000`; targetVariable `pa_state`

Poll projector state

WHEN
- Every 10 s

DOES
- internal · **exec**: path `python3 "/home/samuelbailey/Desktop/AV_Power_scripts/projector_power.py" state`; timeout `15000`; targetVariable `projector_state`

Poll system stats

WHEN
- Every 30 s

DOES
- internal · **exec**: path `awk '{printf "%.0f", $1/1000}' /sys/class/thermal/thermal_zone0/temp`; timeout `5000`; targetVariable `sys_temp`
- internal · **exec**: path `cut -d' ' -f1 /proc/loadavg`; timeout `5000`; targetVariable `sys_load`
- internal · **exec**: path `free | awk '/^Mem:/ {printf "%.0f", $3/$2*100}'`; timeout `5000`; targetVariable `sys_mem`
- internal · **exec**: path `df -P / | awk 'NR==2 {gsub(/%/,"",$5); print $5}'`; timeout `5000`; targetVariable `sys_disk`
- internal · **exec**: path `v=$(vcgencmd get_throttled 2>/dev/null | cut -d= -f2); case "$v" in 0x0) echo OK;; 0x*) echo "$v";; *) a=$(cat /sys/class/hwmon/hwmon*/in0_lcrit_alarm 2>/dev/null | head -1); case "$a" in 0) echo OK;; 1) echo UNDERVOLT;; *) echo n/a;; esac;; esac`; timeout `5000`; targetVariable `sys_throttle`
- internal · **exec**: path `p=$(ping -c 3 -W 2 1.1.1.1 2>/dev/null); if [-z "$p"]; then echo n/a; else l=$(echo "$p" | grep -o "[0-9]*% packet loss" | cut -d% -f1); r=$(echo "$p" | awk -F/ "/rtt|round-trip/ {printf \"%.0f\", \$5}"); if [-z "$r"]; then echo DOWN; elif ["$l" != "0"]; then echo "${r}ms ${l}%"; else echo "${r}ms"; fi; fi`; timeout `5000`; targetVariable `sys_net`
- internal · **exec**: path `awk '{d=int($1/86400); h=int(($1%86400)/3600); m=int(($1%3600)/60); if (d>0) printf "%dd %dh", d, h; else printf "%dh %dm", h, m}' /proc/uptime`; timeout `5000`; targetVariable `sys_uptime`
- internal · **exec**: path `df -BG -P / 2>/dev/null | awk 'NR==2 {gsub(/G/,"",$4); print $4} END {if (NR<2) print "n/a"}'`; timeout `5000`; targetVariable `sys_disk_free`
- internal · **exec**: path `hostname -I 2>/dev/null | awk '{print ($1=="" ? "n/a" : $1)}'`; timeout `5000`; targetVariable `sys_ip`
- internal · **exec**: path `awk '$2=="/" {print ($4 ~ /(^|,)ro(,|$)/) ? "READ-ONLY" : "OK"; exit}' /proc/mounts 2>/dev/null || echo n/a`; timeout `5000`; targetVariable `sys_rw`

Poll PTZ camera

WHEN
- Every 1 s

DOES
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_state.py 10.23.0.181`; timeout `4000`; targetVariable `ptz_state`

Poll PTZ 2 camera

WHEN
- Every 1 s

DOES
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_state.py 10.23.0.196`; timeout `4000`; targetVariable `ptz2_state`

Poll PTZ tracking

WHEN
- Every 5 s

DOES
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.181 get`; timeout `6000`; targetVariable `ptz_track`

Poll PTZ 2 tracking

WHEN
- Every 5 s

DOES
- internal · **exec**: path `python3 /home/samuelbailey/Desktop/AV_Power_scripts/ptz_web.py 10.23.0.196 get`; timeout `6000`; targetVariable `ptz2_track`

Sync PTZ dials

WHEN
- When $(internal:custom_ptz_state) changes

DOES
- internal · **custom_variable_set_value**: name `ptz_gain`; create `yes`; value = `jsonpath($(internal:custom_ptz_state), '$.gain') == "--" || jsonpath($(internal:custom_ptz_state), '$.gain') == "" ? $(internal:custom_ptz_gain) : fromRadix(jsonpath($(internal:custom_ptz_state), '$.gain'), 10)`
- internal · **custom_variable_set_value**: name `ptz_wbk`; create `yes`; value = `includes(jsonpath($(internal:custom_ptz_state), '$.wb'), "K") ? fromRadix(substr(jsonpath($(internal:custom_ptz_state), '$.wb'), 0, 4), 10) : $(internal:custom_ptz_wbk)`
- internal · **custom_variable_set_value**: name `ptz_wbcode`; create `yes`; value = `$(internal:custom_ptz_wbk) == 2400 ? 12 : $(internal:custom_ptz_wbk) == 2500 ? 13 : $(internal:custom_ptz_wbk) == 2600 ? 14 : $(internal:custom_ptz_wbk) == 2700 ? 15 : $(internal:custom_ptz_wbk) == 2800 ? 16 : $(internal:custom_ptz_wbk) == 2900 ? 17 : $(internal:custom_ptz_wbk) == 3000 ? 1 : $(internal:custom_ptz_wbk) == 3100 ? 18 : $(internal:custom_ptz_wbk) == 3200 ? 19 : $(internal:custom_ptz_wbk) == 3300 ? 20 : $(internal:custom_ptz_wbk) == 3400 ? 21 : $(internal:custom_ptz_wbk) == 3500 ? 7 : $(internal:custom_ptz_wbk) == 3600 ? 22 : $(internal:custom_ptz_wbk) == 3700 ? 23 : $(internal:custom_ptz_wbk) == 3800 ? 24 : $(internal:custom_ptz_wbk) == 3900 ? 25 : $(internal:custom_ptz_wbk) == 4000 ? 2 : $(internal:custom_ptz_wbk) == 4100 ? 26 : $(internal:custom_ptz_wbk) == 4200 ? 27 : $(internal:custom_ptz_wbk) == 4300 ? 28 : $(internal:custom_ptz_wbk) == 4400 ? 29 : $(internal:custom_ptz_wbk) == 4500 ? 8 : $(internal:custom_ptz_wbk) == 4600 ? 30 : $(internal:custom_ptz_wbk) == 4700 ? 31 : $(internal:custom_ptz_wbk) == 4800 ? 33 : $(internal:custom_ptz_wbk) == 4900 ? 34 : $(internal:custom_ptz_wbk) == 5000 ? 4 : $(internal:custom_ptz_wbk) == 5100 ? 35 : $(internal:custom_ptz_wbk) == 5200 ? 36 : $(internal:custom_ptz_wbk) == 5300 ? 37 : $(internal:custom_ptz_wbk) == 5400 ? 38 : $(internal:custom_ptz_wbk) == 5500 ? 9 : $(internal:custom_ptz_wbk) == 5600 ? 39 : $(internal:custom_ptz_wbk) == 5700 ? 40 : $(internal:custom_ptz_wbk) == 5800 ? 41 : $(internal:custom_ptz_wbk) == 5900 ? 42 : $(internal:custom_ptz_wbk) == 6000 ? 10 : $(internal:custom_ptz_wbk) == 6100 ? 43 : $(internal:custom_ptz_wbk) == 6200 ? 44 : $(internal:custom_ptz_wbk) == 6300 ? 45 : $(internal:custom_ptz_wbk) == 6400 ? 46 : $(internal:custom_ptz_wbk) == 6500 ? 6 : $(internal:custom_ptz_wbk) == 6600 ? 47 : $(internal:custom_ptz_wbk) == 6700 ? 48 : $(internal:custom_ptz_wbk) == 6800 ? 49 : $(internal:custom_ptz_wbk) == 6900 ? 50 : $(internal:custom_ptz_wbk) == 7000 ? 11 : $(internal:custom_ptz_wbk) == 7100 ? 51 : 30`

Sync PTZ 2 dials

WHEN
- When $(internal:custom_ptz2_state) changes

DOES
- internal · **custom_variable_set_value**: name `ptz2_gain`; create `yes`; value = `jsonpath($(internal:custom_ptz2_state), '$.gain') == "--" || jsonpath($(internal:custom_ptz2_state), '$.gain') == "" ? $(internal:custom_ptz2_gain) : fromRadix(jsonpath($(internal:custom_ptz2_state), '$.gain'), 10)`
- internal · **custom_variable_set_value**: name `ptz2_wbk`; create `yes`; value = `includes(jsonpath($(internal:custom_ptz2_state), '$.wb'), "K") ? fromRadix(substr(jsonpath($(internal:custom_ptz2_state), '$.wb'), 0, 4), 10) : $(internal:custom_ptz2_wbk)`
- internal · **custom_variable_set_value**: name `ptz2_wbcode`; create `yes`; value = `$(internal:custom_ptz2_wbk) == 2400 ? 12 : $(internal:custom_ptz2_wbk) == 2500 ? 13 : $(internal:custom_ptz2_wbk) == 2600 ? 14 : $(internal:custom_ptz2_wbk) == 2700 ? 15 : $(internal:custom_ptz2_wbk) == 2800 ? 16 : $(internal:custom_ptz2_wbk) == 2900 ? 17 : $(internal:custom_ptz2_wbk) == 3000 ? 1 : $(internal:custom_ptz2_wbk) == 3100 ? 18 : $(internal:custom_ptz2_wbk) == 3200 ? 19 : $(internal:custom_ptz2_wbk) == 3300 ? 20 : $(internal:custom_ptz2_wbk) == 3400 ? 21 : $(internal:custom_ptz2_wbk) == 3500 ? 7 : $(internal:custom_ptz2_wbk) == 3600 ? 22 : $(internal:custom_ptz2_wbk) == 3700 ? 23 : $(internal:custom_ptz2_wbk) == 3800 ? 24 : $(internal:custom_ptz2_wbk) == 3900 ? 25 : $(internal:custom_ptz2_wbk) == 4000 ? 2 : $(internal:custom_ptz2_wbk) == 4100 ? 26 : $(internal:custom_ptz2_wbk) == 4200 ? 27 : $(internal:custom_ptz2_wbk) == 4300 ? 28 : $(internal:custom_ptz2_wbk) == 4400 ? 29 : $(internal:custom_ptz2_wbk) == 4500 ? 8 : $(internal:custom_ptz2_wbk) == 4600 ? 30 : $(internal:custom_ptz2_wbk) == 4700 ? 31 : $(internal:custom_ptz2_wbk) == 4800 ? 33 : $(internal:custom_ptz2_wbk) == 4900 ? 34 : $(internal:custom_ptz2_wbk) == 5000 ? 4 : $(internal:custom_ptz2_wbk) == 5100 ? 35 : $(internal:custom_ptz2_wbk) == 5200 ? 36 : $(internal:custom_ptz2_wbk) == 5300 ? 37 : $(internal:custom_ptz2_wbk) == 5400 ? 38 : $(internal:custom_ptz2_wbk) == 5500 ? 9 : $(internal:custom_ptz2_wbk) == 5600 ? 39 : $(internal:custom_ptz2_wbk) == 5700 ? 40 : $(internal:custom_ptz2_wbk) == 5800 ? 41 : $(internal:custom_ptz2_wbk) == 5900 ? 42 : $(internal:custom_ptz2_wbk) == 6000 ? 10 : $(internal:custom_ptz2_wbk) == 6100 ? 43 : $(internal:custom_ptz2_wbk) == 6200 ? 44 : $(internal:custom_ptz2_wbk) == 6300 ? 45 : $(internal:custom_ptz2_wbk) == 6400 ? 46 : $(internal:custom_ptz2_wbk) == 6500 ? 6 : $(internal:custom_ptz2_wbk) == 6600 ? 47 : $(internal:custom_ptz2_wbk) == 6700 ? 48 : $(internal:custom_ptz2_wbk) == 6800 ? 49 : $(internal:custom_ptz2_wbk) == 6900 ? 50 : $(internal:custom_ptz2_wbk) == 7000 ? 11 : $(internal:custom_ptz2_wbk) == 7100 ? 51 : 30`

Sunday countdown

Fires the PP1 "Countdown" key's actions at 10:24:57 on Sunday. The actions are copied from that key, so re-point the key with tools/pp1-countdown.js and re-run this to keep them together. Change TIME in tools/pp1-countdown-trigger.js and re-run to move it.

WHEN
- At 10:24:57 on Sun

DOES
- propresenter · **specificPlaylistOperation**: specific_playlist_operation `trigger_index`; playlist_id `6551D76D-C095-40B9-94C5-9A97B7CE57FB`; index 3; cue_index `0`
- videohub · **route**: source 4; destination 13; ignore_lock `yes`
- hyperdeck · **rec**
- atem · **streamStartStop**: stream `true`

Poll stage layout

Writes the "Stage Screen" stage layout name into pp1_stage_layout every 2s. Exists because the ProPresenter module's own stagescreen_<uuid>_layout variable does not update on this rig. Rebuild with tools/pp1-stage-layout.js.

WHEN
- Every 2 s

DOES
- internal · **exec**: path `curl -s -m 2 http://10.23.0.204:1100/v1/stage/screen/E668F1AE-30EF-4207-BB51-4EE810A660D9/layout | sed -n 's/.*"name" *: *"\([^"]*\)".*/\1/p'`; timeout `5000`; targetVariable `pp1_stage_layout`

Sunday power on

Turns on Projectors and PA at 08:05:00 on Sunday. The commands are copied from the Home page's own keys with their toggle variable replaced by "on", so this can only ever turn things ON — a toggle here would switch off gear that was already on. Rebuild with tools/sunday-power-trigger.js.

WHEN
- At 08:05:00 on Sun

DOES
- internal · **exec**: path `bash -c 'nohup python3 /home/samuelbailey/Desktop/AV_Power_scripts/projector_power.py on >>/tmp/av_power.log 2>&1 &'`; timeout `15000`
- internal · **exec**: path `bash -c 'nohup python3 /home/samuelbailey/Desktop/AV_Power_scripts/speaker_power.py on >>/tmp/av_power.log 2>&1 &'`; timeout `15000`

Sunday tracks

Starts the tracks in Chrome on the PP1 iMac at 10:00:00 on Sunday with "chrome_tracks.sh play", which never pauses a player that is already going. Rebuild with tools/pp1-tracks-trigger.js.

WHEN
- At 10:00:00 on Sun

DOES
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh play`; timeout `8000`; targetVariable `chrome_tracks`

Sunday tracks fade

Fades the tracks out over four seconds and pauses them, so they are silent at 10:30 as the countdown ends; puts the volume slider back for next time. At 10:29:56 on Sunday. Rebuild with tools/pp1-tracks-trigger.js.

WHEN
- At 10:29:56 on Sun

DOES
- internal · **exec**: path `/home/samuelbailey/Desktop/AV_Power_scripts/chrome_tracks.sh fade`; timeout `15000`; targetVariable `chrome_tracks`

Sunday pre-service

Fires the PP1 "Pre-Service" key's actions at 10:00:00 on Sunday, then cuts the switcher to input 7 (PP1). Re-point the key with tools/pp1-preservice.js, then remove both pre-service triggers and re-run tools/pp1-preservice-trigger.js.

WHEN
- At 10:00:00 on Sun

DOES
- propresenter · **specificPlaylistOperation**: specific_playlist_operation `trigger_index`; playlist_id `6551D76D-C095-40B9-94C5-9A97B7CE57FB`; index 1; cue_index `0`
- atem · **program**: mixeffect 1; input 3

Sunday pre-service check

At 10:00:20 on Sunday, fires the "Pre-Service" key's actions again if ProPresenter is not showing "Pre-Service" — a second go when the 10:00:00 start did not land.

WHEN
- At 10:00:20 on Sun

ONLY IF
- internal · **variable_value**: variable `propresenter:active_presentation_name`; op ne; value `Pre-Service`

DOES
- propresenter · **specificPlaylistOperation**: specific_playlist_operation `trigger_index`; playlist_id `6551D76D-C095-40B9-94C5-9A97B7CE57FB`; index 1; cue_index `0`

Custom variables

Read in expressions as `$(custom:name)`. "Kept" values survive a restart.

Name	What it holds	Default	Kept
`pa_state`	FOH speaker outlets state: on/off/mixed/unreachable (polled 10s)		
`projector_state`	Projectors aggregate state: on/off/mixed/unreachable (polled 10s)		
`fader1_level`	House lights knob level 0-100 (Companion-tracked)	0	yes
`fader2_level`	Stage lights knob level 0-100 (Companion-tracked)	100	yes
`fader3_level`	Background lights knob level 0-100 (Companion-tracked)	0	yes
`fader4_level`	Wednesday look knob level 0-100 (Companion-tracked)	0	yes
`projector_action`	A custom variable	off	
`pa_action`	A custom variable	off	
`sys_temp`	CPU Temp on the Companion host (polled every 30s)		
`sys_load`	CPU Load on the Companion host (polled every 30s)		
`sys_mem`	Memory on the Companion host (polled every 30s)		
`sys_disk`	Disk on the Companion host (polled every 30s)		
`sys_throttle`	Power on the Companion host (polled every 30s)		
`sys_net`	Internet on the Companion host (polled every 30s)		
`sys_uptime`	Uptime on the Companion host (polled every 30s)		
`vh_dest`	A custom variable		
`sys_disk_free`	A custom variable		
`sys_ip`	A custom variable		
`sys_rw`	A custom variable		
`ptz_speed`	PTZ pan/tilt drive speed 1-24, set by the Speed knob	24	yes
`ptz_zspeed`	PTZ zoom speed 0-7, derived from ptz_speed	7	yes
`ptz_fspeed`	PTZ focus speed 0-7, derived from ptz_speed	7	yes
`ptz_preset`	PTZ preset number on the Preset dial, 1-254	200	yes
`ptz_last`	PTZ preset most recently recalled or saved from the deck	1	yes

Name	What it holds	Default	Kept
`ptz_armed`	1 while the next PTZ preset press saves instead of recalling	0	yes
`ptz_atem_input`	ATEM input number the PTZ camera is on, for tally on the Stop key	2	yes
`ptz_state`	PTZ camera state as JSON, polled every second		
`ptz_tspeed`	PTZ tilt drive speed 1-20, derived from ptz_speed	20	yes
`ptz_track`	PTZ AI-tracking settings as JSON from the camera web API: tracking, body, mode, speed, sensitivity, placement, headroom, lost		
`ptz2_speed`	PTZ 2 pan/tilt drive speed 1-24, set by the Speed knob	24	yes
`ptz2_tspeed`	PTZ 2 tilt drive speed 1-20, derived from ptz_speed	20	yes
`ptz2_zspeed`	PTZ 2 zoom speed 0-7, derived from ptz_speed	7	yes
`ptz2_fspeed`	PTZ 2 focus speed 0-7, derived from ptz_speed	7	yes
`ptz2_preset`	PTZ 2 preset number on the Preset dial, 1-254	1	yes
`ptz2_last`	PTZ 2 preset most recently recalled or saved from the deck	1	yes
`ptz2_armed`	1 while the next PTZ 2 preset press saves instead of recalling	0	yes
`ptz2_track`	PTZ 2 AI-tracking settings as JSON from the camera web API: tracking, body, mode, speed, sensitivity, placement, headroom, lost		
`ptz2_atem_input`	ATEM input number the PTZ 2 camera is on, for tally on the Stop key	1	yes
`ptz2_state`	PTZ 2 camera state as JSON, polled every second: online, pan, tilt, zoom, focus, ae, wb, backlight, power, menu		
`chrome_tracks`	Chrome tracks player on the iMac: Playing, Paused, Toggled or -- (from chrome_tracks.sh)		
`ptz_gain`	PTZ gain dial 0-15, set by the Gain knob and kept in step with the camera	0	yes
`ptz_wbk`	PTZ white balance dial in kelvin, 2400-7100, set by the WB knob and kept in step with the camera	4700	yes
`ptz_wbcode`	PTZ VISCA white balance code for ptz_wbk, derived whenever the dial moves	31	yes
`ptz_match`	PTZ result of the last Match press as JSON from ptz_web.py: online, from, copied, left		
`ptz2_gain`	PTZ 2 gain dial 0-15, set by the Gain knob and kept in step with the camera	0	yes

Name	What it holds	Default	Kept
`ptz2_wbk`	PTZ 2 white balance dial in kelvin, 2400-7100, set by the WB knob and kept in step with the camera	4700	yes
`ptz2_wbcode`	PTZ 2 VISCA white balance code for ptz_wbk, derived whenever the dial moves	31	yes
`ptz2_match`	PTZ 2 result of the last Match press as JSON from ptz_web.py: online, from, copied, left		
`ptz_look`	PTZ saved look as JSON from ptz_web.py look: online, saved, applied, left, and its wb, ae, shutter, iris, gain, sharp	{"online": "OK", "saved": "2026-10-11 09:42:46", "applied": 0, "left": 0, "wb": "4700K", "ae": "Auto", "shutter": "1/60", "iris": "F1.8", "gain": "0", "sharp": "0"}	yes
`ptz2_look`	PTZ 2 saved look as JSON from ptz_web.py look: online, saved, applied, left, and its wb, ae, shutter, iris, gain, sharp	{"online": "OK", "saved": "2026-10-10 09:31:15", "applied": 4, "left": 0, "wb": "4700K", "ae": "Auto", "shutter": "1/60", "iris": "F1.8", "gain": "0", "sharp": "0"}	yes
`pp1_main_label`	Which PP1 preset last fired ProPresenter Main Timer — names the left strip readout	Speaker	yes
`pp1_stage_layout`	ProPresenter "Stage Screen" stage layout name, polled every 2s	Full	yes
`chrome_tracks_volume`	YouTube Music's volume slider on the PP1 iMac, as NN% or -- (from chrome_tracks.sh)		
`chrome_tracks_title`	The song YouTube Music is on, on the PP1 iMac, or -- (from chrome_tracks.sh)		
`ptz_pan_at`	PTZ time of the last Pan knob click, in ms	0	
`ptz_tilt_at`	PTZ time of the last Tilt knob click, in ms	0	
`ptz_zoom_at`	PTZ time of the last Zoom knob click, in ms	0	
`ptz_focus_at`	PTZ time of the last Focus knob click, in ms	0	
`ptz_pan_dir`	PTZ pan direction the knob is driving: 1 left, 2 right, 3 stopped	3	
`ptz_tilt_dir`	PTZ tilt direction the knob is driving: 1 up, 2 down, 3 stopped	3	
`ptz2_pan_at`	PTZ 2 time of the last Pan knob click, in ms	0	
`ptz2_tilt_at`	PTZ 2 time of the last Tilt knob click, in ms	0	
`ptz2_zoom_at`	PTZ 2 time of the last Zoom knob click, in ms	0	
`ptz2_focus_at`	PTZ 2 time of the last Focus knob click, in ms	0	
`ptz2_pan_dir`	PTZ 2 pan direction the knob is driving: 1 left, 2 right, 3 stopped	3	
`ptz2_tilt_dir`	PTZ 2 tilt direction the knob is driving: 1 up, 2 down, 3 stopped	3	
`ptz_recall`	PTZ result of the last preset recall as JSON from ptz_web.py: online, preset, kept, left		
`ptz2_recall`	PTZ 2 result of the last preset recall as JSON from ptz_web.py: online, preset, kept, left		

Connections

atem bmd-atem 4.4.0

Setting	Value
`host`	(empty)
`modelID`	0
`presets`	0
`fadeFps`	10
`enableCameraControl`	true
`pollTimecode`	false
`bonjourHost`	10.23.0.34:9910
`autoModelID`	26
`autoModelName`	Television Studio HD8

videohub bmd-videohub 3.2.1

Setting	Value
`host`	192.168.10.150
`take`	false
`inputCount`	16
`outputCount`	16
`monitoringCount`	0
`serialCount`	0
`bonjourHost`	10.23.0.21:9990
`port`	9990

propresenter renewedvision-propresenter-api 1.1.1

Setting	Value
host	10.23.0.204
port	1100
use_sd	no
sdport	1025
indexOfClockToWatch	0
GUIDOfStageDisplayScreenToWatch	0
sendPresentationCurrentMsgs	no
typeOfPresentationRequest	auto
clientVersion	701
looksPolling	disabled
timerPolling	disabled
control_follower	no
followerhost	10.23.0.11
followerport	1100
pass	(credential: see the vault)
sdpass	(credential: see the vault)
midi_base_page	1
timeout	2000
exta_debug_logs	false
custom_timer_format_string	hh:mm:ss

SQ allenheath-sq 3.1.0

Setting	Value
host	10.23.0.191
model	SQ7
faderLaw	LinearTaper
talkbackChannel	0
midiChannel	1
retrieveStatusAtStartup	full
verbose	false

grandma2 malighting-grandma2 2.3.0

Setting	Value
`host`	10.23.0.99
`user`	administrator
`pass`	(credential: see the vault)

MA2osc generic-osc 3.0.0

Setting	Value
`protocol`	udp
`listen`	true
`host`	10.23.0.99
`targetPort`	8000
`feedbackPort`	9000

novastar novastar-controller 2.6.4

Setting	Value
`host`	10.23.0.23
`modelID`	vx6s
`port`	5200

visca ptzoptics-visca 4.0.0

Setting	Value
`host`	10.23.0.181
`port`	5678
`debugLogging`	false

visca2 ptzoptics-visca 4.0.0

Setting	Value
`host`	10.23.0.196
`port`	5678
`debugLogging`	false

pco planningcenter-serviceslive 2.7.0

Setting	Value
`applicationid`	(credential: see the vault)
`secretkey`	(credential: see the vault)
`parentfolder`	(empty)
`servicetypeid`	919088
`perpage`	4
`polling`	true
`pollingRate`	3000
`servicetypeid_polling`	919088
`verbose`	false

dante audinate-dantecontroller 1.1.2

Setting	Value
`ip`	10.23.0.242
`interval`	5000
`timeoutInterval`	15000
`verbose`	false

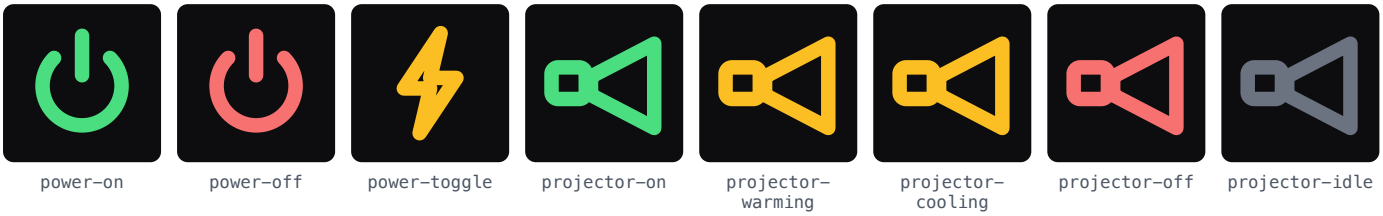
hyperdeck bmd-hyperdeck 3.2.0

Setting	Value
`host`	10.23.0.206
`modelID`	hdStudioMini
`reel`	A001
`timecodeVariables`	notifications
`pollingInterval`	500
`warnRemoteDisabled`	true

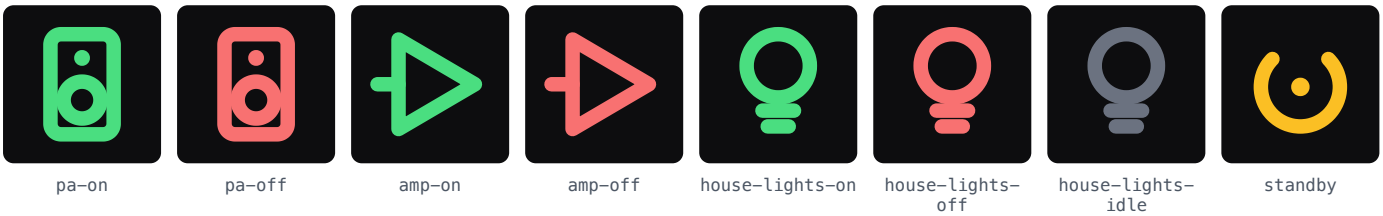
Image library

Each is published as `$(image:name)`; a key's image layer names one, so changing an image here changes every key that uses it. Built by `npm run build` in this repo.

power



power-on power-off power-toggle projector-on projector-warming projector-cooling projector-off projector-idle



pa-on pa-off amp-on amp-off house-lights-on house-lights-off house-lights-idle standby

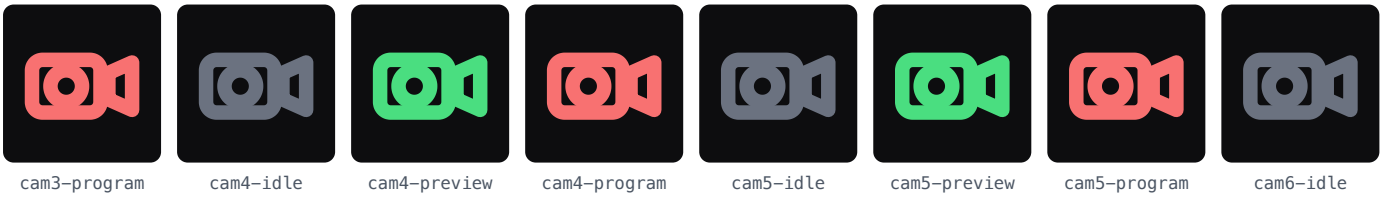


mains-power

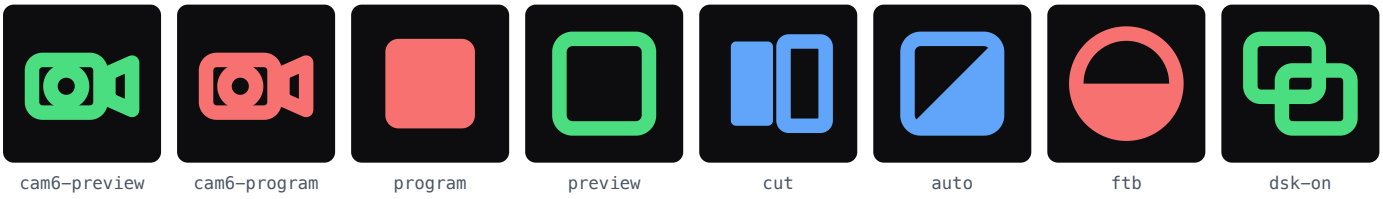
video



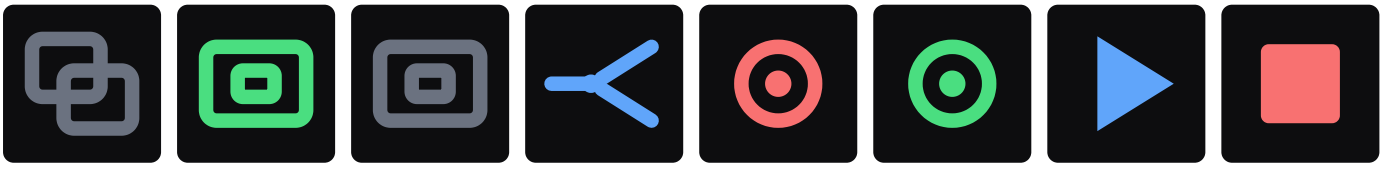
cam1-idle cam1-preview cam1-program cam2-idle cam2-preview cam2-program cam3-idle cam3-preview



cam3-program cam4-idle cam4-preview cam4-program cam5-idle cam5-preview cam5-program cam6-idle



cam6-preview cam6-program program preview cut auto ftb dsk-on



dsk-off key-on key-off aux tally-live tally-preview macro-run macro-stop


transition


still


brightness

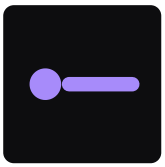
routing

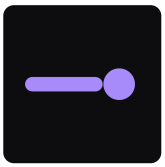

route


take


route-lock


unlock


source


destination


matrix

present


slide-next


slide-prev


slide-first


slide-last


clear-all


clear-slide


clear-props


clear-audio


logo


stage-display


message


timer-start


timer-stop


timer-reset


countdown


media


prop


playlist


focus-next


focus-prev


cue-next


cue-back


executor


clear-messages


clear-announce


clear-media


clear-video


green-wall


water-calm


water-storm


thunder

folders


folder-power


folder-present


folder-lighting

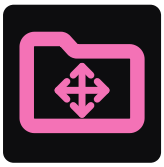

folder-video


folder-audio

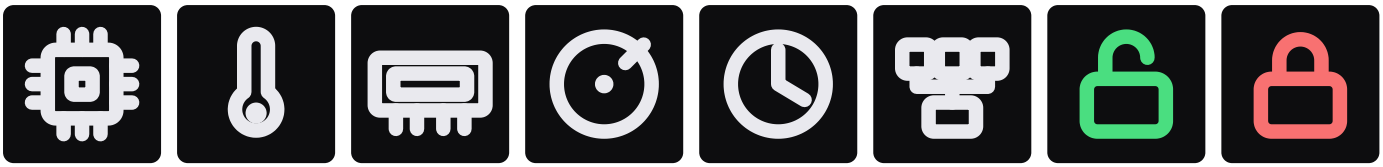

folder-routing


folder-system


folder-wireless

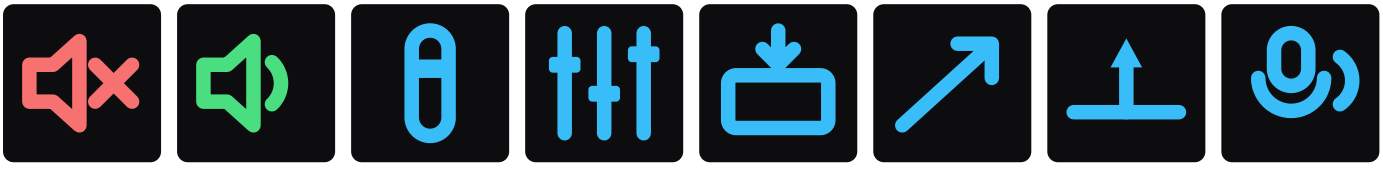

folder-ptz

system



cpu temperature memory disk uptime network storage-ok storage-locked

audio



mute-on mute-off fader mix scene-recall gain aux-send talkback



pfl phantom meter dca mono mains

wireless

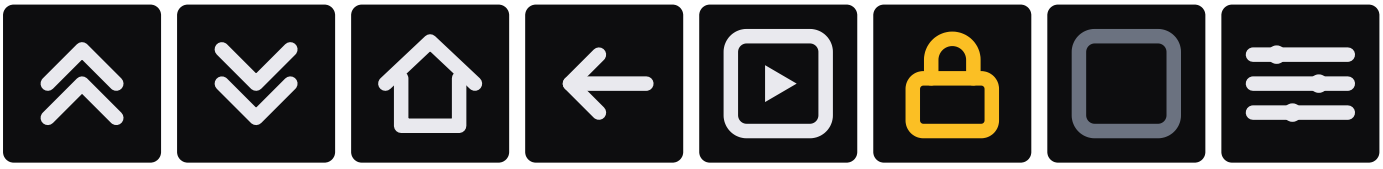


mic-on mic-off mic-muted mic-alert battery-0 battery-1 battery-2 battery-3



battery-4 rf-0 rf-1 rf-2 rf-3 rf-idle tx-fault

utility

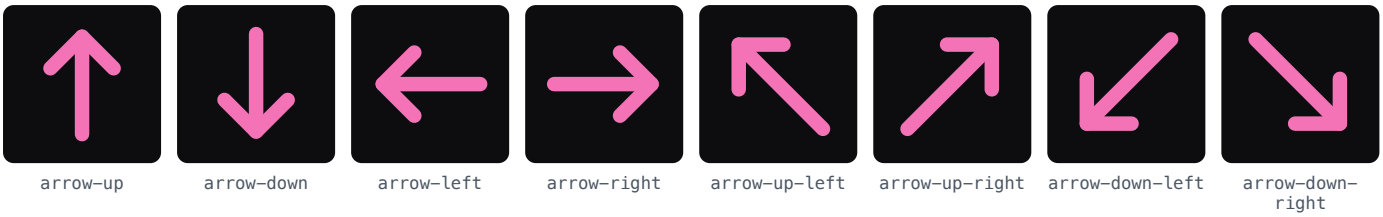


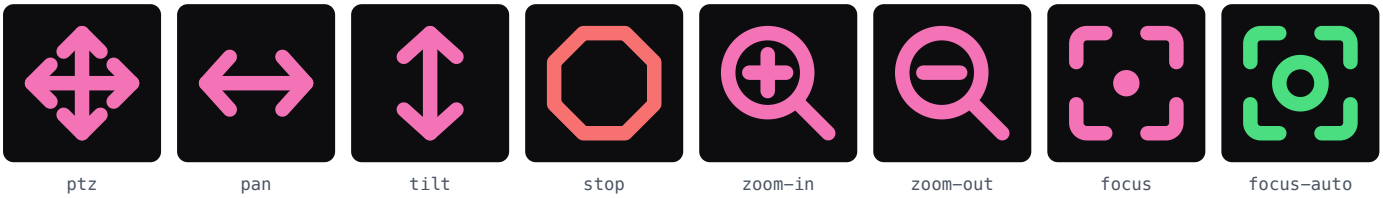
page-up page-down home back macro lock-surface blank settings

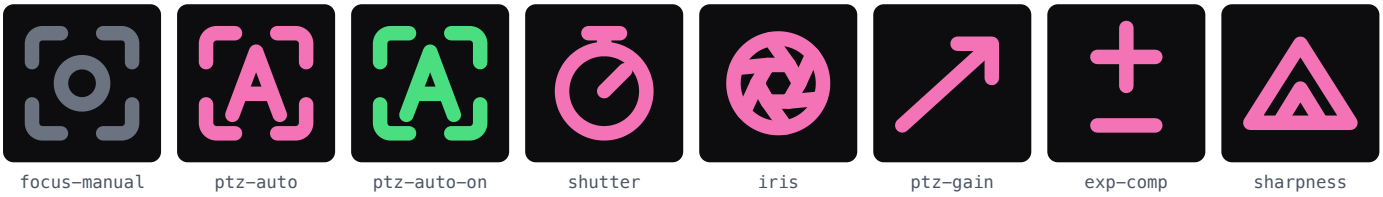


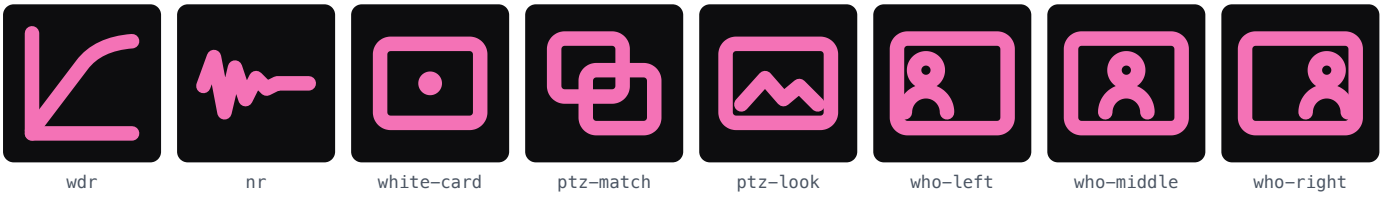
alert

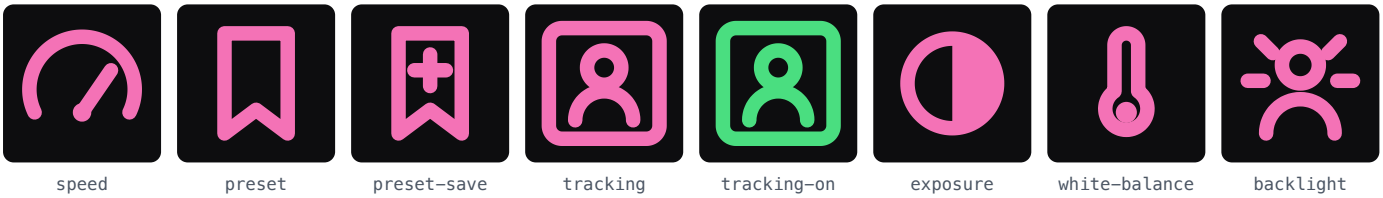
camera

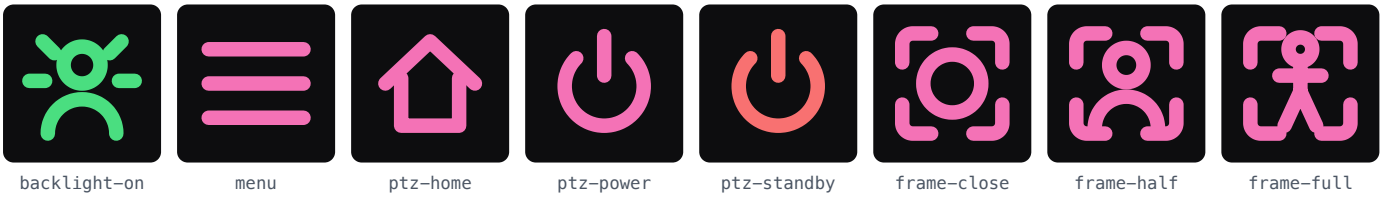


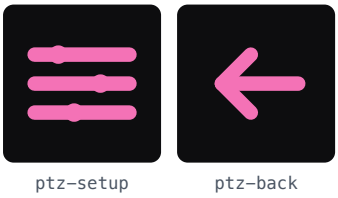






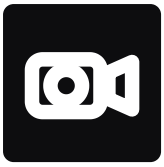






contrast




camera–paper


camera–ink


media–paper


media–ink


macro–run–paper


macro–run–ink


message–paper


message–ink


cut–paper


cut–ink


auto–paper

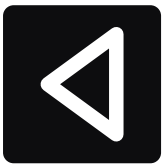

auto–ink


clear–paper


clear–ink


stage–display–paper


stage–display–ink


slide–prev–paper


slide–prev–ink


slide–next–paper


slide–next–ink


clear–messages–paper


clear–messages–ink


clear–props–paper


clear–props–ink


clear–announce–paper


clear–announce–ink


clear–media–paper


clear–media–ink


clear–video–paper


clear–video–ink


ftb–paper

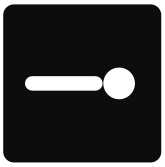

ftb–ink


executor–paper

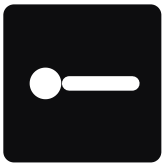

executor–ink

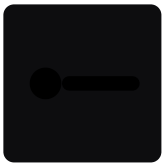

fader–paper


fader–ink


destination–paper


destination–ink


source–paper


source–ink


dca–paper


dca–ink


stop–paper


stop–ink


preset–save–paper


preset–save–ink

Engineering notes: The rig

What is in the room, how it is addressed, and which Companion page drives it.

Addresses, pages and device identities here were **read off the live rig on 2026-08-28**, not reconstructed — where this file and a `tools/` builder disagree, this file is what the rig actually does. Anything still marked `TODO(sam)` cannot be read over the network at all and needs hands on the equipment.

Control

Companion	Build `5.0.7+9763-stable-cec2f88e6f` (2026-10-27) on a Raspberry Pi 5, 8 GB, Debian 13 CcompanionPi image (`internal:hostname` reads `CompanionPi`). Runs as user `companion`; settings in `/etc/companion/config.yaml`, which must keep `enableShellCommandSupport: true`
Address	**http://10.23.0.242:8000** on the AV LAN — see Network below. Not reachable off-site
Updates	**Companion updates itself unattended.** On 2026-08-28 it went 5.0.3 → 5.0.4 and restarted on its own; Pi uptime was 40d, so the Pi did not reboot. The same thing landing on a Sunday morning takes the deck down mid-service
Admin API	tRPC over a WebSocket at `/trpc`, **unauthenticated** — anyone on the AV LAN can read the config and the log
Surfaces	Stream Deck + and Stream Deck XL. The XL grid is 9 columns; row 0 is the folder row on every page, row 4 the touchstrip, row 5 the encoders
Config in/out	`node tools/rig.js export <out.json>` reads the live config; `import` pushes selected sections. See the header of that file for why it exists and why it never touches `connections` or `userconfig`.
Reading it without the tool	The export half is one plain HTTP GET: `curl -s http://10.23.0.242:8000/int/export/full | gunzip > rig.json`. Useful when you are on the LAN with no checkout
Backup	`tools/rig.js export` **is** the backup. Run it before any change. **The file carries the ProPresenter and grandMA2 passwords in clear text** — treat an export as a credential, not as a document: keep it in the private Drive folder, never here. The Pi's nightly `companion-backup.timer` (03:17) writes to a USB drive labelled `storage`, which **was not plugged in on 2026-10-27** (fault 7). The Recovery Guide in `manual/` has the restore steps

`tools/rig.js` talks tRPC over a WebSocket at `/trpc`, the same way Companion's own web UI does. The rig address is hardcoded there on purpose so the tool can never be pointed at the wrong Companion by a typo.

Pages

Eighteen. The folder row across the top of every page but Videohub holds nine: Home, Power, PP1, MA2, ATEM, SQ7, VW, PTZ, System. The other nine are sub-pages, opened from a key. Read off the live rig on 2026-10-27; the Reference in `manual/` draws every one of them, key by key, and is regenerated from the rig, so it wins wherever this table has gone stale.

#	Page	Drives	Opened from	Notes
1	Home	nothing — status only	folder row	Projectors, PA, Dante, PROGRAM, PREVIEW, Stream, Record, three ATEM macros. The deck resumes on its last page, so check where it is after a restart
2	Power	Projectors, PA	folder row	One state-aware toggle per system; direction is worked out from polled state, not a step counter

#	Page	Drives	Opened from	Notes
3	PP1	ProPresenter	folder row	Clear keys, Next/Previous, **Freeze Slide** at r1c8 (`tools/pp1-freeze.js`, released by **All Screens** at r2c7), **Pre-Service** at r2c3 and **Countdown** at r2c6 (both trigger by playlist index: re-run their tools if items are added above them), Stage layout swap, and the Hosting / Speaker / Worship timers with their readouts on the strip
4	MA2	grandMA2 lighting	folder row	Executors 101–109 across row 1, Go Next / Go Back / BLACKOUT below, four encoders on row 5
5	ATEM	Television Studio HD8	folder row	Program bus over preview bus, CUT and AUTO at column 8, guarded Stream, Record (the HyperDeck), three macros
6	SQ7	Allen & Heath SQ7	folder row	DCA 1–8 with level and mute state, Mute DCAs key, Dante at r2c8. Stream / Foyer / MAIN readouts on the touchstrip
7	VW	Blackmagic Videohub + LED wall	folder row	Five destinations over four sources: tap a destination to select it, tap a source to route. Two direct routes, four Videohub readouts, a brightness encoder for the wall, and **Videohub** at r1c8
8	System	the Pi itself	folder row	CPU temp, load, memory, disk, power, internet, address, uptime, Companion, storage. Fed by an `internal: exec` trigger
9	PTZ	nothing: a chooser	folder row	Opens CAM 1, CAM 3, their Setup pages, Worship and Sermon. The camera keys are captioned with each camera's ATEM input
10	CAM 1	the PTZ on 10.23.0.196	PTZ	Presets, save, nudge, zoom/focus, tracking, and pan/tilt/zoom/focus/speed/preset encoders. Fed by `ptz_state.py` every second and `ptz_web.py` every 5 s
11	CAM 1 Setup	the same camera	PTZ, CAM 1	Exposure, white balance, backlight, power, tracking mode and body framing
12	CAM 3	the PTZ on 10.23.0.181 (the switcher's "Cam 2" since 2026-10-26)	PTZ	As CAM 1, built by the same code
13	CAM 3 Setup	the same camera	PTZ, CAM 3	As CAM 1 Setup
14	Tracks	walk-in music on the PP1 iMac	PP1 r2c4	Play/pause, previous, skip, fade out, title and volume (`tools/pp1-tracks.js`)
15	Worship	both PTZs	PTZ, Sermon	Both cameras side by side: presets, save, speed, autofocus, pan/tilt/zoom knobs
16	Sermon	both PTZs	PTZ, Worship	Both cameras for the sermon: tracking, framing, who to follow, and pan/tilt/zoom by hand
17	Dante	nothing: read only	Home r1c2, SQ7 r2c8	Every Dante source into the SQ7 and whether its routes are up (`tools/dante-page.js`)
18	Videohub	the whole router	VW r1c8	Every output over every input, captioned with the router's own labels. **No folder row**: a Back key at r0c8 returns to VW (`tools/switch-page.js`)

Both cameras run one page layout. Pages 11 and 12 are emitted by the same `buildPage` / `buildSetupPage` calls as 9 and 10, and the build asserts they are identical once the connection, host, page numbers and variable prefix are normalised away — so they cannot drift apart. See `tools/ptz2-page.js`.

The four PTZ pages share one folder-row slot. The row is exactly nine columns for nine pages and was already full, so pages 10, 11 and 12 are sub-pages reached from page 9. The swap key at row 1 column 8 of each run page

moves between the two cameras, and is captioned with that camera's **ATEM input number** so the deck and the switcher agree. It took the slot that held preset 6, which was unnamed and saved at preset 5's identical position.

There is **no Mics page any more.** The four Shure connections still run, but nothing on the deck reads them.

Planning Center

Nothing on the deck drives or shows Planning Center any more. The NEXT / PREV step keys on Worship and Sermon and the four readouts on Home row 3 were added on 2026-10-06 and taken off on 2026-10-10 (Sam: "take it off the stream deck"). Their builder is in git history at `bf094f0` (`src/pco-live.js`).

Planning Center now lives on the Service Health board's Services page (http://10.23.0.242:8080/services, Decima-Labs/service-health): the current item, the whole order of service with Planning Center's times, notes and teams.

Connection	`pco` — Bitfocus `planningcenter-serviceslive` v2.7.0, polling every 3 s, pinned to service type **919088** (Sunday Service). **Kept:** the board reads its variables for the current item and time left
Advancing LIVE	With the step keys gone, LIVE moves only when someone advances it in Planning Center itself
Credential	Sam's Personal Access Token "pco-lint (read-only plan linter)", shared with the `pco-lint` report (Decima-Labs/pco-lint). It lives in `~/.config/pco/lint.env` on Sam's Mac, in that repo's secrets, in this Companion connection, and since 2026-10-10 in `~/.config/service-health/pco.env` on the Pi (0600) for the board. **Like every connection here it is readable in Companion's export and log by anyone on the AV LAN** (fault 6 below). Revoke it at api.planningcenteronline.com/personal_access_tokens if the Pi is ever compromised
Planning Center's own settings	Two things found on 2026-10-10 that are fixes there, not here: the organisation's time zone is `America/Guatemala` (no daylight saving, so absolute times are an hour late for half the year), and "Prayer Works" is marked as part of the service, which puts Worship at 11:00 in Planning Center's times

Dante

Page 17, Dante, a sub-page of SQ7, opened by the **Dante** folder at Home r1c2 (beside Projectors and PA, so the Stream Deck + and the iPad deck show it too) and at SQ7 r2c8. Both go red when the patch is not normal. On the rig since 2026-10-11, `tools/dante-page.js`, logic in `src/dante.js`. It reports and changes nothing: no key on it has an action beyond the folder row.

Connection	`dante` — Bitfocus `audinate-dantecontroller` **1.1.2**, bound to the Pi's 10.23.0.242, 5 s sweep, a device dropped after 15 s unheard. The module can make and clear routes; nothing on the deck asks it to
Row 1	One key per source into the SQ7: Leads, BGV 1-4, BGV+Host, Lav 1-2, Tracks, PP1, Devan, Josue, then **Routing**. Green = on the network with every route subscribed; amber = on the network, a route down; red = a source the desk needs is missing. The three laptops show "Not here" in slate when away, and BGV+Host shows "On XLR" (below). Routing is green only when every required route is up, and the SQ7's Dante key goes red with it
Row 2	Each device's sample rate and Dante latency under its source; the SQ7's own card at r2c8
r3c0	Sample rate: red if any device present is not at 48 kHz
Not shown	The PTP clock leader — module 1.1.2 cannot read it. On 2026-10-11 the SQ7 card led and the rest followed

The routes are the desk's own subscriptions, read off the network on 2026-10-11, not a design. Repatch in Dante Controller and `SOURCES` in `src/dante.js` has to follow, or the page reports the new patch as broken.

Dante device	Address	Into SQ7
`AllenHth-0e0184` — SQ7 Dante card, 64×64, clock leader	10.23.0.70	—
`ULXD4Q-LEADS-1-4`	10.23.0.14	1-4

Dante device	Address	Into SQ7
`ULXD4Q-BGV-1-4`	10.23.0.107	5-8
`ULXD4Q-BGV-5-6-HOST-1-2`	not on the network — feeds the desk over XLR	9-12 subscribed but unused
`ULXD4D-LAV-1-2`	10.23.0.143	13-14
`TCBs-MacBook-Pro-Tracks` — Dante Virtual Soundcard	10.23.0.230	15-22
`iMac-PP1` — Dante Virtual Soundcard	10.23.0.204	23-24 and 45-46
`Devans-Mac-mini`, `Josues-MacBook-Pro`	away on 2026-10-11	33-41 and 51-64; 42-44 and 47-49

The receivers' Dante interfaces have their own addresses, separate from the :2202 control addresses in Devices below. **The BGV+Host receiver is not on Dante, by design**: it reaches the desk over its XLR outputs (Sam, 2026-10-11). Its control port answers at 10.23.0.253 and names a Dante interface `ULXD4Q-BGV-5-6-HOST-1-2` that is not cabled, and the SQ7 still holds Dante subscriptions on inputs 9-12 to it, left over and harmless. The page shows it "On XLR" in slate and leaves it out of the Routing check; if it is ever cabled for Dante it lights up like the others.

Dante Controller will not see these devices over WiFiman or any routed VPN: it finds them by multicast. Off site, use this page through Companion's web UI, or remote into a Mac on the LAN.

Devices

Read off the live Companion on 2026-08-28 and confirmed against each device on the LAN.

Device	Role	Address
ATEM Television Studio HD8	Vision mixer and **stream** (recording moved to the HyperDeck on 2026-10-26)	**10.23.0.34** (2026-10-26), control on UDP 9910. Also answers the Blackmagic routing protocol on TCP 9990, which is how its identity was confirmed: unique ID `1947523eab1a4015aa81b6a4ae2dfe43`, the same uuid stamped into the recordings. That same port reports its video standard — `1080p59.94` — and its aux routing, and accepts routing changes
Blackmagic Micro Videohub, 16×16	Routing to the screens **and to the switcher's inputs 1–6**	**10.23.0.21**:9990. Re-patched on 2026-10-26: see Video routing below
HyperDeck Studio Mini	**Records the service** since 2026-10-26, from Videohub OUT 13	**10.23.0.206**:9993, Companion connection `hyperdeck` (bmd-hyperdeck 3.2.0). SD cards only. See recorder.md
Allen & Heath SQ7	Audio console	**10.23.0.191** (2026-10-26)
grandMA2	Lighting	**10.23.0.99** (2026-10-26) for the console login; OSC out to :8000 and feedback in on :9000
Shure ULXD4Q × 3 + ULXD4D × 1	Wireless, **14** channels — not 16	`10.23.0.20` "BGV 1-4" (Q), `10.23.0.23` "Lead 1-4" (Q), `10.23.0.253` "BGV+Host" (Q), `10.23.0.214` "Preach" — **a ULXD4D, two channels only**, Lav 1 and Lav 2. All on :2202
ProPresenter	Presentation, on the PP1 iMac	**10.23.0.204**:1100 (2026-10-27; it was 10.23.0.111). A follower is configured at 10.23.0.11 but `control_follower` is off, and nothing is listening there
NovaStar VX6S	LED wall processor	**10.23.0.23**:5200 (was .19 until August 2026)

Device	Role	Address
FoMaKo UV602 PTZ — **CAM 3** pages, shown as **CAM 2**	Camera, VISCA over IP, **ATEM input 2 since 2026-10-26** (was 3). The chooser, tally and captions read `ptz_atem_input`; the pages keep their CAM 3 names	**10.23.0.181**:5678, with an unauthenticated HTTP API on :80 that `ptz_web.py` drives
FoMaKo UV602 PTZ — **CAM 1**	Camera, added 2026-08-28, ATEM input 1	**10.23.0.196**:5678, same API
JVC GY-HM600 static — **CAM 2**	Fixed, unmanned. **Not on the switcher since 2026-10-26**: both camera inputs are the PTZs "for this moment" (Sam)	Not on the network: it has no network option at all, and its only remote input is LANC. Goes HD-SDI into the switcher
Raspberry Pi	Companion host	10.23.0.242

Addresses drift. Every one of these except the Pi had moved at least once by 2026-08-28, and three of the four Shure receivers were pointing at dead hosts. The ATEM and the Videohub survived it only because their modules re-discover over Bonjour; the Shure module has no such fallback, which is why those were the ones that broke. The Videohub's stored `host` is still `192.168.10.150`, an address on a subnet that does not exist here — it works purely on Bonjour. **Check the addresses whenever a page goes quiet, and prefer a pinned address over a discovered one.**

Network

10.23.0.x is a dedicated AV LAN — church production gear only, nothing else on it. It covers roughly half the building, with a direct line to backstage.

That is the right design, and it is also why none of this is readable from off-site: being on church wifi is not enough, you have to be on that LAN. Anything that needs the live Companion (`tools/rig.js export`, reading connection addresses, reading the log) has to happen on site.

Being on the LAN is still not enough for everything. Companion exposes no variable for the ATEM's video standard, so the 59.94 check cannot be automated from here at all — it needs ATEM Software Control or the switcher's front panel. The same goes for camera menus and for the SQ7's output routing. On-site splits into two jobs, and only one of them is scriptable.

Open configuration faults

Found by reading the live config on 2026-08-28. None of these are network problems, so none of them will fix themselves.

1. **The Record key has no two-step guard.** Home `2,3` and ATEM `3,1` are a single press straight into the HyperDeck's `recordStopToggle` (the ATEM's `recordStartStop` before 2026-10-26). The Stream key beside it has the guard — two steps, arm then fire — and Record does not. This is exactly the 7.6 s gap from 2026-08-23: one stray press stops the recording. The record *state* indicator did ship, on both keys, so the other half of that fix is in.
2. ~~**Four VW feedbacks never evaluate.**~~ **Fixed 2026-10-26.** The source row's highlight built a variable name out of `custom_vh_dest`, which Companion's expression parser rejects. VW and the Videohub page now use the bmd-videohub module's own selected destination (`select_destination`, `route_source`, and the `selected_*` feedbacks), so there is no expression to fail, and both pages share one selection. `vh_dest` is unused. After a Companion restart nothing is selected, and an input press then sends nothing: checked on the rig by watching the router while pressing one.

3. ~~**The PTZ cannot be cut to from the deck.**~~ **Gone with the 2026-10-26 re-patch**: the two PTZs are ATEM inputs 1 and 2 now, and the ATEM page's buses carry inputs 1–7 and Media Player 1. Inputs 7 and 8 are not fed by the router.
4. ~~**Two pages depend on scripts that are not backed up.**~~ **Backed up 2026-10-27** in `pi/`, passwords swapped for their 1Password entries; refresh with `node tools/pi-backup.js` whenever a script on the Pi changes. They still live in a personal Desktop folder on the Pi, `/home/samuelbailey/Desktop/AV_Power_scripts`, which every key calls by full path.
5. **The PTZ poll runs at 1 Hz, always.** `trigger-ptz-poll` fires every second whether or not anyone is on the PTZ page. The Pi is not struggling (load 0.02, 62 °C, 16% memory, 3% disk), so this is a tidiness point rather than an urgent one.
6. **Credentials are readable by anyone on the AV LAN.** Companion echoes each module's config into its log on connect, passwords included, and both the log and the config export are served without authentication. Record where a credential lives, assume anything in a Companion connection is visible on this LAN, and keep export files somewhere that reflects that.

Found by the manual build on 2026-10-27:

1. **The nightly Companion backup is not running.** `companion-backup.timer` fires at 03:17 and fails with "Dependency failed": it needs `/mnt/storage`, a USB drive labelled `storage`, and none is plugged in. Until one is, the only backups are exports taken by hand.
2. **The PA relay's password is effectively public.** It is the same word as the PP1 iMac's account name, which `chrome_tracks.sh` (in this public repo) has to name. Change it on the relay and in 1Password.
3. **The grandMA2 connection's password is a short dictionary word.** Change it on the lighting PC and in the Companion connection.
4. **Two notes on the rig have gone stale.** The *Sunday pre-service* trigger's note says it cuts to input 7; it cuts to input 3 (PP1) since 2026-10-26. The PP1 *Countdown* key's note says it starts the ATEM recording; it records on the HyperDeck and starts the stream.
5. **Companion lists 25 surfaces besides the two Stream Decks**: one Service Health deck session per browser that has opened the board's Deck tab, and four "timing probe" satellites left from testing. Harmless, but the stale ones clutter Surfaces.

Video routing

The rack moved into the back on 2026-10-26 and the Videohub was re-patched. It now sits in front of the switcher as well as behind it: router outputs 1–6 feed ATEM inputs 1–6, so a route to OUT 1–6 changes what the switcher sees. The ports, by the labels Sam gave them that day (read over :9990):

In	Label	Out	Label
1	Aux 1 (ATEM)	1–6	ATEM 1–6
2	Multiview	7	Output 7 (unlabelled)
3	Aux 2 (ATEM)	8	Backwall Projector
4	Program	9, 10	Mini Screen 1, 2
5, 6	Input 5, 6 (unlabelled)	11	Seetec Mono
7	Camera 1	12	Output 12 (unlabelled; the off-air test port)
8	Camera 2	13	Deck Recorder
9–13	Input 9–13 (unlabelled)	14	LED Wall
14	Words Overlay	15	Left Projector
15, 16	PP1B, PP1	16	Right Projector

On the ATEM that makes input 1 "Cam 1", 2 "Cam 2", 3 "PP1", 4 "PP1B", 5 "Words Overlay". The Sunday 10:00 pre-service cut moved from input 7 to 3 the same day. On 2026-10-26 the wall and both projectors were taking PP1 straight from the router, not from an ATEM aux.

The deck reads these labels live, so renaming a port in Videohub Setup renames its key. VW's curated keys are by number (`src/vw.js`), so a re-patch means re-running `tools/switch-page.js`, which refuses a VW page it does not recognise.

Credentials do not go in this file. Record **where** they live — a password manager entry name — not what they are.

The ISO problem

The HD8 is the **plain** variant, and the HyperDeck that records since 2026-10-26 takes one SDI feed, the program — so there is still no second angle. That has a direct consequence for post, and it is the single biggest constraint on making clips:

> Every clip inherits whatever the operator cut to at that moment. There is no second angle to fall back on, and no way to recover a shot that was framed badly or was on the wrong camera.

This bit us on 2026-08-23: the best line in the sermon landed while the switcher was on the house-wide, so the vertical crop for that clip had to follow the speaker across 1074 px of stage instead of sitting on a clean close-up.

The cameras cannot help here: all three go SDI straight into the switcher and none of them records locally. So the only routes to a clean tight angle are:

1. **Free AUX 2 via the VideoHub** (reroute the back-wall projector, as above), park a locked tight shot on it — and record it. **That last part needs a device that does not currently exist in the rig.** The Pi could potentially do it with a capture input, or a cheap standalone SDI recorder would.
2. **Cut better.** Free, but every future clip stays dependent on live decisions.

Be honest about the cost: option 1 is a reroute plus a recorder plus a camera dedicated to not being on program. That is real money and a real constraint for a benefit that is mostly about social clips. It is worth doing if clips matter; it is not worth doing to fix one bad week.

Engineering notes: Cameras

Three cameras, none of them manned. Since 2026-08-28: **two PTZs and one fixed static**, not the other way round. The second PTZ went in on ATEM input 1 and took the place of a static camera.

ATEM input	Camera	Address
1	**CAM 1** — FoMaKo UV602 PTZ	10.23.0.196
2	CAM 2 — JVC GY-HM600, fixed static, unmanned	not on the network
3	**CAM 3** — FoMaKo UV602 PTZ	10.23.0.181

They are FoMaKo UV602s (OEM product name `K30NS`), not PTZOptics. An earlier version of this file said PTZOptics — that was inferred from the Companion module name `ptzoptics-visca`, which is a generic VISCA module and says nothing about the make. Both cameras report identical firmware on every component.

That shapes everything below. There is no camera operator to give framing notes to — the static shots are whatever they were left pointing at, for the whole service, every service.

Every camera, every week

Setting	Value	Why
Output format	**1080p59.94** exactly	Not 29.97, not 1080i59.94, not 60.00. This is one of only two boxes that can be causing the frame doubling — see switcher-atem.md
Shutter	**1/125** at 59.94	172.6°, the standard film angle. **The PTZs cannot do 1/120** — their 60Hz shutter table is 1/100, 1/125, 1/180 and the 50Hz values are silently remapped onto it whatever the anti-flicker setting, so a request for 1/120 comes back as 1/125. Never drag it slower to gain exposure — if it needs to, that is a lighting problem
Auto shutter	**OFF**	
Gain	**0 dB**, manual	Gain is the noise. See the exposure section below — whether 0 dB is *achievable* is currently a lighting question, not a camera one
Iris	**F5.6** on the PTZs; match the statics by result, not by f-number	Metered, but unverified against a lit face — see below. Also the sharp end of the lens: at F8 on a 1/2.8" sensor the Airy disk is ~3.7 px wide and the picture is visibly diffraction-softened
White balance	**~4640K**, or better a One Push, locked	Metered at the preaching position on 2026-08-28 with a colour meter: **4640K**. That agrees to within 40K — about a tenth of a mired, invisible — with the figure derived independently from the 2026-08-23 recording (U 127.7–130.1, V 128.0–131.1 on a white shirt; neutral is 128/128). Two unrelated methods agreeing is why this number can be trusted
Auto white balance	**OFF**	The LED wall changes colour constantly and AWB drifts with it
Zebras	**On, 100%**	The white shirt must not trip them
Knee / highlight compression	**On**	Anything above ~90 IRE should roll off rather than clip flat

Better than dialling 4640K in as a number: **One Push** all of them off the same white card held at the preaching position, then lock. A number matches a camera to the *light*; one card matches the cameras to *each other*, which is the actual problem — see fault 3 in the README. The metered 4640K is then the sanity check that the One Push landed right, not the setting itself.

Blank the LED wall while you do it. The wall is behind the subject, so it contributes rim and floor bounce rather than key — the 4640K above was metered with the wall live and is demonstrably unaffected — but blanking costs nothing and removes the one variable that could quietly ruin a One Push on the card.

Metered exposure, and why it does not currently work

Measured at the preaching position on 2026-08-28 with a colour meter, incident, dome toward the camera:

Colour temperature	**4640 K**
Correct exposure	**F8 at ISO 200, 1/60 s** (358° at 60p)
Converted to the 1/125 the camera can actually do	**F5.6 at ISO 200** — halving the exposure time is one stop, so open one stop

Both PTZs are set to F5.6, 1/125, 0 dB. **That setting has never been verified against a correctly lit face, and the one attempt to verify it failed** — see below. Treat F5.6 as the metered starting point, not as a proven value.

The shutter is 1/125, not 1/60. 180° is 1/(2 × frame rate), so 59.94 wants 1/120 and the camera's nearest is 1/125. 1/60 would be right only at 29.97, and at 59.94 it is a 360° shutter — double the blur. The meter's own 358° reading is not a recommendation: it is the meter opening the shutter as far as it will go to find exposure in a dim room.

The face is not lit

Measured on 2026-08-28 with the lighting desk on its **preaching** state and a person sitting in the chair at the preaching position, LED wall blanked so only stage light fell on them:

Iris	Face
F5.6	6.8 IRE
F2.8	7.8 IRE
F1.8, wide open	**10.0 IRE**
F1.8 + maximum gain	64.8 IRE

The target is 65–70 IRE. **Wide open at no gain the face sits at 10 IRE**, and the only combination that reaches target is maximum aperture with maximum gain, which is unusably noisy. With the wall live the face reached about 21 IRE — meaning what light the camera sees on a face at that spot is mostly **spill from the LED wall behind them**.

No aperture fixes this. It is a lighting problem, which is exactly the case this file's shutter rule anticipates.

It also contradicts the recording. The 2026-08-23 file measured lit skin at 62 IRE across the sermon. Faces cannot land at 62 IRE off a state that meters at 10, so either a service runs different lighting from the state recalled on the 28th, or the preaching position under service conditions is not where it was measured. **Resolve that before changing any aperture** — re-metering against a real service is the check that settles it, and until then F5.6 stands because it is what a properly lit face would want.

One trap worth keeping: **do not stop down to tame the LED wall.** The wall is a light you control — the VW page has a brightness encoder on the NovaStar (`$(novastar:brite)%`, and it was found sitting at 10). Turn the wall down rather than closing the iris and taking the face down with it.

The f-number transfers between the two PTZs, which are the same model, and **does not transfer** to the static camera, which is different glass. Set that one by result — face at 65–70 IRE on a waveform — not by copying F5.6 across.

Both PTZs were on auto, and are now manual

Read off the cameras on 2026-08-28. Both were on **auto exposure, auto white balance and auto focus** — the prediction in this file, confirmed on the hardware. Both are now:

Exposure	**Manual**
Shutter	**1/125**
Iris	**F5.6**
Gain	**0 dB**
White balance	**4600K fixed**
Focus	**Auto** — deliberately left, it is the one auto worth keeping

Their full configurations were dumped and diffed group by group. **Twenty-five settings differed; they now differ in three**, all of them intended: the two device names (`CAMERA-1` and `CAMERA-2`), the NDI name that derives from the device id, and one inert register. Re-run that diff after any change — it is the only way to be sure they still cut together.

CAM 1 also had four things simply wrong, which are fixed: it was outputting **4K at 30 fps** into a 1080p59.94 switcher, running **50Hz anti-flicker** in a 60Hz country, had **auto slow-shutter on**, and carried **noise reduction at 8/8** against CAM 3's 2/4.

The remaining static camera is a JVC GY-HM600, identified on 2026-10-11 from a photo: three 1/3" CMOS sensors, a 23× Fujinon lens, and no network option at all (no USB HOST port, no Network menu), so not the 650 or 660 that share its body. It goes SDI straight into the switcher, so nothing about it can be read or set remotely; its only remote input is the 2.5mm LANC jack. White balance measured stable across the 23rd, so AWB at least appears locked.

It cannot output 1080p59.94. Its SDI is HD-SDI (1.5 Gb/s), and 1080p59.94 needs 3G-SDI. Sam believes it is set to 59.94, which over HD-SDI means 1080i59.94 (or 720p59.94) converted by the switcher. Confirm in its menu that it is not a 30p mode: that would double frames exactly as CAM 1's 4KP30 did.

Exposure targets

Subject	Target	Measured 2026-08-23
Face	**65–70 IRE**	62 IRE median on the sermon close shots
White shirt	**~90 IRE** (Y≈215)	96 IRE, with 79% of it clipped flat at 235

Faces are close to right. **The problem is the highlights, not the exposure level.** Do not stop down — faces would go genuinely dark. Use the knee, and put more light on the face than the chest.

An earlier version of this note claimed the face was at 49 IRE. That was a bad measurement — the sample box landed on the speaker's beard and mouth — and the advice it produced would have made the picture worse. Correction in the findings.

Framing — the cheap win

Because the static cameras never move, **a badly framed static camera is badly framed all service, every service.** On 2026-08-23 one of them carried an off-stage black band up to 161 px into frame, appearing on roughly a fifth of sampled shots — that is that camera, every time it was live.

The corollary is the good news: **frame them once, carefully, and it is fixed permanently.** That is probably the highest value-per-minute job on this whole list. Worth doing with a monitor, checking the edges rather than the

middle, and checking what the shot looks like when the speaker walks to the far side of the stage.

For the PTZ:

- **Move it off-air only.** On 2026-08-23 it ran a continuous four-minute zoom live during worship. Cut away, reframe, cut back.
- **Use presets.** Recalled off-air they are repeatable week to week and impossible to drift. The PTZ page is the natural place for them.
- Tally is **not** wired to the cameras, so nothing on the camera itself shows it is live — but the ATEM page on the deck does carry program and preview feedback, so the deck is where to look before moving it.

Adding another camera

The second PTZ went in on 2026-08-28. What it needed, in order, for next time:

- **Check the output format first.** CAM 1 arrived set to **4KP30** and went straight onto ATEM input 1 like that. Set `emVoutFormat` to `1080P59.94` and `VideoIn.nFrameRate` to 60 before anything else — changing the output format restarts the camera's video pipeline and drops every other write made in the same minute.
- **Set it to manual iris, shutter, white balance and low noise reduction.** Out of the box it is on auto everything. Numbers: 1/125, F5.6, 0 dB, 4600K.
- **Then dump both cameras and diff them.** Guessing at parity does not work; twenty-five settings differed on arrival and several were invisible from the UI.
- **Sweep its moiré band** — it will not be the same as the other camera's.
- **Save its presets last**, after the exposure is right, or they capture the auto settings.
- **Leave WDR / DRC off.** With a wall that swings whole-frame luma 42→99, wide-dynamic-range modes pump visibly and lift noise while they do it. A flatter gamma is the right lever on a PTZ; a true knee is a static-camera control.

The look, as saved on 2026-10-10

Re-saved 2026-10-11 11:03, mid-service: Sam corrected both cameras live, raising luminance from 0 to 80 and lowering the gain limit from 4 to 0, and both looks were saved from the cameras as they stood. The two cameras are identical. The table below predates that.

Each PTZ has a saved **look**: its whole picture block, kept on the Pi as `ptz_look_<address>.json` beside `ptz_web.py`. The Look key puts it back, and every preset key re-applies it straight after a recall. **The look is whatever was saved last, per camera; it is not a shared standard.**

Sam declared both cameras right on 2026-10-10, after the stage lighting was adjusted, and both looks were saved from the cameras as they stood (09:31). The looks they replaced, from 2026-09-10, are kept on the Pi in `~/ptz-look-backup-20261010/`.

Setting	CAM 1 (10.23.0.196)	CAM 3 (10.23.0.181)	Changed from the September look
Exposure mode	Auto	**Iris priority**, iris fully open (F1.8)	CAM 3 changed 19:14, see below
Autofocus zone	All	**Center**	CAM 3 changed 19:14, see below
White balance	4700K fixed	4700K fixed	
Red / blue gain	40 / 57	50 / 50	
Analogue gain	0	1	
DRC	0	0	CAM 3 was 8
Gamma	2	2	both were 1
Contrast	50	40	CAM 1 was 40

Setting	CAM 1 (10.23.0.196)	CAM 3 (10.23.0.181)	Changed from the September look
Luminance	0	50	CAM 1 was 50
Saturation, hue, sharpness	30, 4, 0	30, 4, 0	
Noise reduction 2D / 3D	0 / 0	0 / 0	

Since 2026-10-11 09:42 the two cameras are identical: Sam asked for CAM 3 to match CAM 1, so CAM 3 now carries CAM 1's column above (auto exposure, autofocus zone All, CAM 1's colour gains, gain, contrast and luminance). The CAM 3 column is what it held before. Match again only when Sam asks.

Brightness still follows the light on both cameras; the look fixes the colour and the image settings. The older sections below describe the Manual / 1/125 / F5.6 / 4600K standard of 2026-08-28, which the looks have not held since September.

Presets: CAM 3's six presets were re-saved at 19:15 and recall the look above by themselves, positions unchanged. **CAM 1's presets still carry the September picture**; a recall shows it for the moment before the deck re-applies the look.

CAM 3's wall-blur test (2026-10-10 evening)

The 2026-10-04 sermon close-up was focused on the **LED wall**, not the speaker: every LED crisp, his hair and the water-bottle label soft. Autofocus on zone **All** picks the highest-contrast thing in frame, and that is the wall. Auto exposure also closes the iris against the bright wall, which deepens focus and darkens faces.

So CAM 3 now differs from CAM 1 in two settings, to compare on Sunday:

- **Iris priority with the iris fully open.** Shutter and gain still follow the light, so wide shots do not blow out, but the lens stays as wide as it goes and the wall as soft as it can be.
- **Autofocus zone Center.** Focus comes from the middle of the frame, where the framed speaker is, not from the wall around him.

Expected: the grid on the sermon close-up mostly gone on CAM 3, still there on CAM 1. If it works, copy both settings to CAM 1 and re-save its presets. To undo: put the 09:31 look back (`ptz_look_10.23.0.181.prev.json`, also in `~/ptz-look-backup-20261010-evening/`) and re-save the presets. The deck's Exp and Auto keys change the exposure mode, so a press on either undoes the iris priority until the next preset recall.

Ended 2026-10-11, the wrong way round. Before the service CAM 3's close shots glowed: at F1.8 this lens halates around anything bright, and the thin focus landed on the wall behind the people, not on them (the wall's edge crisp, faces and white clothes soft with a halo; CAM 1 crisp on the same people). Sam: "just put it back make it match cam 1". At 09:42 CAM 3 was matched to CAM 1 (`ptz_web.py 10.23.0.181 match 10.23.0.196`) and its look re-saved; at 10:05 all six presets were re-saved with it (recall, `look apply`, save), and a bare recall of preset 1 came back on auto exposure and zone All. Backups of the test look and the camera as it stood are in `~/ptz-look-backup-20261011-cam3/` on the Pi. **Lesson:** a fully open iris costs more than the wall grid it hides; if the wall is tried again, stop down part way, never wide open.

A preset carries its exposure with it

Since 2026-10-11 the deck works round this. Every recall from a preset key or the dial is `ptz_web.py <host> recall <n>`: it reads the picture, recalls, and puts that picture back at once and again when the camera stops moving, so a preset is a shot and nothing else. Proven on both cameras from the real keys that afternoon. What follows still applies to a recall made anywhere else (the camera's web page, its IR remote), and to what a preset holds.

This is the one that silently undoes everything else. On these cameras a preset stores the exposure and white balance settings alongside the pan, tilt and zoom. Recall a preset and the camera is put back to whatever those

settings were when the preset was saved.

Proved on 2026-08-28: the camera was set to Manual / 1/125 / F5.6 / 0 dB / 4600K, preset 2 was recalled, and it came back Auto / 1/180 / F1.8 / 1 / Auto. Every preset on CAM 3 had auto-everything baked in, because they were all saved before the cameras were set to manual.

So: after changing exposure or white balance, re-save every preset. Recall it, re-apply the settings, save it back. The position is preserved because the recall put the camera there. All six presets on CAM 3 were re-saved this way on the 28th and now recall as Manual / 1/125 / F5.6 / 0 dB / 4600K. CAM 1 has no presets saved, so anything saved on it from now on captures the manual settings automatically.

The failure mode if this is skipped: everything looks right at the desk, then the first preset an operator presses in the service quietly puts that camera back on auto, and nothing on the deck says so.

Moiré on the LED wall

The rainbow interference across the wall is **moiré** — the camera's sensor grid beating against the wall's pixel pitch. It appears in a narrow band of zoom positions and is clean either side.

Measured by sweeping the zoom range and taking the high-frequency chroma energy over the wall, on static content:

	Moiré band	Clean
CAM 3 (10.23.0.181)	**52%–60% zoom**	below 50%, above 62%
CAM 1 (10.23.0.196)	**56%–60% zoom**	below 52%, above 65%

Inside the band it measures 10–25× the clean level, and it is unmistakable on screen. The bands differ between the two cameras because they sit at different distances and angles to the wall — **each camera has its own band, and a new camera needs its own sweep.**

"Clean" here means no rainbow, not no lines. The sweep measured colour only. The wall's plain LED grid, in brightness, shows at nearly every zoom: on the 2026-10-04 recording it was clearly visible on about half the service, worst on the sermon close-up (about 4 px per LED). Zoom does not remove that; blurring the wall does — see CAM 3's wall-blur test above.

Zoom is the only lever that works. Tested and found to do nothing:

- **Sharpness** — swept 0 to 15, moiré moved from 1.549 to 1.498. No effect.
- **Shutter** — swept 1/60 to 1/250 against static content; all within noise of each other.
- **Wall brightness** — swept 30 to 90 and got no consistent trend.

Do not spend time on those three. Frame outside the band instead.

CAM 3's preset 4 "Stage" was sitting at 57%, dead centre of its band, which is why the wall rainbowed on the shot most likely to be used. It was moved to 62% and re-saved; it measures 0.20 where it measured 2.14. All five presets now read clean on a worship background.

Reading and setting these cameras

The web UI is only a front end for an HTTP API that needs **no authentication at all** on the AV LAN:

```
# read everything
curl -s -X POST 'http://10.23.0.181/ajaxcom?szCmd={"GetEnv":{"VideoParam":{"nChannel":0}}}'
# what model, and the video output format
curl -s -X POST 'http://10.23.0.181/ajaxcom?szCmd={"GetEnv":{"SysAttr":{"nChannel":-1}}}'
curl -s -X POST 'http://10.23.0.181/ajaxcom?szCmd={"GetEnv":{"VideoOut":{"nChannel":-1}}}'
```

Groups worth knowing: `SysAttr`, `VideoOut`, `VideoIn`, `VideoParam`, `MonoTracking`, `ArmPtz`, `NetWork`. `{"QueryState":{"QueryVersion":{}}}` returns the component firmware versions.

Two traps in the write path, both of which cost a pass on the 28th:

1. `SetEnv VideoParam` **takes an ARRAY**, not an object: `{"SetEnv":{"VideoParam":[{"stExp":{...},"nChannel":0}]}}`. Send an object and it returns `{"nRetVal":0}` — success — and does nothing at all.
2. `antiflicker` **is only writable in auto exposure mode.** Set it before switching to manual, or dip back to auto to change it.

Always read the value back after writing. Both cameras' full configuration can be dumped and diffed this way, which is how they were proved identical.

TODO(sam)

Everything left needs hands on the equipment; none of it is on the network.

☐ **Why does the preaching lighting state put nothing on a face?** See the exposure section — this is the biggest open image-quality question on the rig.

☑ Make and model of the remaining static camera on ATEM input 2 — JVC GY-HM600

☐ Its output format, read off its menu: 1080i59.94 is the best it can send, and a 30p mode would double frames

☐ Are iris, gain and white balance manual or auto on that static camera?

☐ Its shutter speed, which should read 1/125 at 59.94

Engineering notes: Switcher

The **plain** HD8, not the ISO variant. It records the program feed only; see the ISO section in rig.md for what that costs and how to work around it.

At **10.23.0.31**, control on UDP 9910. It also answers the Blackmagic routing protocol on TCP 9990, which reports its unique ID as `1947523eab1a4015aa81b6a4ae2dfe43` — the same uuid stamped into the recording files, so there is no doubt this is the box that records the service.

Driven from the ATEM page: program bus over preview bus, CUT and AUTO at column 8, and two-step guarded Stream and Record keys (first press arms, second fires). The model spec in the `bmd-atem` module declares `streaming: true` and `recording: true`, which is what those keys rely on.

The one that matters — answered

Settings → Video → Video Standard must read `1080p59.94`. On 2026-08-28 it does.

And it turns out this **is** readable over the network after all, which this file previously said it was not. The HD8 answers the Blackmagic routing protocol on TCP 9990 and its status block ends with a `CONFIGURATION` section:

```
nc 10.23.0.31 9990 </dev/null | grep -A1 CONFIGURATION
#   CONFIGURATION:
#   Video Mode: 1080p59.94
```

So the switcher is not the frame-doubling fault. That leaves the cameras, and one of them was: the PTZ that went onto input 1 on 2026-08-28 was outputting **4K at 30 fps** into this 59.94 chain. It is now `1080P59.94` like the other. The remaining static camera on input 2, a JVC GY-HM600, is the last unverified box in the chain. Its HD-SDI cannot carry 1080p59.94, so the best it can send is 1080i59.94; a 30p mode would double frames. See cameras.md.

Evidence it is a single point in the chain rather than one bad camera: at a hard cut in the 2026-08-23 recording the duplicate-pair phase ran straight through unbroken, and the cut landed exactly on a pair boundary. A switcher cannot cut mid-pair, so the program feed reaching the recorder was already 29.97.

Check in this order and find the one box that is not 59.94:

1. Each camera's own output format — in the camera menu, not the ATEM's display. **Both PTZs are readable over the network**: `GetEnv VideoOut` returns `emVoutFormat`, where 20 is `1080P59.94` and 47 is `4KP30`. See cameras.md
2. ~~ATEM video standard~~ — **confirmed 1080p59.94 on 2026-08-28**, see above
3. Any converter, scaler or extender in the path

There is **no separate recorder** — the HD8 records its own program feed — so the fault is in one of only two places: a camera's output format, or the switcher's video standard. That is a much shorter hunt than it first looked.

If the chain genuinely cannot do 59.94 end to end, then **record 29.97 deliberately** — same motion, half the file size. What you must not do is record doubled 59.94.

Input mapping

Read off the switcher on 2026-08-28.

Input	Label	On the deck
1	Camera 1	program + preview
2	Camera 2	program + preview

Input	Label	On the deck
3	Camera 3	program + preview
4	Camera 4	program + preview
5	Words Overlay	program + preview
6	PP1B	program + preview
7	PP1	program + preview
8	Camera 8	**nothing**
3010	Media Player 1	program + preview, in the eighth slot

The labels are stale. As of 2026-08-28 the cameras are: input 1 = a PTZ (10.23.0.196), input 2 = the fixed static, input 3 = the other PTZ (10.23.0.181). The switcher still calls input 8 "Camera 8" and inputs 3 and 4 "Camera 3" and "Camera 4"; relabelling them on the ATEM updates the deck automatically, because those keys read `$(atem:long_N)`.

Inputs 1–8 all report BNC connected. The deck's tally variables were set to match the patch on the 28th — `ptz_atem_input` = 3 and `ptz2_atem_input` = 1 — but that came from the patch as described rather than from tracing cable, so **it is worth eyeballing that the Stop key lights on the right camera.**

TODO(sam)

☑ ~~Current video standard setting~~ — `1080p59.94`**, read over port 9990 on 2026-08-28**

☐ Can it be set to H.264 High profile, and a higher audio bitrate? Companion exposes no variable for the encoder settings and the 9990 status block does not carry them, so this one really does need ATEM Software Control

Tally

Not wired. No camera shows whether it is live. Since two cameras are static and unmanned that matters less than it sounds — but it is exactly the gap behind the PTZ being zoomed for four minutes on air.

The deck covers it in practice: the ATEM page already carries program and preview feedbacks, so whoever is driving the PTZ can see what is live before touching it. Worth knowing that is the only indication there is.

Outputs

AUX 1 feeds the LED wall and the two flanking projectors; AUX 2 feeds the back-wall projector. Both in use, but AUX 2 can be freed by rerouting the back-wall projector through the VideoHub — see rig.md.

Engineering notes: Audio

Driven from the SQ7 page: DCA 1–8 across row 1 with level and mute state, plus the Stream / Foyer / MAIN readouts on the touchstrip. The DCAs are what a service is actually mixed on.

Targets

	Target	Measured 2026-08-23
Integrated loudness	**−14 LUFS**	−8.0 / −11.8 / −9.5 across the three files
True peak	**−1.5 dBTP**, never above	+1.8 / +1.4 / **+3.5** dBTP
Speech loudness range	~8–11 LU	13.2 LU on the sermon — too wide

All three files exceeded 0 dBFS, so **real distortion is baked into the recording** — the decoded audio reaches ±1.367 on file 03. Post cannot undo that.

Being 6 LU hot buys nothing. Every platform normalises it back down and you keep the distortion.

The stream is affected too

The livestream and the recording come off the same audio path. So the +3.5 dBTP clipping measured on the 23rd did not just land in the recording — it went out live as well, and everyone watching heard it.

That raises the priority of the limiter: it is not a post-production convenience, it is the only thing standing between a hot desk and distortion reaching the congregation online. It also means one fix covers both outputs.

At the desk

- Pull the feed to the recorder down about **6 dB**.
- Put a **limiter at −1.5 dBTP** on that feed so nothing can reach 0 dBFS.
- More compression on the pulpit mic — 13.2 LU of range is too wide for speech.
- Check the SQ7's output meters are not pinned. If they are, the problem is upstream of the recorder and the limiter is only hiding it.

Note for post

Encoding to AAC adds roughly **0.8 dB** of true peak. Audio measuring −1.1 dBTP going in comes out at −0.3. The post pipeline limits to −2.5 dBFS before the encoder to absorb this — see `Services/_tools/02-grade-join.sh` on the media drive.

TODO(sam)

Attempted on 2026-08-28 and not answerable: the SQ7 was powered down, and in any case Companion's SQ module publishes DCA and matrix levels but nothing about output routing or insert processing. Both of these need the desk itself.

- ☐ Which SQ7 output feeds the recorder, and at what level
- ☐ Is there a limiter on that path already?

Engineering notes: Recording

Since 2026-10-26 the service records on a HyperDeck, and the ATEM only streams (Sam: "we added a disk recorder to the network so let's use that. we will be recording on that and streaming on the HD8"). Everything below the HyperDeck section is how the ATEM recorded before that, kept because the board still tends the ATEM's drive and recording could move back.

The HyperDeck

Read off the deck over its control port (TCP 9993) on 2026-10-26.

Model	Blackmagic **HyperDeck Studio Mini**, unique id `7c2e0d0cbd95`
Address	**10.23.0.206** on the AV LAN (DHCP: give it a reservation, or it will drift like the rest)
Feed	Videohub OUT 13 "Deck Recorder", which carried IN 4 "Program" that night. SDI in, 1080p59.94 in and out
Media	**Two SD card slots, nothing else**: it cannot record to the 4 TB USB drive. Both slots were **empty** on 2026-10-26
Format	H.264 High, AAC, 2 channels embedded, timecode from the clip. That is High profile where the ATEM gave Main
Remote	Enabled on the deck's front panel; Companion cannot record without it

What starts it. The Record keys on Home r2c3 and ATEM r3c1 toggle it, the PP1 Countdown key starts it, and the Sunday 10:24:57 trigger (which copies the Countdown key) starts it. The Countdown key and the trigger first route Videohub IN 4 "Program" to OUT 13 "Deck Recorder" (Sam, 2026-10-27), so a route left on that output cannot become the recording. Then they start the ATEM's **stream** (Sam, same day: "I'd like the stream to be automatic"), with the module's On, not its toggle, so a second Countdown press cannot end a live stream. The guarded Stream key still starts and stops it by hand. Nothing on the deck starts the ATEM's recording any more. Built by `tools/record-hyperdeck.js`; the trigger's action was swapped on the rig in place, because the trigger copies the key only when it is built.

What the deck shows. The Record keys are red with the clip's running time while it records. Idle, they show the time left on the card, or go amber with **NO CARD** when neither slot has one. The Service Health board's recording card reads the same through Companion (it has no session of its own: some HyperDecks take one control connection, and Companion's must win).

Not yet proven with a card in. On 2026-10-26 a Record press reached the deck and was refused with `105 no disk`, as it should be. The module's `slotId`, `recordingTime` and its spelling of the recording status cannot be read until a card is in, so check on the first one: the key should show hours left, a press should turn it red and count up, and the board should go to REC.

Cards. At the 40–52 Mb/s the ATEM's H.264 ran at, an hour is about 23 GB, so a 128 GB card holds roughly five hours. The deck's own rate is unmeasured until the first file. Use cards from Blackmagic's supported media list for the Studio Mini and format them in the deck.

Before 2026-10-26: the ATEM recorded

The ATEM Television Studio HD8 recorded its own program feed directly to disk.

Media

A **4 TB SanDisk** drive — the same physical drive that mounts as `/Volumes/Hard Drive` for post. It holds hundreds of hours and sits at well under 1% used, so **space is not a consideration** and was not a factor in anything that went wrong.

Note the drive therefore travels between the church and the edit desk. Nothing durable should live on it for that reason: it is media and scratch, not storage. Documentation lives in git; the render pipeline lives at `Services/_tools/` on the drive alongside the media it operates on.

Which disk

Read off the switcher on 2026-10-09 and tested on it, idle, on 2026-10-10.

Disk	Id	Free
`Hard Drive` — the USB drive above	301	115 h
`Internal` — the HD8's own SSD	1	62 h

What records is the disk the switcher flags Active. That is usually the first working-set slot, but not always, and three things move it:

What happens	Slots	Active
Power reset with the drive in	Internal, then the drive	Internal
Drive unplugged	empty, then Internal	Internal
Drive plugged into a running switcher	the drive, then Internal (Sam, 2026-10-09)	the drive
Switch Disk (the module's `recordSwitchDisk`), idle	unchanged	moves to the other slot
Setting the slot order	as set	stays on the slot position it was on

The rig power-cycles every Sunday morning, so left alone **every Sunday starts with Internal recording**. That was the "records to internal" fault.

The fix is in the Service Health board (Decima-Labs/service-health, `driveCommand` in `src/atem.ts`), not on the deck. It already holds a session to the switcher; while the switcher is idle and a formatted drive is present it makes the drive the first slot and then the Active disk, checking the state after each step. Proven on the switcher on 2026-10-10: with the power-reset order set by hand, the drive was back first and Active within 20 ms. It cannot be done from Companion: `bmd-atem` 4.4.0 has only the Switch Disk toggle and reports only the first slot's name, as `$(atem:record_disk_volume)`, so a deck-side "switch if Internal" would toggle back on the next press and its own readout could name the drive while Internal records.

What the deck shows (`tools/record-disk.js`, on the rig since 2026-10-09): the Record keys on Home r2c3 and ATEM r3c1 name the first slot while idle and turn amber when it is `Internal` or there is `NO DRIVE`. The board keeps the first slot and the Active disk the same, so the key reads true; if the board is down after a power reset the key is amber, which is also true.

What it produced

2026-08-23: 1080p59.94 container, H.264 **Main** profile, yuv420p, bt709 limited range, 40–52 Mbps, AAC-LC 128 kbps stereo. Metadata carries uuid `1947523EAB1A4015AA81B6A4AE2DFE43-0`.

Two things worth changing if the HD8 exposes them:

- **Main profile → High.** A free quality win at the same bitrate (8×8 transform).
- **128 kbps AAC → higher.** Low for a music-heavy service.

The 7.6 second gap

Cause: known. A manual stop/start mid-service. Not the recorder rolling over, not media, not disk space — all three are cleared.

The useful fix is not "be more careful". It is making the mistake visible and hard to make, and there are two cheap moves:

1. **A record-state indicator on the deck. — Shipped.** Both Home `2,3` and ATEM `3,1` carry a Record key showing `$(atem:record_duration_hm)` with the module's own `recordStatus` feedback colouring it, so a stopped recording is now visible at a glance on two pages.
2. **The Record key's two-step guard did not ship.** Checked against the live config on 2026-08-28: the Stream key has two steps — arm, then fire — and the Record key beside it has one, going straight into `recordStartStop` on a single press. An accidental single press is exactly this failure, and it is still one press away. **This is the open half of the fix.**

TODO(sam)

Needs ATEM Software Control — Companion publishes nothing about the encoder.

- ☐ Can the HD8 be set to High profile? A higher audio bitrate?

Engineering notes: Checklists

Pre-service

The first block matters most — every item in it is destroyed at capture if you get it wrong, and no amount of post fixes it.

Format — do this first

- ☐ Every camera menu reads `1080p59.94` (check the camera, not the ATEM display)
- ☐ ATEM video standard reads `1080p59.94`
- ☐ Recorder **input** format reads `1080p59.94`
- ☐ Recorder **record** format reads `1080p59.94`
- ☐ Any converter / scaler / extender in the path is 59.94

Both PTZs answer from a laptop on the LAN, so the camera half is one command:

```
for ip in 10.23.0.181 10.23.0.196; do
  curl -s -X POST "http://$ip/ajaxcom?szCmd={\"GetEnv\":{\"VideoOut\":{\"nChannel\":-1}}}"
done   # emVoutFormat must be 20 = 1080P59.94. 47 is 4KP30, which is how CAM 1 arrived
```

Then prove it. Record 10 seconds and run:

```
ffmpeg -i test.mp4 -an -filter_complex \
  "[0:v]split=2[a][b];[b]trim=start_frame=1,setpts=PTS-STARTPTS[b2];[a][b2]psnr" \
  -f null - 2>&1 | tail -3
```

Then read the per-frame log. **Good:** consecutive-frame PSNR is roughly even. **Bad:** it alternates high/low with a gap of 10 dB or more — that is every second frame carrying no new picture, and it is what is wrong today.

Cameras

Do each line across **all** cameras before moving to the next. Matching is the point, and taking one camera all the way through then starting the next is how they end up unmatched.

- ☐ Shutter `1/125`, auto-shutter off, on all cameras. **Not 1/60** — that is 180° only at 29.97, and this chain is 59.94
- ☐ Gain `0 dB` and iris manual on all cameras — `F5.6` on the PTZs, the static matched by result (face 65–70 IRE on a waveform), not by copying the f-number
- ☐ White balance fixed at `4600K`, AWB off. Better still, blank the wall and **One Push** every camera off one white card at the preaching position without moving the card
- ☐ **Recall each PTZ preset and check it still reads Manual.** A preset stores the exposure it was saved with, so any exposure change has to be re-saved into every preset or the first press in the service undoes it
- ☐ **No preset sits inside its camera's moiré band** — 52–60% zoom on CAM 3, 56–60% on CAM 1. Look at the wall on each preset; rainbow banding means reframe and re-save
- ☐ Wall brightness set so it sits *under* the face — the VW encoder, never the iris
- ☐ Zebras at 100% — the white shirt must not trip them
- ☐ Knee / highlight compression enabled
- ☐ Pan/tilt locks engaged on static shots
- ☐ The off-stage black edge is out of frame on every camera

Audio

- ☐ SQ7 output meters not pinned

- ☐ Feed to the recorder pulled down ~6 dB
- ☐ Limiter engaged at −1.5 dBTP on the recorder feed
- ☐ Pulpit mic compression increased

Recording and control

- ☐ Seamless file splitting confirmed on
- ☐ Enough space on the media, and the media is healthy
- ☐ `node tools/rig.js export` run, so there is a current Companion backup
- ☐ **Every connection on the deck is green** — addresses drift, and on 2026-08-28 three of the four Shure receivers were silently pointing at dead hosts. Companion's status dots lie about this, so check a readout on each page rather than the connections list
- ☐ Deck is on the page you want to start from (it resumes wherever it was left)

Operators

- ☐ Move the PTZ **off-air only** — cut away before reframing
- ☐ Hold locked frames at least 10 seconds; do not micro-reframe
- ☐ Face is the priority for light and focus, not the chest

Last thing

- ☐ Record 10 s and confirm the white shirt still holds fabric detail

---

Post-service

Getting the files off

- ☐ Copy the recorder files to `Services/YYYY-MM-DD/originals/` on the media drive
- ☐ Name them `YYYY-MM-DD Sunday Service NN.mp4`
- ☐ **Never modify anything in** `originals/` — every later step reads from it
- ☐ Note any stops or restarts during the service, and roughly when

Run the pipeline

Scripted at `Services/_tools/` on the media drive; its README explains each step and why.

```
./01-fix-source.sh YYYY-MM-DD   # de-duplicate frames, normalise loudness
./02-grade-join.sh ...          # lift the face, dissolve any recording gap
./03-title-card.sh ...          # 3s card into a 1.5s dissolve
./04-transcribe.sh ...          # whisper transcript, for chapters and clips
./05-clips.sh ...               # 16:9 and 9:16 clips
python3 cover.py                # thumbnail
python3 cover.py card.png 1.5   # title card
```

Each script prints a verification command when it finishes. **Run them.**

Before uploading

- ☐ Loudness reads −14 LUFS integrated, true peak under −1.5 dBTP
- ☐ Full decode is clean: `ffmpeg -v error -i final.mp4 -f null -`
- ☐ Opening title card and dissolve look right
- ☐ Any recording gap is covered by a dissolve, not a jump cut

- [ ] Chapter times match the file being uploaded — the title-card version is ~3 s offset from the no-card master

Write down what went wrong

Anything noticed during the service — a stop, a bad cut, a camera that drifted — goes in `docs/production/findings/YYYY-MM-DD.md`. That file is what makes next week better.

Engineering notes: Findings, 2026-08-23

Measured off the actual recording, not estimated.

Corrections to earlier notes

Two claims in the first pass were wrong. Both are recorded here because the advice they produced would have made the picture **worse**, not better.

The face was not underexposed

The first pass reported the speaker's face at 49 IRE, 47 IRE below his shirt, and concluded "the key light is on his chest, not his face". That measurement used a fixed sample box which landed on his **beard and mouth**, not lit skin.

Measured properly — forehead patch, then skin-tone detection across 13 frames spread through the sermon:

	Measured
Forehead, single frame	**74 IRE**
Lit skin, median across the sermon	**62 IRE**
Same, after the post grade	**69 IRE**

62 IRE is slightly under the 65–70 target, not a fault. **The problem is the highlights, not the exposure level** — the shirt still clips 79% flat at 235.

The corrective action inverts with the correction: enable the camera's **knee**, do **not** stop down. Stopping down would have pushed faces genuinely dark.

Lesson: never measure a moving subject with a fixed box. Detect the subject, or verify the box visually first. Skin detection that worked: chroma window U 95–127, V 138–172, luma > 60, upper ¾ of frame only, then take the 75th percentile of luma across many frames.

Whole-frame median luma is not scene brightness

The first pass used "median luma 38–43 out of 235" as evidence of a very dark stage. That figure is dominated by the near-black LED wall filling most of the frame and says almost nothing about light falling on the subject. The stage lighting is in fact reasonably bright — it had recently been upgraded.

PTZ behaviour during worship

Confirmed: it makes long moves live on air. Between 04:00 and 25:00 there are only about ten cuts in twenty-one minutes, with unbroken shots of 295 s, 266 s, 213 s and 203 s. The 14:13–18:39 shot is a continuous four-minute zoom from a close-up of one singer out to the whole team, entirely on air.

Ruled out, with evidence

Recorded so nobody chases them again.

Hypothesis	Verdict	Evidence
Auto-iris pumping	**No**	Whole-frame luma swings 42→99 as the LED wall cycles, while the singers' skin holds at 35–52. If iris were chasing the wall, skin would move with it

Hypothesis	Verdict	Evidence
Auto white balance drifting	**No**	Skin chroma holds at U 118–124 / V 143–153 across four minutes of the wall cycling pink→blue→purple. That is a locked WB
Frozen / dropped frames	**No**	5.7% of frames looked near-duplicate, concentrated after minute 26. False alarm: during announcements the camera is locked, the speaker nearly still, and the image clean enough (noise sigma 0.68) that consecutive frames are legitimately near-identical. Real frame doubling **alternates** every other frame; this was uniform across all of them

Measurement techniques that do not work

- **Scene detection to find shots when a PTZ is zooming.** A zoom is not a cut, so `select='gt(scene,...)'` reported the four-minute zoom as one static shot. It also fires ~125×/minute on the pre-service countdown video.
- **Temporal per-pixel noise as a camera fingerprint.** It is dominated by camera *motion*, not sensor noise — a locked shot of a still speaker measured 0.00 sigma, which is impossible for a real sensor. Any camera comparison has to control for motion first.

What post did

Pipeline and reasoning in `Services/_tools/README.md` on the media drive.

	Before	After
Lit skin	62 IRE	**69 IRE**
Shirt clipping	78.9%	78.8% (deliberately unchanged)
Background luma	28.1	28.2 (deliberately unchanged)
Loudness	−14.5 LUFS / −1.0 dBTP	**−14.5 LUFS / −1.7 dBTP**
Frame timing	0.118 s PTS gap at the join, marginally variable	true CFR 29.97
The 7.6 s gap	hard jump cut at 13:27	0.5 s cross dissolve

Delivered: graded master, title-card version, 6 clips × 16:9 and 9:16, SRT captions, 14 chapters.

The vertical-clip trap

Worth its own note. Auto-reframing 16:9 → 9:16 by tracking the subject each frame **visibly shakes**, because the luma centroid follows the speaker's *arms*. The first attempt had one clip's crop travelling 2301 px inside a 432 px span.

What works: split at shot cuts, take the **median** position per shot, and decide whether a shot needs to move at all from the 10th–90th percentile of subject position rather than min/max. Most shots then lock outright, and a crop that never moves cannot shake. Only genuinely traversing shots get a follow, sampled at 12.5/s so each step is ~5 px.

For auto-reframe, stillness beats accuracy. An imperfectly centred locked frame looks professional; a perfectly centred moving one looks broken.

Resolved since

The 7.6 s recording gap was a manual stop/start by the operator mid-service. Not the recorder rolling over, not media. See `../recorder.md` for the two cheap changes that make it visible rather than relying on care: a record-state indicator on the deck, and confirming the Record key's two-step guard shipped.

Still unresolved

- The 29.97-inside-59.94 fault is **not traced to a box**. Needs someone walking the chain at the desk — camera menus, then the ATEM video standard, then the recorder.
- Cameras are not matched to each other; skin level varies between wide and close shots.
- The HD8 is the plain variant, so there is no ISO recording. Every clip inherits the live cut — see the ISO section in ../rig.md for the ways around it.